

**Treatment Success and Failure Prediction Using Machine Learning Models
for Pulmonary Tuberculosis Patients Under the Department of Health-Cordillera
Administrative Region (DOH-CAR) Tuberculosis-Directly Observed Treatment,
Short-course (TB-DOTS) Program**

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Abstract

Pulmonary Tuberculosis (PTB) is still lingers as a big public health concern in the Cordilleras, where differences in patients and gaps in the treatment program still contributes to treatment failure. This study develops and evaluates machine learning models to predict treatment success or failure among patients in DOH-CAR TB-DOTS facilities for the years from 2015 to 2025. A preprocessing pipeline performs the following: schema harmonization, patient pseudonymization (for anonymization), and a diagnostic-first missing data handling protocol. The protocol classifies each variable's missingness mechanism as Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR), then routes the variable to one of four strategies: column exclusion, listwise deletion, missing indicator imputation, or stochastic Multivariate Imputation by Chained Equations (MICE). Supervised tree-based classifiers (XGBoost, LightGBM, and Random Forest) are trained alongside Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) temporal architectures to model longitudinal treatment dynamics over the 6- to 12-month treatment course. Because clinical decision support needs more than an accurate risk score, Shapley Additive Explanations (SHAP) are used to show which patient-level factors contributed to each prediction so that clinicians can review the system's outputs. The validated models are embedded in a Clinical Decision Support System (CDSS) that provides dynamic risk estimates, SHAP-grounded factor explanations, and bounded narrative summaries. The CDSS is intended to support early clinical intervention while preserving clinician judgment.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

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Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by *Mycobacterium tuberculosis* (MTB), a slow-growing, acid-fast bacillus. Because the disease progresses gradually and treatment can fail, control is hard (Tobin & Tristram, 2024). Effective treatments exist, but PTB remains widespread. It tracks closely with poverty, undernutrition, and comorbidities such as diabetes and HIV, all of which weaken patients' immune defenses and raise susceptibility to the disease (Tobin & Tristram, 2024). These conditions also delay diagnosis and complicate treatment. TB is one of the leading causes of death worldwide — in 2018 alone, there were around 10 million new cases and 1.5 million deaths, and about half a million of those were resistant to rifampicin.

In the Philippines, TB still remains as a big problem with regards to public health, and the World Health Organization (WHO) lists the country as one of the biggest contributors to the global TB burden, estimated at around 6.8 percent of cases coming from the Philippines. Before the COVID-19 pandemic, Philippine TB rates were also among the highest in Asia: over 554 cases per 100,000 people in 2019 (Florentino et al., 2022). In 2023, about 37,000 Filipinos died of TB out of 739,000 diagnosed cases (World Health Organization, 2023), and the WHO classifies the country as a high-burden setting for both MDR-TB and TB/HIV co-infection (The Borgen Project, 2025). The DOH recorded 575,770 TB cases that year through its Integrated TB Information System, a figure that points to substantial under-registration and underdiagnosis (Baron, 2025). The DOH applies CDC-recommended regimens for active and latent TB through the TB-DOTS strategy, which was expanded in 2003 to cover private clinics, pharmacies, and hospitals (Bendre et al., 2021; Department of Health [DOH], 2014). Gaps in diagnosis, treatment monitoring, and patient support nonetheless remain, particularly in the management of MDR-TB (Stop TB Partnership, 2021).

In Baguio City, the DOH-CAR reported 437 confirmed TB cases in 2022 along with

over 4,405 missing or unreported cases, making the city a priority for surveillance and treatment improvement (Department of Health - Republic of the Philippines, 2022). The gap is driven by passive case-finding limitations, stigma, and financial barriers that let undetected patients keep transmitting the disease within their communities (Kaur & Sharma, 2021).

The large population and a lot of daily traffic (both foot and vehicular) in the city may increase opportunities for tuberculosis to spread. The colder climate can also create more ideal conditions for the transmission of *Mycobacterium tuberculosis*. The city also has a more inflated local load of cases because it takes in more complex tuberculosis cases from neighboring provinces because of it being a tertiary referral center (Khalid et al., 2022). Patient-level factors also play an important role in treatment outcomes such as comorbidities like diabetes mellitus, HIV infection, and malnutrition, all of which weaken their immune response and may reduce treatment completion. Food insecurity, poor housing conditions, and transportation costs, can also interrupt adherence to directly observed treatment short-course (DOTS). (Austin et al., 2021; Dong & Peng, 2013; Khalid et al., 2022; Sisk et al., 2023).

All in all, these constraints argue for strategies that work on more than one facet of assessment. Treatment failure not only worsens the outcome for the singular patient, it also extends the period for which transmission can happen in the community and in-turn speeds up the emergence of multidrug-resistant tuberculosis (MDR-TB), defined as resistance to at least rifampicin and isoniazid. MDR-TB requires regimens that are longer, more toxic, and more expensive, and it carries much lower success rates. The result is a wider threat to regional disease control in the Philippines (Aracri et al., 2025; Groenwold et al., 2012; World Health Organization, 2022).

Risk Factors for Treatment Failure

The Department of Health (DOH) established standardized criteria for classifying treatment outcomes through the National Tuberculosis Program (NTP), so programs can compare findings across settings and track what predicts outcomes. Treatment success and failure are not random

— they tend to follow identifiable patient, clinical, and program-level patterns. Mortality is the primary outcome most studies on treatment failure examine, though some also look at disease progression, drug-resistant TB, and loss to follow-up (Department of Health [DOH], 2014).

One of the risk factors is close or direct contact with patients who have active TB infection where patients who acquired the infection through close contact were at higher risk of delayed diagnosis and more likely to be non-adherent, either by refusing or quitting the regimen (Lamsal et al., 2009). Another risk factor is recurrence of the infection or a history of receiving TB treatment because patients who missed follow-ups had a higher risk of reactivation and mortality, and were harder to treat than patients with no prior history of TB (Tok et al., 2020). Frequent missed or interrupted treatment also leads to drug resistance (Ndambuki et al., 2020).

Comorbidities push TB treatment outcomes in a consistent direction across settings. In Eastern Ethiopia, HIV co-infection was a substantial risk factor for treatment failure because the weakened immune system makes treatment less effective (Amante & Abdosh, 2015). TB and HIV drug interactions also accelerate each other's progression and increase the occurrence of side effects, which can compromise treatment success (Azeez et al., 2018). A study in Davao City, found a higher risk of treatment failure in comorbid patients: 58.82% of the 51 patients who experienced treatment failure had at least one comorbidity (Hinay et al., 2025). Another major factor is Diabetes Mellitus (DM). TB patients with diagnosed DM have a higher risk of treatment failure, relapse, and death than non-diabetics (Noubiap et al., 2019). In a systematic review and meta-analysis, DM patients had a higher risk of treatment failure for Drug-resistant (DR) TB and mortality for MDR-TB, with odds ratios of 1.56 and 1.57 respectively (Xu et al., 2023). Another study reported that patients with both PTB and chronic conditions, especially DM, had reduced rates of sputum conversion and a higher probability of poor treatment outcomes, including death and progression to MDR-TB (Siddiqui et al., 2016).

These findings show that TB treatment failure cannot be driven by any single factor, it comes from how comorbidities, programmatic gaps, and socioeconomic vulnerabilities interact. Conventional single-factor analyses do not capture this. An integrative machine learning framework

that can model these multidimensional interactions is therefore needed to enable earlier and more targeted clinical intervention (Amante & Abdosh, 2015; Department of Health - Republic of the Philippines, 2022; Hinay et al., 2025; Khalid et al., 2022; Noubiap et al., 2019; Peng et al., 2024; Rajpurkar et al., 2022; World Health Organization, 2022; Xu et al., 2023).

Machine Learning for TB Outcome Prediction

Previous studies on tuberculosis treatment outcomes have focused on sociodemographic characteristics, comorbidities, medical history, and other patient-level risk factors. The findings remain inconsistent and fragmented because methodological constraints limit how these factors are analyzed together. Most studies rely on traditional statistical analyses such as statistical tests or regression models, which examine one factor at a time, or a limited set of factors, while assuming linear relationships among variables (Amante & Abdosh, 2015). These approaches provide important evidence on individual risk factors, but they may miss the combined and non-linear effects that shape treatment success or failure.

Algorithms that learn from high-dimensional data can pick up subtle dependencies that affect whether treatment succeeds or fails, and this is where machine learning (ML) has an edge over conventional statistical analysis. The interactions among demographic, clinical, diagnostic, and treatment-related variables are complex and non-linear, and those are exactly the patterns single-factor methods are most likely to miss. Logistic regression, Support Vector Machines (SVM), and ensemble methods like Extreme Gradient Boosting (XGBoost) have all been applied to this kind of problem (Rajpurkar et al., 2022). In TB specifically, earlier work has used ML to predict drug resistance, flag high-risk patients, and optimize treatment strategies (Kaur & Sharma, 2021; Kossakov et al., 2024). What this shows is that predictive models can take risk factor analysis further by pulling together several dimensions of patient data at once.

Clinical decision support systems have to explain themselves. An ML model that returns a risk score with no indication of what drove it gives TB-DOTS staff nothing to act on — a high

failure-risk flag without contributing factors is hard to trust and harder to follow up on. The study therefore uses Shapley Additive Explanations (SHAP), a post-hoc technique that quantifies each variable's contribution to a prediction and produces both patient-level explanations and a global feature importance summary (Lundberg & Lee, 2017a; Lundberg et al., 2020). This is useful for tree-based clinical models such as XGBoost, Random Forest, and LightGBM, where predictive performance can be high but the internal decision structure is not transparent. In the healthcare ML literature, SHAP is among the most widely used methods for explaining prediction models, particularly for tabular data (Allgaier et al., 2023). Studies on clinical ML explainability have used it to identify meaningful contributors to treatment failure, including drug resistance, TB history, and lab markers (Peng et al., 2024; Sibanda & Ndlovu, 2026; Wang et al., 2025).

Fragmented risk factor studies have clear limits, and interpretable ML offers a way past them: analyzing what drives TB treatment outcomes calls for combining several dimensions of patient data rather than examining them one at a time. The present study uses a multi-layer framework over demographic, clinical, diagnostic, and treatment-related variables. Descriptive analysis characterizes patient profiles and treatment patterns, supervised models estimate treatment outcome risk, and SHAP-based explanations show which factors drive each risk estimate. This improves predictive performance while keeping the outputs transparent for healthcare providers (Shickel et al., 2018; Smith et al., 2021), helping them spot patients at higher risk of failure, see the factors behind a given estimate, and act through early intervention, closer monitoring, and better patient management.

Contributions

The DOH-CAR TB-DOTS program follows the National Tuberculosis Program (NTP) protocol. Case finding is passive and starts at the facility. People with TB symptoms such as a long-lasting cough, weight loss, and fever are sent for testing, including sputum smear microscopy, the GeneXpert MTB/RIF test, and chest X-ray (Bendre et al., 2021; Department of Health [DOH],

2014). Once diagnosed, a patient is recorded in the Treatment Card and Master List, given a registration group, and assigned to a health worker or trained treatment partner who watches them take each dose over the six- to twelve-month course (Department of Health [DOH], 2014). Monthly visits track sputum results, record vital signs and weight, and check whether the patient is keeping up with treatment. Outcomes are later classed under NTP criteria as Cured, Treatment Completed, Failed, Died, or Lost to Follow-up (Ting, 2011). The process is mostly paper-based, and risk is judged in hindsight. Problems such as a worsening condition, missed doses, or complications from other illnesses usually surface only after they have already affected treatment, not before. High-risk patients are not flagged in time for early action, which leads to avoidable treatment failures, more deaths, and continued spread of drug-resistant (MDR) TB in the region (Khalid et al., 2022; World Health Organization, 2022).

The research presents six major contributions. The first contribution is developing and validating an Integrated Predictive Model utilizing machine learning models that combine training sets based upon demographic, clinical, diagnostic and treatment related variables to forecast therapy success or failure of PTB patients in CAR. Therefore giving clinicians the opportunity to identify potential problem patients earlier.

Secondly, a reproducible Data Preprocessing & Harmonization Pipeline which takes raw patient information from TB-DOTS clinics throughout the CAR and transforms the datasets for machine learning algorithms' use, using a variety of techniques such as schema normalization; pseudonymous identification of individual patients; automatic correction of errors; and a diagnostic-first approach to handle missing values.

Thirdly, a Descriptive/Exploratory Analysis that describes the study population; provides insight into how the various attributes relate to one another; and identifies the most important predictors of treatment outcomes.

Fourthly, Temporal Modeling to examine time dependent relationships in the monthly monitoring data using Recurrent Neural Networks (RNN), Long Short Term Memory (LSTM) architectures and therefore capture variations that occur over time during treatment. Therefore,

temporal modeling is expected to generate predictive accuracy that improves upon prior studies which did not account for temporal changes in monitoring data.

Fifthly, Shapley Additive Explanation (SHAP) were used to measure the degree to which each patient-specific factor contributed to a prediction. SHAP are also useful because they allow clinicians to see what specific factors led to each predicted risk's score and thus help evaluate whether or not a particular recommendation should be implemented for each patient.

Lastly, the validated model was embedded into a Clinical Decision Support system (CDSS) that displays a patient's current level of risk; lists all the factors that contribute to each risk assessment via SHAP; and updates a patient's risk estimate when new follow-up data becomes available.

Research Objectives

This study aims to develop and validate machine learning models that predict treatment success or failure among TB patients in the CAR, Philippines. Combining demographic, clinical, and laboratory variables supports early identification of patients at risk of unsuccessful treatment and provides evidence-based tools for improving TB program outcomes. The broader goal is to enable personalized clinical interventions and efficient allocation of healthcare resources. The research aims to achieve the following objectives:

1. To document and pre-process patient records of PTB cases from TB-DOTS facilities in the CAR through data cleaning, feature engineering, and reporting, so that the dataset is ready for ML applications.
2. To describe the demographic, clinical, diagnostic, and treatment characteristics of PTB patients in the dataset and provide an overview of the study population.
3. To identify variables that influence treatment outcomes over the course of therapy, so that the most important factors affecting patient recovery can be recognized and used to guide treatment adjustments.

4. To add temporal modeling that analyzes time-dependent patterns in patient data, so that changes and trends throughout treatment can be detected, improving the accuracy of outcome predictions and supporting timely clinical interventions.
5. To train and validate supervised learning models for predicting treatment success or failure among PTB patients, so that individuals at risk of poor outcomes can be identified early, while the model maintains high predictive accuracy and reliability when applied to incoming TB cases.
6. To develop a decision support tool that visualizes patient risk levels and predictive outputs.

Scope and Limitations

The dataset includes PTB patients from 2015 to 2025 in the CAR, specifically Baguio City, and excludes those still undergoing treatment at the time of data extraction. Only patients with a defined treatment outcome of either success or failure are included. The data covers demographic information, laboratory tests, diagnostic data, clinical data, treatment regimen, follow-up examinations, drug administration data, and treatment outcomes. The study has limitations tied to the sufficiency and quality of the dataset. The data is sourced from existing records, so completeness, consistency, and representativeness may be limited by incomplete clinical parameters. This may affect which risk factors can be included and weaken the analysis. The study also may not fully represent the current TB patient population, since patients currently undergoing treatment are excluded. Difficulties are also expected during data extraction from DOH-CAR. Record-keeping over the 10-year period from 2015 to 2025 may be inconsistent due to changes in the healthcare system and shifts in documentation standards. Reliance on self-reported indicators may also introduce bias and inconsistencies that affect the predictive capabilities of the model.

Significance of the Study

The models are built to help healthcare providers spot patients at higher risk of treatment failure early enough to act. Clinical adoption depends on whether users can read what the model is doing, so the framework is built around explainable prediction rather than risk scoring alone. SHAP-based explanations show the patient-level factors behind each risk estimate, and the CDSS presents those factors in a form a clinician can interpret and use. The practical goal is to strengthen TB control in the Cordillera Administrative Region (CAR) by turning patient data the TB-DOTS program already collects into tools that support earlier intervention, better monitoring, and more informed public health decisions. The study also aims to inform health administrators about the models, which may be developed further into a local TB-DOTS support application. A preprocessed and pseudonymized derivative of the DOH-CAR PTB Treatment Outcome Dataset will be published alongside the study to support future clinical, epidemiological, and machine learning research.

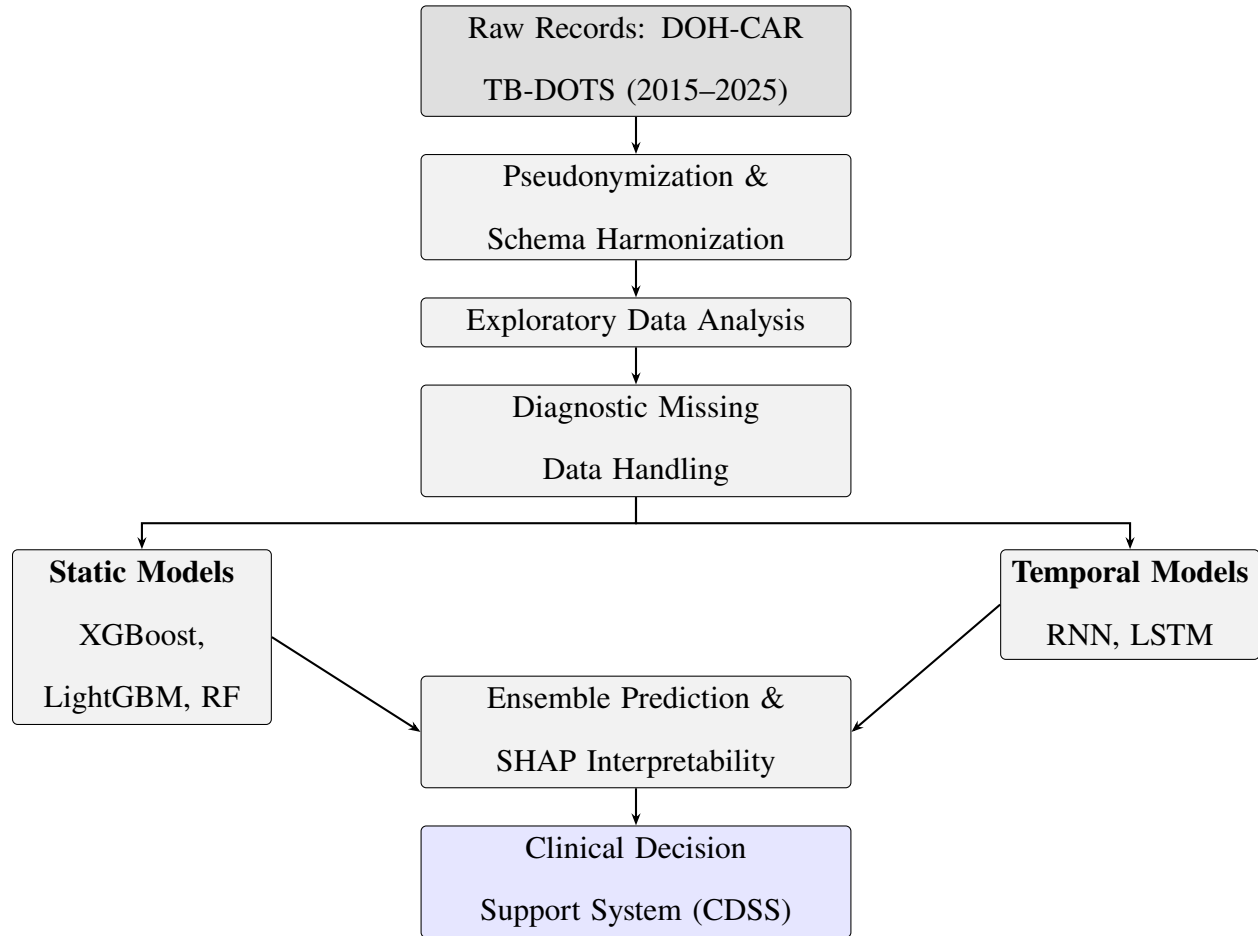
Method

This study uses PTB patient records from the Cordillera Administrative Region (CAR), Philippines, covering 2015 to 2025 and consolidated by the Department of Health-CAR (DOH-CAR). Once ethical approval was granted, records were pseudonymized and run through a standardized preprocessing pipeline (schema harmonization, error correction, feature scaling) and then the diagnostic-first missing data protocol described in the Missing Data Handling subsection below. Exploratory and cluster analyses characterize the cohort and yield subgroup-level features. Ensemble tree-based classifiers (XGBoost, LightGBM, Random Forest) are then trained to predict treatment outcomes defined by National Tuberculosis Program criteria. Because treatment unfolds over months, we also evaluate temporal models built on Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) alongside the static ones, with ensembling used to improve robustness. Performance is measured with standard classification metrics, and Shapley Additive Explanations (SHAP) handle interpretability. The final models feed a clinical decision support

framework that updates risk estimates as treatment progresses. Figure 1 gives the overview.

Figure 1

Data Management and Analysis Process



Data Acquisition and Preprocessing

Two datasets from CAR feed this study, both spanning 2015 to 2025. The first is a non-temporal (static) consolidated registry maintained by the Department of Health-CAR, which trains the static tree-based classifiers. The second is a temporal monitoring dataset that the study's Pharmacy Co-authors compiled by hand, which trains the longitudinal deep learning and temporal tree-based models. Table 1 sets them side by side.

Table 1*Comparison of the Non-Temporal and Temporal Monitoring Datasets*

Attribute	Non-Temporal Dataset	Temporal Monitoring Dataset
Records (patients)	7,389	599
Coverage period	2015–2025	2015–2025 (patient subset)
Data source	DOH-CAR centralised TB-DOTS registry	Facility treatment monitoring forms (35 raw Excel workbooks)
Collection method	Routine national TB-DOTS programme reporting	Manual transcription and encoding by Pharmacy Co-authors
Structure	Cross-sectional: one row per patient	Longitudinal: one row per patient-month (M0–M12)
Static features	Demographics, diagnostics, outcomes	Demographics, diagnostics, baseline vitals, outcomes
Temporal features	None	Monthly adherence, doses taken/missed, weight, height, smear microscopy, Xpert MTB/RIF (M0–M12)
Post-preprocessing columns	28	269
Post-preprocessing rows retained	7,388	599 (zero deleted)
Primary model use	Static tree-based classifiers (XGBoost, LightGBM, Random Forest)	Temporal deep learning models (Bi-LSTM) and tree-based temporal baselines

Non-Temporal Dataset

DOH-CAR consolidates this dataset centrally, pulling reports from government TB-DOTS facilities and from private practitioners under the Public-Private Mix DOTS (PPMD) scheme (Department of Health [DOH], 2014). It holds 7,389 patient records with baseline features only: demographics (age, sex, civil status), clinical indicators (BMI, blood pressure), and diagnostic results (GeneXpert, Smear Microscopy). Each record is a single patient captured at treatment registration, with no follow-up monitoring. It is the main training source for the static classifiers (XGBoost, LightGBM, Random Forest).

The non-temporal pipeline harmonizes variable formats and units and pseudonymizes records for privacy. It then defines the binary target from standardized National Tuberculosis

Program (NTP) criteria: patients marked ‘Cured’ or ‘Treatment Completed’ become ‘Success’, and those marked ‘Failed’, ‘Died’, or ‘Lost to Follow-up’ become ‘Failure’. Extreme outliers in continuous clinical variables are capped so they do not distort training. Missing data goes through the diagnostic-first protocol described in the Missing Data Handling subsection below.

The dataset is split into training and testing sets before any transformation, which keeps test information out of the model. A stratified split holds the success and failure proportions steady across both partitions. Everything that comes after, feature scaling, normalization, and the multivariate imputation of missing values, is fit on the training set alone. Those fitted parameters and imputation rules are then applied to the held-out test set, so no model ever sees the statistical distribution of the test data.

Temporal Monitoring Dataset

The temporal monitoring dataset is a longitudinal subset of 599 patients with complete month-by-month records. Nothing like it sat in the DOH-CAR registry. The study’s Pharmacy Co-authors *gathered and encoded it by hand* from paper Treatment Cards and Master Lists at TB-DOTS centers in Baguio City, Benguet. They accessed the facility records in person and transcribed each patient’s monitoring data: demographics, monthly vital signs, percentage adherence, doses taken and missed, smear microscopy (TB-LAMP) and Xpert MTB/RIF results, and treatment outcomes. These went into a master RAW DATA.xlsx workbook with one sheet per barangay or facility unit, covering 35 facility-year cycles from 2015 to 2025. A consolidation script (`combine_raw_dataset.py`) then merged the per-barangay sheets into one dataset, normalizing facility names against a canonical alias map so inconsistent entries across sources lined up before the main pipeline ran.

The dataset is wide: one row per patient, with columns encoding monthly observations from treatment Month 0 (M0) through Month 12 (M12). Each monthly visit tracks percentage adherence (`pct_adherence`), doses taken and missed that month, cumulative doses taken, body

weight, height, smear microscopy (TB-LAMP), and Xpert MTB/RIF result. Each record also carries static baseline features alongside those monthly columns: demographics, baseline vital signs (heart rate, blood pressure, oxygen saturation, temperature, respiratory rate), comorbidities, starting treatment regimen, and diagnostic classification. After preprocessing, the cleaned dataset is 599 rows by 269 columns.

The temporal pipeline runs in eight stages, built around the wide longitudinal layout of this dataset:

1. **Schema Validation & Column Standardization.** All column names are standardized to snake_case; duplicate and entirely null columns are flagged and deduplicated.
2. **Identifier Removal.** Four patient- and facility-identifying columns (no, source_file, name_of_diagnosing_facility, name_of_treatment_unit) are retained in the export file for traceability but are explicitly excluded from all model inputs to enforce patient privacy.
3. **Data Cleaning & Type Coercion.** All columns are parsed to correct data types with clinical validation. Specialized parsers handle domain-specific formats: weight strings (45kg, 56.5 kg) are converted to numeric kilograms; height strings are parsed to centimeters; blood pressure entries in 120/80 format are split into separate systolic and diastolic fields; oxygen saturation values in decimal-fraction (0.95), percent-suffix (95%), and raw-integer formats are unified to a 0 to 100 scale; and adherence percentages are stripped of trailing % signs. Date fields undergo a dedicated correction pass: future dates of birth are corrected (e.g., year 2063 $\rightarrow$ 1963), transposed-digit errors are detected (e.g., 2990 $\rightarrow$ 1990), and chronological ordering is enforced (treatment start $\geq$ diagnosis date). All continuous clinical variables are winsorized to physiologically plausible bounds. Temperature is clipped to 30 to 45°C, systolic BP to 60 to 220 mmHg, diastolic BP to 30 to 140 mmHg, heart rate to 20 to 250 bpm, oxygen saturation to 70 to 100%, weight to 10 to 180 kg, height to 80 to 220 cm, and respiratory rate to 4 to 60 breaths/min.
4. **Categorical Harmonization.** Inconsistent categorical entries are mapped to canonical

labels. Outcome variants, sex designations, regimen types, comorbidity flags, and civil status are standardized to a fixed set of valid categories, reducing per-field cardinality from hundreds of free-text variants to the expected small canonical sets.

5. **Drop Uninformative Columns.** Near-entirely null or redundant columns (e.g., `nationality_raw` after merging into `nationality`) are flagged in the output rather than silently deleted, maintaining auditability of the cleaning decisions.
6. **Wide → Long Temporal Restructuring.** The wide-format dataset is melted into two structures: a static baseline table (`df_static`, one row per patient) and a temporal table (`df_temporal`, one row per patient-month), with columns [`patient_id`, `month`, `weight`, `height`, `pct_adherence`, `monthly_doses_taken`, `monthly_missed_doses`, `cumulative_doses_taken`, `smear_tb_lamp`, `xpert_mtb_rif`].
7. **Diagnostic-First Missing Data Handling.** The same four-pathway diagnostic pipeline described in the Missing Data Handling subsection is applied independently to both the static and temporal feature sets. Temporal features are never hard-excluded on sparsity grounds; their structural missingness (later months having fewer observations as patients complete or abandon treatment) is preserved via missingness indicators.
8. **Post-Imputation Clinical Clipping & Export.** Clinical bounds are re-enforced on all imputed values (e.g., `weight` to 10 to 180 kg, `pct_adherence` to 0 to 100). A monotonicity repair pass corrects any MICE-introduced decreases in `cumulative_doses_taken` per patient. The pipeline exports `cleaned_human_readable.csv` (599 rows × 269 columns) with all original clinical units and categorical labels intact. Encoding, scaling, and tensor preparation are deliberately deferred to model-specific scripts post-train-test split to prevent data leakage.

As in the non-temporal case, the temporal train-test split happens before any scaling, one-hot encoding, outlier capping, or 3-dimensional tensor construction for the recurrent networks. Both pipelines thus guard against leakage the same way, whether the model is a static classifier or a temporal network. Table 2 summarizes per-patient observed means of the temporal features across

M0 to M7.

Table 2

Per-Patient Observed Means of Temporal Features (M0–M7)

Feature	N	Mean	SD	Min	Max
monthly_doses_taken	573	23.36	4.32	0.0	64.29
cumulative_doses_taken	535	70.09	23.39	0.0	127.38
monthly_missed_doses	458	0.91	2.47	0.0	25.0
pct_adherence	342	98.23	8.35	0.0	100.0
weight	361	55.69	21.55	7.2	189.69
height	315	155.49	20.45	31.0	181.0
smear_tb_lamp	25	0.12	0.33	0.0	1.0
xpert_mtb_rif	41	0.0	0.0	0.0	0.0
last_month_attended	599	5.23	2.06	-1.0	7.0
total_observed_months	599	6.21	2.06	0.0	8.0

Exploratory Data Analysis

Before any modeling, an exploratory analysis profiled the study population, checked how complete the data was, and measured the associations among variables. Its findings shaped the preprocessing and modeling decisions that followed.

Table 3

Missing Data Summary by Column

Column	Missing_Count	Missing_Percentage
Days_To_Microscopy_Result	7297	98.75
Days_To_RDT_Result	6638	89.84
Days_Screening_To_Diagnosis	4444	60.14
Days_To_Treatment	4378	59.25
Diagnosis_Season	3953	53.5

Column	Missing_Count	Missing_Percentage
Diagnosis_DayOfYear	3953	53.5
Diagnosis_Quarter	3953	53.5
Diagnosis_Month	3953	53.5
Microscopy Result	2792	37.79
Brgy	1026	13.89
City/Municipality	105	1.42
Province	71	0.96
Treatment Health Facility	29	0.39
Region	15	0.2
Date of Outcome/Status	2	0.03
Bacteriologic Status	1	0.01
Type	0	0
Validation Status	0	0
Age	0	0
Sex	0	0
Anatomical Site	0	0
Source of Patient	0	0
Screening/Diagnosing Health Facility	0	0
RDT Result	0	0
Year	0	0
Outcome/Status	0	0
Registration Group	0	0
Computed_Age	0	0

A global missingness audit across all 28 columns mapped the dataset's sparsity (Table 3), ranking each column by its missing rate. Several variables were heavily incomplete: microscopy result at 37.8%, diagnosis-derived temporal features at 53.6%, and clinical delay intervals anywhere from 59.3% up to 98.8%.

These figures set the missingness thresholds used later in the Missing Data Handling subsection.

Pairwise Pearson's r on available-case observations covered the continuous variables: age, four derived clinical delay intervals (days from screening to diagnosis, to treatment, to microscopy result, and to RDT result), and diagnosis day-of-year. The point was to decide whether MICE was worth using. When continuous predictors share real correlation structure, MICE can borrow signal from the observed features to estimate what's missing; when they don't, a simple univariate mean-fill does just as well.

For categorical variables, association strength came from Cramér's V , a chi-square-based measure scaled to the $[0, 1]$ interval. Columns with fewer than two unique non-null values, or fewer than ten non-null observations, were dropped from the matrix. The categorical set covered patient demographics (sex, type of TB, anatomical site) and clinical classifications (registration group, bacteriologic status, microscopy results).

The exploratory phase also clusters patients across demographic, clinical, and diagnostic dimensions to surface latent subgroups. K-prototypes handles the dataset's mix of categorical and continuous variables. The cluster count is chosen iteratively from the Silhouette score and the elbow method, which lets the analysis pick out groups that differ in comorbidity profiles, adherence behavior, and how they respond to a regimen.

The cluster assignments then serve two purposes: descriptive interpretation, and engineered features for the supervised models. Cluster labels are one-hot encoded as extra categorical inputs for the tree-based classifiers (XGBoost, LightGBM, Random Forest) and added to the feature space for the temporal networks (RNN, LSTM). A model can then pick up on subgroup-level patterns tied to treatment success or failure, not just individual predictors.

The temporal monitoring dataset got the same treatment before preprocessing: a global missingness audit over every feature, with columns ranked by missing rate. Anything above 80% missingness was routed to column exclusion. Anything between 50% and 80% went to the missing-indicator-with-fill pathway, weighed against its predictive importance. This kind of threshold triage is standard practice; it matches the handling strategy to how severe the missingness is instead of forcing one rule on everything (Emmanuel et al., 2021).

Patient retention was traced by the share of patients with non-missing adherence data at each interval from M0 through M12. A raw per-month count says little, since retention naturally falls over a six- to twelve-month course, so the decay was expressed as a percentage of the starting cohort at each successive month. This sets the temporal modeling scope. Once fewer than 25% of patients still have data, that month

and everything past it drops out of the feature-level correlation analyses, which keeps the estimates from being driven by a small, non-representative surviving group.

The same Pearson's r check, on available-case observations, was run over the vital signs in the temporal dataset. Strong correlation among vitals is what justifies a multivariate imputation over a univariate one: when vitals move together, Multiple Imputation by Chained Equations (MICE) can borrow signal from the observed features to fill gaps, and when they don't, MICE buys nothing over mean- or median-fill (Aracri et al., 2025). The vital set was age, heart rate, oxygen saturation, systolic and diastolic blood pressure, weight, height, respiratory rate, and temperature, wherever the temporal records carried them.

To see how the temporal features relate, Pearson correlations were taken on per-patient observed means across Months 0 through 7. For each feature, whether a dose metric (monthly doses taken, cumulative doses taken, percentage adherence), an anthropometric one (weight, height), or a diagnostic result (smear microscopy, Xpert MTB/RIF), the values across a patient's available visits were averaged. Retention characteristics, the last month attended and the total months observed, came from the same window. The point of averaging this way is that it keeps the dropout signal intact. A patient who stops after M4 contributes means from M0 to M4 only, and that short window is itself informative rather than a gap to impute.

Categorical variables in the temporal dataset again used Cramér's V , the chi-square-based measure scaled to the $[0, 1]$ interval. We did not mix Pearson and Cramér's V into one matrix. Pearson's r assumes interval-level measurement and returns misleading values on nominal or ordinal categories whose spacing is uneven (Kossakov et al., 2024), so computing Cramér's V on its own for the categorical features sidesteps that and gives interpretable estimates. Columns with fewer than two unique non-null values, or fewer than ten non-null observations, were left out, since association can't be estimated meaningfully there.

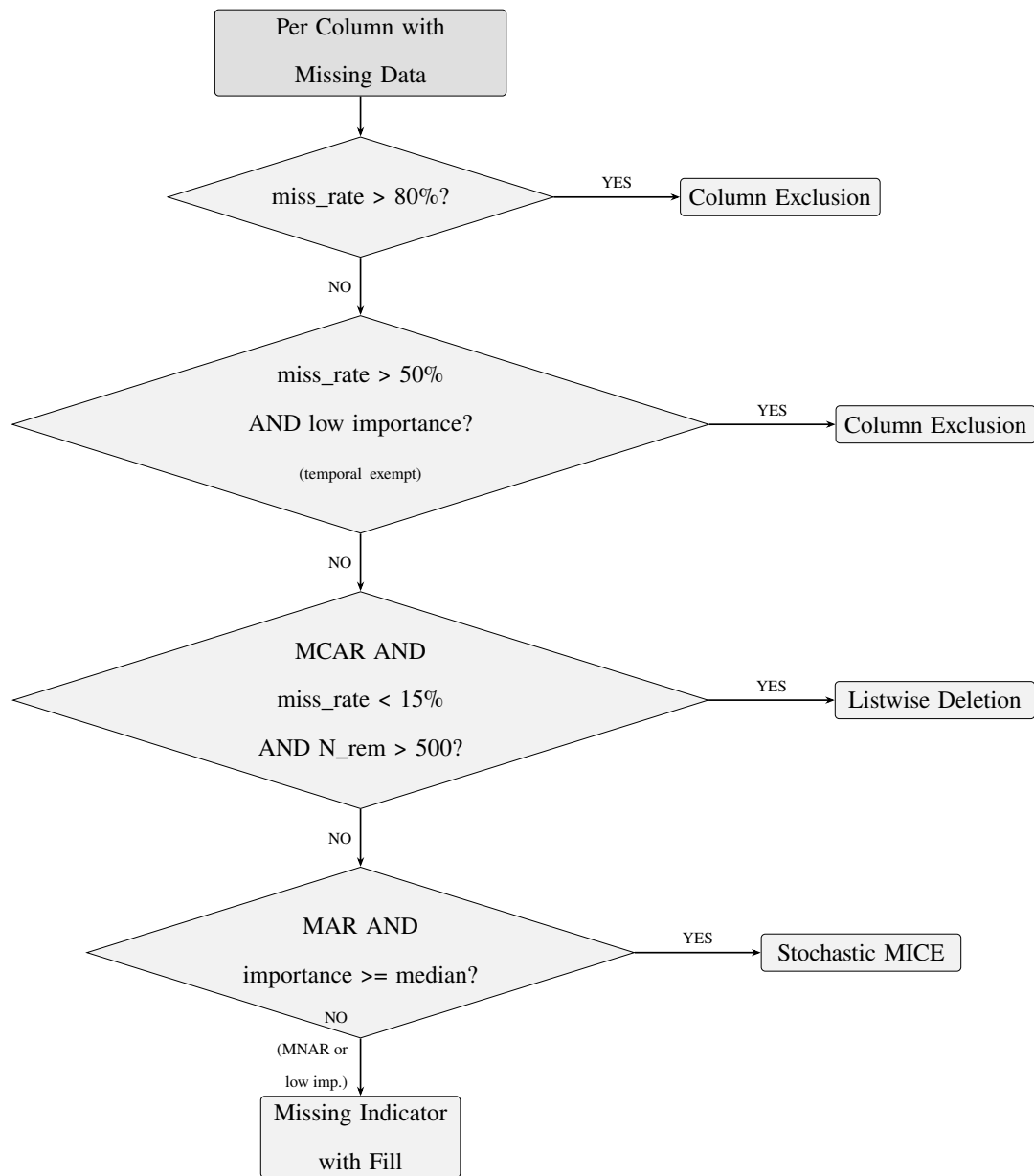
A cardinality audit over every categorical feature in the raw dataset showed how much inconsistent coding there was to fix. For each feature, counting the unique raw values quantified the entry inconsistency. When a feature's cardinality ran far past its expected number of valid categories, say a clinical observation field that should hold a handful of standardized codes but instead held hundreds of free-text variants, it was flagged for harmonization in Stage 4. The audit gives the harmonization targets an empirical baseline, measuring the data entry problem before any cleaning touched it.

Whether adherence stays stable over the course was checked with lag-1 Pearson correlations of percentage adherence across consecutive month pairs (M0 to M1 through M7 to M8). That month-to-month

strength is what decides if a recurrent architecture, which assumes one time step depends on the last, is even worth it. Weak correlation between consecutive months means a static model on aggregated patient means would tell you just as much; strong correlation is what earns the added complexity of an RNN or LSTM (Laine & Aila, 2016; Nayebi et al., 2022).

Missing Data Handling

Missingness in clinical TB registries is uneven. It comes from inconsistent documentation, facilities with different capacities, and the longitudinal nature of treatment monitoring. When the mechanism behind the gaps varies from feature to feature, forcing one imputation strategy on all of them is statistically wrong. This study instead uses a diagnostic-first pipeline that classifies each variable's missingness mechanism and then routes it to one of four strategies: column exclusion, listwise deletion, missing indicator with fill, or stochastic Multiple Imputation by Chained Equations (MICE) (Figure 2) (Dong & Peng, 2013; Emmanuel et al., 2021).

Figure 2*Missing Data Handling Decision Logic*

The first diagnostic step runs Little's MCAR test at the dataset level to ask whether data are Missing Completely at Random (MCAR). If they were, missingness would be independent of every observed and unobserved value, the complete cases would form an unbiased subsample, and listwise deletion would be fine. The test came back significant ($p < 0.05$), which rejects MCAR: the missingness is systematically tied to the observed data. That rules out deleting cases wholesale (Dong & Peng, 2013).

Little's test gives one omnibus verdict for the whole dataset, so classifying each column's mechanism needs its own step. Each variable's missingness indicator is then checked against the observed data to assign a column-level mechanism. For temporal (monthly) variables, Spearman rank correlations relate the column's monthly missing rate to treatment month; a correlation that rises monotonically marks a dropout pattern, which is Missing Not at Random (MNAR). For static baseline variables, point-biserial correlations relate the binary is-missing indicator to the observed features. A significant association says other observed variables explain the missingness, so it is Missing at Random (MAR); a non-significant one points to MNAR. Classifying column by column instead of accepting one dataset-level verdict is what lets the pipeline fit each handling strategy to the statistics of each variable (Dong & Peng, 2013; Emmanuel et al., 2021).

The column exclusion and MICE eligibility decisions lean on a Random Forest classifier trained on available cases, meaning rows where the treatment outcome is observed. For ranking only, features are filled with column medians, a coarse fill but enough to estimate importance. Variables in the bottom quartile of importance become candidates for soft exclusion, while those above the median get priority for stochastic imputation (Aracri et al., 2025).

A hard threshold drops any variable over 80% missingness outright, importance aside, since nothing reliable can be imputed at that sparsity. A soft threshold also drops variables past 50% missingness whose importance sits in the bottom quartile, because imputing a low-value sparse variable adds more synthetic variance than information. Temporal variables escape both thresholds: their structural missingness is the longitudinal design of TB monitoring at work, not a data quality fault (Junaid et al., 2025).

Listwise deletion stays available for a narrow case: a variable confirmed MCAR at the column level, under 15% missing, and leaving more than 500 patients after deletion. There the complete cases are an unbiased subsample, so deletion gives unbiased estimates and avoids the variance distortion imputation adds. No variable in this dataset met all three at once, so the pathway never fired (Dong & Peng, 2013).

Variables classified MNAR, plus every temporal feature whatever its mechanism, get a two-part treatment. A binary indicator (`is_missing_[variable]`) is added to encode the absence itself. The original variable is then filled: last observed value carried forward for temporal variables, column median for continuous static ones, column mode for categorical static ones. The absence is treated as a signal, not noise to paper over. A missing Month 9 weight in TB monitoring often means the patient dropped out, and the indicator hands that clinical pattern to the model. Backward-fill is barred from temporal imputation,

since letting a future measurement bleed into an earlier time step would leak information the model can't legitimately have at inference (Austin et al., 2021; Groenwold et al., 2012; Sisk et al., 2023).

Four vital signs cleared both bars, confirmed MAR and ranked above median importance: oxygen saturation, heart rate, and systolic and diastolic blood pressure. These are imputed with stochastic MICE using an ExtraTreesRegressor as the conditional model. ExtraTreesRegressor is a nonparametric tree-based estimator that picks up non-linear relationships in mixed-type clinical data without needing standardized variables, much like the missForest framework (Aracri et al., 2025), and it tends to beat linear and k -nearest-neighbor imputers on such data. To track the fraction of missing information per variable, the MICE iteration count scales as $\max(20, \min(100, \lfloor \text{miss_rate} \times 100 \rfloor))$, giving high-sparsity features room to converge while capping compute. The idea mirrors Von Hippel's rule that the number of imputations should roughly match the percentage of missing information (von Hippel, 2018), applied here to iterations rather than to the count of imputed datasets. Stochastic rather than deterministic imputation keeps the imputed variables' natural spread instead of collapsing them onto conditional means (Alruhaymi & Kim, 2021).

Data Quality Assurance

A layered quality assurance protocol keeps the supervised pipeline reliable, spanning schema validation, clinical bounds checking, temporal consistency, and post-processing validation.

Schema and Column Validation

Column names are normalized to snake_case: whitespace stripped, text lowercased, special characters mapped to word equivalents, so M0 %Adherence becomes m0_pct_adherence. Duplicate and entirely null columns are flagged and deduplicated before modeling. Four patient- and facility-identifying columns stay in the export file for traceability but are kept out of all model inputs for privacy.

Date Integrity Correction

A dedicated correction pass handles the recording errors common in clinical registries. A future date of birth (year past the data year) is pulled back 100 years. A date of birth before 1920 is moved forward 100 years when that lands the age in a plausible 0 to 110 range. Transposed digits, such as 2990 for 1990, are caught against adjacent records. Date pairs are forced into order: treatment start cannot precede diagnosis, and an outcome date cannot precede treatment start. Any missing date takes the cohort median date for that variable.

Temporal Consistency

Every fill in the longitudinal structure is forward-only, which is what keeps the temporal data from leaking. A Month 4 observation can carry forward to Month 5, but a Month 6 measurement can never inform an earlier step the model is supposed to predict without it. After MICE, a monotonicity repair pass undoes any drop in `cumulative_doses_taken` the multivariate imputer introduced, so each patient's cumulative dose never decreases across treatment months.

Categorical Harmonization Audit

A cardinality audit of the raw categorical features confirmed how widespread the entry inconsistency was. The `chest_x_ray_at_case_notification` field held over 300 unique free-text entries and `co_morbidities` more than 150, while even nominally binary fields like `sex` and `outcome` showed 5 to 12 raw representations apiece, the residue of treatment centers abbreviating things their own way. That is what prompted a 17-field harmonization pass mapping free-text and inconsistently coded entries to standardized categories before analysis.

Missing Data Diagnostics and Feature Importance

The diagnostic-first missing data pipeline (detailed in the Missing Data Handling subsection) adds one more validation layer. Little's MCAR test runs at the dataset level. Per-column mechanisms come from Spearman rank correlations to catch temporal dropout and point-biserial correlations to tell MAR from MNAR. A Random Forest trained on available cases ranks predictive importance, which decides where each variable gets routed.

Post-Processing Verification

Before the cleaned dataset goes to any model, an automated routine checks four things: (1) identifier columns are still present for traceability, (2) no scaled or normalized values have crept in, so everything stays in original clinical units, (3) categorical labels remain human-readable strings rather than integer codes, and (4) no one-hot columns have appeared. Passing all four confirms that encoding, scaling, and tensor construction were left entirely to the model-specific scripts that run after the split.

Leakage Prevention and Reproducibility

Every validation step exists to keep test-set information out of training. The patient-level stratified split comes before any transformation. Scalers, imputers, and encoders are fit on the training set alone, then pushed to the test and validation sets through `reindex()` so the column sets match exactly. Temporal fills are forward-only, with backward-fill explicitly disabled. For reproducibility, the split is fully deterministic under a seeded random state, and the fitted scaler parameters are checked by SHA-256 hash to confirm independent runs preprocess identically. An audit trail (`missing_data_report.json`) logs the decision pathway, mechanism classification, and rationale for every column.

Identification of Variables Associated With Treatment Outcomes

For the objective of finding which variables track with treatment outcomes, the study pairs descriptive association analysis with model-derived explanation. The descriptive side summarizes which demographic, clinical, diagnostic, and temporal variables separate the success and failure groups. The supervised models then estimate how those variables work together to predict the outcome. Both are needed because risk in TB treatment is multidimensional: a variable can look weak on its own yet matter once it sits alongside adherence, diagnostic, geographic, or monitoring information.

Bridging predictive performance and clinical interpretability falls to Shapley Additive Explanations (SHAP), a post-hoc method that measures each variable's marginal contribution to the model's output (Lundberg & Lee, 2017a). SHAP shows which features pushed a patient-level prediction toward "Success" (Cured, Completed) or "Failure" (Failed, Died, Lost to Follow-up; Peng et al., 2024). For clinical review, that means seeing what the model leaned on, whether drug resistance, comorbidities, adherence gaps, diagnostic results, or a delayed diagnosis. These explanations describe how the model behaved; they do not prove clinical causality, and that line between computational attribution and medical judgment is kept deliberately clear (Kaur & Sharma, 2021; Kossakov et al., 2024).

Temporal Modeling

TB treatment runs 6 to 12 months, so this study adds temporal modeling for the longitudinal monitoring data. A static model trained on baseline features alone can miss how adherence, follow-up attendance, and lab measurements shift over the course. The temporal models are evaluated to give month-by-month risk estimates from treatment Month 0 (M0) through Month 12 (M12).

These models run only on the 599-patient longitudinal dataset from the Data Acquisition and Preprocessing subsection (see also Table 1), the one the Pharmacy Co-authors compiled by hand from facility monitoring forms. It is a separate dataset from the 7,389-patient DOH-CAR registry the static classifiers use. After the eight-stage temporal pipeline, each patient has up to 8 usable monthly timesteps, M0 to M7. Anything past M7 is left out of the model inputs: fewer than 25% of patients had observations there (Table 5), and a surviving subset that small and skewed gives unreliable estimates. As before, encoding,

scaling, and 3-dimensional tensor preparation wait for the post-split, model-specific scripts so nothing leaks.

Temporal modeling went through two iterations, v1 and v2. Both share the same dataset lineage; v2 is the corrected, modeling-ready preparation that a methodological review of v1 produced. The v1 pipeline served as a modeling baseline, with patient-level stratified splitting and progressive temporal expansion across M0 to M12. Review then flagged several things that could weaken its clinical defensibility: late-treatment fields raising the risk of outcome leakage, inconsistent handling of imputed or unreliable outcome labels, and cleaning and temporal feature construction that had not been audited as carefully.

Version 2 (v2) fixed those issues and retrained. It builds on the audited output of the eight-stage cleaning pipeline from Data Acquisition and Preprocessing, limits supervised learning to patients whose outcome is actually observed, and strips out leakage-prone and administrative fields such as outcome-date fields and late timeline variables that wouldn't exist at early inference. Temporal predictors are encoded two ways: as sequence inputs for the deep models and as engineered summaries for the tree-based ones. The deep architecture is a Hybrid Bidirectional LSTM (Bi-LSTM), chosen to model long-term dependencies without the vanishing gradients of a plain RNN (Lu et al., 2022); its recurrent structure fits the strong month-to-month adherence dependence the lag-1 analysis confirmed (Nayebi et al., 2022). The tree-based baselines instead flatten the monthly values and bolt on aggregates, trends, latest-observed values, and missingness indicators. Class imbalance is handled by training-only resampling such as SMOTE-ENN. The hybrid framing stays so temporal predictions can be set beside strong static learners like XGBoost, which holds up better across the different patient subgroups (Laine & Aila, 2016).

Supervised Learning Model Development and Validation

At the core, supervised models are trained and validated to classify treatment outcomes. The study uses ensemble tree-based algorithms that handle complex non-linear interactions among mixed-type clinical variables and deal with class imbalance natively: XGBoost (Chen & Guestrin, 2016; Rajpurkar et al., 2022), LightGBM (Ke et al., 2017), and Random Forest (Breiman, 2001). These three came out of a preliminary run over six algorithms that also tried Logistic Regression, SVM, and Gradient Boosting. The three that were dropped either discriminated poorly ($CV\text{-}AUC \leq 0.63$) or had near-zero recall on the minority failure class, which makes them unsafe to deploy in a CDSS. Each classifier is run under two sampling strategies:

no resampling with native class-weighting, and SMOTE with Edited Nearest Neighbors (SMOTE-ENN), which grows the training failure class by 433% to a near-balanced 0.70:1 ratio.

Feature Set Versions

Three feature sets are evaluated in sequence to see how much each extra clinical and temporal group of features actually adds. The *baseline* set (11 features) is the core demographic and diagnostic block: age, sex, anatomical site, bacteriologic status, microscopy result, registration group, source of patient, TB type, province, days to treatment, and year of registration. The *improved* set (14 features) adds three facility-level geographic variables to the baseline: city/municipality, treatment health facility, and screening/diagnosing health facility. The *extended* set (21 features) further adds seven temporal and diagnostic delay features derived from date engineering: diagnosis month, diagnosis quarter, diagnosis season (using the Philippine three-season meteorological calendar: *Tag-lamig* for cool dry December to February, *Tag-init* for hot dry March to May, and *Tag-ulan* for the rainy season June to November), days from screening to diagnosis, days from diagnosis to microscopy result, days from diagnosis to rapid diagnostic test (RDT) result, and RDT result (binned from 42 raw variants into three classes: Detected, Not Detected, and Not Done). The date of outcome was excluded from all feature sets to prevent causal leakage, since this field is generated after the treatment result.

Data Partitioning and Class Imbalance

A stratified 70/20/10 patient-level split divides the data into training (70%), test (20%), and holdout validation (10%), holding the class proportions across all three (Hastie et al., 2009). The heavy imbalance, roughly 86% success against 14% failure, is corrected on the training set only, using one of three oversamplers. SMOTE (Chawla et al., 2002) interpolates synthetic minority-class points. SMOTE-ENN (Batista et al., 2004; Nishat et al., 2022) runs SMOTE and then Edited Nearest Neighbors undersampling to clear out ambiguous borderline cases. SMOTE-Tomek (Batista et al., 2004) runs SMOTE and then removes Tomek links to clean the boundary. Resampling never touches the test or validation sets, which keep their original distributions so the evaluation stays unbiased (Salmi et al., 2024).

Cross-Validation and Model Selection

Each configuration is scored by five-fold stratified cross-validation on the training data, which estimates how well it generalizes and yields a mean and standard deviation of ROC-AUC across folds (Hastie et al., 2009). Since the clinical goal is catching at-risk patients early, selection uses a composite metric, $\text{score} = 0.6 \times \text{AUC} + 0.4 \times \text{Recall}(\text{Failure})$, that favors discrimination while rewarding sensitivity to the minority failure class. Whichever configuration scores highest across the feature versions and resamplers is then checked on the held-out 10% validation set for an unbiased real-world estimate.

Alongside the composite metric, each configuration also reports the Matthews Correlation Coefficient (MCC) as a supplementary check. Where accuracy or a class-specific F1 can mislead, MCC reads all four quadrants of the confusion matrix at once and stays informative under heavy imbalance. It lands in $[-1, +1]$: +1 for perfect prediction, 0 for random, -1 for complete inversion (Chicco & Jurman, 2020). Seeing MCC next to the composite score makes the trade-off from aggressive resampling visible, since something like SMOTE-ENN can lift failure recall while adding false positives that drive up overall error.

Performance Evaluation

Performance is read across several metrics, each catching a different facet of classification quality under imbalance. Treatment failure is only 13.8% of the cohort, so leaning on any one metric, accuracy above all, can mislead: a classifier that simply says “Success” every time hits 86.2% accuracy and catches not a single failure. The framework therefore computes discriminative, threshold-dependent, and calibration-aware metrics together.

Every metric comes from the confusion matrix, with treatment failure as the positive class. On the held-out test set, each model gives four quadrants: true positives (TP, failures correctly predicted), false negatives (FN, failures missed), false positives (FP, successes wrongly flagged as failures), and true negatives (TN, successes correctly identified). The metrics below follow from these:

Accuracy ($\frac{TP+TN}{TP+TN+FP+FN}$) measures the overall proportion of correct predictions. It is intuitive but is reported with the caveat that it is dominated by the majority class under 86:14 imbalance.

Precision ($\frac{TP}{TP+FP}$), also called positive predictive value, quantifies the proportion of failure alerts

that correspond to actual failures. Low precision means a high rate of false alarms, which erodes clinician trust in the CDSS.

Recall ($\frac{TP}{TP+FN}$), also called sensitivity or true positive rate, measures the proportion of actual failures the model detects. In the TB-DOTS context, a missed failure carries higher clinical cost than a false alert, because undetected treatment breakdown can lead to disease progression, drug resistance, and continued community transmission.

Specificity ($\frac{TN}{TN+FP}$), the true negative rate, measures the proportion of actual successes correctly identified. High specificity keeps patients on track from being unnecessarily flagged for review.

F1 Score ($2 \times \frac{Precision \times Recall}{Precision + Recall}$) is the harmonic mean of precision and recall. It provides a single metric that penalizes extreme imbalance between the two. F1 is useful when comparing models under different resampling strategies.

Balanced Accuracy ($\frac{Recall + Specificity}{2}$) averages the sensitivity and specificity, giving a metric that is robust to class imbalance by construction. A balanced accuracy of 0.50 is equivalent to random guessing regardless of the class distribution.

Area Under the Receiver Operating Characteristic Curve (ROC-AUC) evaluates the model's discriminative ability across all possible decision thresholds. The ROC curve plots the true positive rate against the false positive rate at each threshold, and the AUC summarizes the curve as a single scalar. An ROC-AUC of 1.0 indicates perfect separation; 0.5 indicates random chance. ROC-AUC is threshold-independent and serves as the primary ranking metric for model comparison, but it can be optimistic under severe class imbalance because the false positive rate is dominated by the large majority class.

Area Under the Precision-Recall Curve (PR-AUC) addresses this limitation by plotting precision against recall across thresholds, ignoring true negatives entirely. PR-AUC is more informative than ROC-AUC when the positive class is rare (Davis & Goadrich, 2006). The no-skill baseline for PR-AUC equals the prevalence of the positive class (0.138), giving a natural reference point.

Matthews Correlation Coefficient (MCC) is computed as:

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

MCC accounts for all four confusion-matrix quadrants at once and remains informative under class imbalance.

It gives a coefficient in $[-1, +1]$ where $+1$ is perfect prediction, 0 is random, and -1 is complete inverse prediction (Chicco & Jurman, 2020). Unlike accuracy or F1, MCC is symmetric with respect to class swapping and does not inflate under majority-class dominance.

Threshold optimization. A 0.50 cutoff need not match clinical priorities under heavy imbalance, so each model gets a threshold search. Candidate thresholds from 0.10 to 0.90 are scored by composite objectives that encode different priorities: *balanced* (40% F1, 30% recall, 30% specificity), *failure_recall* (55% failure recall, 25% balanced accuracy, 20% F1), and *specificity* (50% specificity, 30% F1, 20% recall). A model’s reported metrics use whichever threshold maximizes the relevant objective.

Model selection uses a composite score weighting discriminative performance and clinical utility:

$$S = 0.6 \times \text{ROC-AUC} + 0.4 \times \text{Recall(Failure)}$$

This puts most of the weight on ranking ability (AUC) but still rewards sensitivity to the minority failure class. The top configuration by this score becomes the recommended model.

Validation strategy. Performance is estimated at three levels to guard against overfitting and gauge generalization. Five-fold stratified cross-validation on the training set gives the mean and standard deviation of ROC-AUC across folds, a stability check. The 20% held-out test set carries the primary evaluation behind all the metrics above, on data the model never saw in training or preprocessing. A final 10% holdout set is reserved for the recommended model alone, to confirm the test-set numbers aren’t just an artifact of one split. Scaling, encoding, and imputation are all fit on the training set and then applied to the test and validation sets, so nothing from the evaluation data feeds back into training.

For reproducibility, every experiment runs through one centralized pipeline that standardizes training, cross-validation, and evaluation. It exports the metrics and visualizations straight to native $\text{\LaTeX}$ for this manuscript.

Development of a Clinical Decision Support Tool

The validated models are embedded in a Clinical Decision Support System (CDSS) built for health-care providers, with responsibilities kept distinct. The predictive model estimates the probability of treatment

success or failure, the SHAP layer names the factors behind that prediction, the interface shows the risk and those factors, and the clinician reads it all in the context of the patient. In practice the provider sees a dashboard with the failure and success probabilities, a risk-factor view that surfaces what drove the score (adherence gaps, lab results, diagnostic delays), and live updates: feed in new follow-up data like a monthly weight or smear result and the prediction refreshes, so review can continue across the whole treatment. The CDSS is meant for risk stratification and explanation, not as an autonomous system that diagnoses or prescribes.

Small Language Model (SLM)-Based Narrative Explanation Layer

To make the explanations easier for clinical users to read, the CDSS adds a small language model (SLM) as a constrained narrative layer that turns model outputs into plain clinical text. The SHAP feature contributions are still the explanation of record. The SLM is only a presentation layer; it is not a predictor, a diagnostic engine, or a source of new clinical inference (Lundberg & Lee, 2017b; Zeng & Zhu, 2024). It runs after prediction and changes nothing upstream: not the training, not the inference probabilities, not the risk thresholds, not the SHAP values. In the implemented pipeline, the web client assembles a structured interpretation request from selected patient context fields, the predicted treatment-failure probability, and the ranked contribution signals, then posts it to a dedicated endpoint that streams the narrative back into the interface.

The SLM is confined to explanation. It does not infer new clinical causes, write prescriptions, suggest regimen changes, or stand in for clinician judgment. All it does is make the SHAP-grounded attribution easier to read, so a provider can see why the CDSS flagged a patient and decide whether a closer look is warranted. Keeping the roles this separate lines up with how medical-domain SLMs are currently framed and with health AI governance expectations for bounded decision support, and it keeps interpretability anchored to the original attribution signal (Collins et al., 2024; Sellergren et al., 2025; World Health Organization, 2021). The final call still belongs to the treating professional.

Results and Discussion

This section presents the results of the machine learning analyses for predicting treatment success and failure among PTB patients under the DOH-CAR TB-DOTS program. Descriptive statistics characterize the patient cohort. Model performance is then evaluated on the test and validation sets, comparing the static tree-based models (XGBoost, LightGBM, Random Forest) under two sampling conditions against the temporal models. Feature importance analyses identify which variables drove the predictions. The discussion puts these findings in the context of clinical relevance and CDSS usability, and explains how risk estimates and explanations can support earlier intervention and better patient management.

Missing Data Handling Outcomes

Of the 44 features identified with missing data, 15 were excluded via column exclusion. All 15 exceeded the hard missingness threshold of 80%, including near-complete variables such as tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). One more variable, the raw blood pressure string field, was soft-excluded at 67.3% missingness and below-median predictive importance because it had already been parsed into the structured systolic and diastolic blood pressure variables. All 599 patient records were kept in the dataset, with zero row deletion. Full sample retention matters given the small cohort size and the severe class imbalance of 86.2% treatment success versus 13.8% failure. Listwise deletion under these conditions would have made the minority class representation problem worse (Dong & Peng, 2013).

The listwise deletion pathway was not activated. The dataset-level MCAR test was rejected, and no individual variable met all three conditions of confirmed MCAR mechanism, fewer than 15% missing observations, and a post-deletion sample size above 500. Twenty-five variables were routed to the missing indicator with fill strategy: eight temporal monthly features (including weight, percentage adherence, cumulative doses taken, and monthly smear results) and 17 static baseline variables with confirmed MNAR patterns. Four variables (oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure) were imputed via stochastic MICE after confirmation of a MAR mechanism. Their missingness indicators were significantly associated with visit attendance under the point-biserial correlation procedure described

above. This is a clinically plausible MAR mechanism, since vital sign recording depends on whether the patient attended the monthly monitoring visit rather than on the unobserved values of the measurements themselves. The concentration of stochastic imputation on this small set of high-value features reflects the diagnostic-first design of the pipeline, which reserves computationally intensive probabilistic imputation for variables where the assumptions required for its validity are met (Austin et al., 2021; Emmanuel et al., 2021).

Table 4

Global Feature Missingness Rates in the Raw Temporal Dataset

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m8_weight	599	100.0	Yes
m12_weight	599	100.0	Yes
m11_height	599	100.0	Yes
m9_xpert_mtb_rif	599	100.0	Yes
m9_smear_tb_lamp	599	100.0	Yes
m7_xpert_mtb_rif	599	100.0	Yes
m7_smear_tb_lamp	599	100.0	Yes
m10_smear_tb_lamp	599	100.0	Yes
m10_weight	599	100.0	Yes
risk_factor_s_for_drug_ _resistance_tuberculosis	599	100.0	Yes
m12_height	599	100.0	Yes
m12_smear_tb_lamp	599	100.0	Yes
m12_xpert_mtb_rif	599	100.0	Yes
m9_weight	599	100.0	Yes
m8_xpert_mtb_rif	599	100.0	Yes
m8_smear_tb_lamp	599	100.0	Yes

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m1_xpert_mtb_rif	599	100.0	Yes
m11_weight	599	100.0	Yes
m10_xpert_mtb_rif	599	100.0	Yes
m11_smear_tb_lamp	599	100.0	Yes
m11_xpert_mtb_rif	599	100.0	Yes
m3_xpert_mtb_rif	599	100.0	Yes
m4_xpert_mtb_rif	598	99.83	Yes
m10_height	598	99.83	Yes
tuberculosis_culture	598	99.83	Yes
m12_monthly_missed_doses	598	99.83	Yes
other_lab_test	598	99.83	Yes
m12_pct_adherence	598	99.83	Yes
m8_height	598	99.83	Yes
m9_height	598	99.83	Yes
m12_monthly_doses_taken	597	99.67	Yes
m4_smear_tb_lamp	597	99.67	Yes
m1_smear_tb_lamp	597	99.67	Yes
others	597	99.67	Yes
m11_monthly_missed_doses	597	99.67	Yes
m11_pct_adherence	597	99.67	Yes
m12_cumulative_doses_taken	597	99.67	Yes
m10_monthly_missed_doses	596	99.5	Yes
m9_pct_adherence	596	99.5	Yes
m10_pct_adherence	596	99.5	Yes
m11_monthly_doses_taken	595	99.33	Yes
m9_monthly_missed_doses	595	99.33	Yes

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m11_cumulative_doses_taken	595	99.33	Yes
m8_pct_adherence	593	99.0	Yes
m3_smear_tb_lamp	593	99.0	Yes
m8_monthly_missed_doses	592	98.83	Yes
m10_monthly_doses_taken	592	98.83	Yes
m10_cumulative_doses_taken	592	98.83	Yes
m6_smear_tb_lamp	590	98.5	Yes
m9_cumulative_doses_taken	588	98.16	Yes
m7_height	588	98.16	Yes
m5_smear_tb_lamp	588	98.16	Yes
m9_monthly_doses_taken	588	98.16	Yes
m6_xpert_mtb_rif	587	98.0	Yes
m2_smear_tb_lamp	587	98.0	Yes
m0_smear_tb_lamp	586	97.83	Yes
m7_weight	586	97.83	Yes
m5_xpert_mtb_rif	585	97.66	Yes
m8_cumulative_doses_taken	582	97.16	Yes
m8_monthly_doses_taken	581	96.99	Yes
dat_supported_dup	576	96.16	Yes
m2_xpert_mtb_rif	572	95.49	Yes
other	570	95.16	Yes
m5_height	563	93.99	Yes
m4_height	562	93.82	Yes
m6_height	562	93.82	Yes
m6_weight	561	93.66	Yes
tuberculin_skin_test	559	93.32	Yes

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m3_height	558	93.16	Yes
m5_weight	557	92.99	Yes
m2_height	555	92.65	Yes
m0_xpert_mtb_rif	554	92.49	Yes
m4_weight	554	92.49	Yes
m1_height	554	92.49	Yes
nationality_raw	551	91.99	Yes
m3_weight	550	91.82	Yes
dat_supported	549	91.65	Yes
prior_history_of_tb	549	91.65	Yes
m2_weight	543	90.65	Yes
m1_weight	540	90.15	Yes
m7_pct_adherence	519	86.64	Yes
regimen_type_at_6th_month_of_ _treatment	518	86.48	Yes
smear_microscopy	502	83.81	Yes
m7_monthly_missed_doses	502	83.81	Yes
respiratory_rate	501	83.64	Yes
height	492	82.14	Yes
temperature	481	80.3	Yes
weight	476	79.47	No
m7_cumulative_doses_taken	465	77.63	No
o2_sat	462	77.13	No
m7_monthly_doses_taken	453	75.63	No
heart_rate	441	73.62	No
blood_pressure	403	67.28	No

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
m6_pct_adherence	398	66.44	No
regimen_type_at_end_of_ _treatment	355	59.27	No
m6_monthly_missed_doses	347	57.93	No
m5_pct_adherence	336	56.09	No
m4_pct_adherence	319	53.26	No
m3_pct_adherence	312	52.09	No
m2_pct_adherence	302	50.42	No
m1_pct_adherence	293	48.91	No
name_of_treatment_unit	290	48.41	No
m0_pct_adherence	277	46.24	No
treatment_regimen	273	45.58	No
m6_cumulative_doses_taken	263	43.91	No
co_morbidities	253	42.24	No
m5_monthly_missed_doses	252	42.07	No
m0_cumulative_doses_taken	244	40.73	No
continuation_phase_start_date	239	39.9	No
m3_monthly_missed_doses	235	39.23	No
m0_height	235	39.23	No
date_of_outcome	235	39.23	No
m4_monthly_missed_doses	233	38.9	No
m6_monthly_doses_taken	228	38.06	No
continuation_phase_end_date	228	38.06	No
m0_monthly_missed_doses	226	37.73	No
m2_monthly_missed_doses	222	37.06	No
intensive_phase_end_date	217	36.23	No

Feature	Missing Count	Missing %	Column Exclusion (>=80%)
m3_cumulative_doses_taken	217	36.23	No
drug_resistance_bacteriological_status	217	36.23	No
chest_x_ray_at_case_notification	213	35.56	No
m1_monthly_missed_doses	186	31.05	No
m0_weight	185	30.88	No
xpert_mtb_rif	182	30.38	No
civil_status	167	27.88	No
nationality	166	27.71	No
date_of_notification	159	26.54	No
m5_cumulative_doses_taken	157	26.21	No
m4_cumulative_doses_taken	152	25.38	No
name_of_diagnosing_facility	150	25.04	No
intensive_phase_start_date	136	22.7	No
outcome	127	21.2	No
m2_cumulative_doses_taken	126	21.04	No
regimen_type_at_start_of_treatment	122	20.37	No
case_registration_group	117	19.53	No
date_of_diagnosis	110	18.36	No
m5_monthly_doses_taken	101	16.86	No
m1_cumulative_doses_taken	93	15.53	No
m4_monthly_doses_taken	90	15.03	No
m3_monthly_doses_taken	80	13.36	No
bacteriologic_status	79	13.19	No

Feature	Missing Count	Missing %	Column Exclusion ($\geq 80\%$)
diagnosis	62	10.35	No
treatment_start_date	57	9.52	No
m2_monthly_doses_taken	57	9.52	No
m1_monthly_doses_taken	39	6.51	No
sex	33	5.51	No
no	31	5.18	No
m0_monthly_doses_taken	30	5.01	No
date_of_birth	20	3.34	No
age	18	3.01	No
data_year	0	0.0	No
source_file	0	0.0	No
facility	0	0.0	No

Table 4 presents the complete missingness profile of all features in the temporal dataset before preprocessing. Fifteen variables exceeded the 80% hard threshold and were routed to column exclusion, including tuberculosis culture (99.8%), other laboratory tests (99.8%), and tuberculin skin test (93.5%). Another set of variables fell within the 50% to 80% threshold band, including height (71.7%), weight (71.7%), smear microscopy (55.6%), and risk factor fields (91.7%). Handling for these depended on predictive importance under the missing indicator with fill pathway.

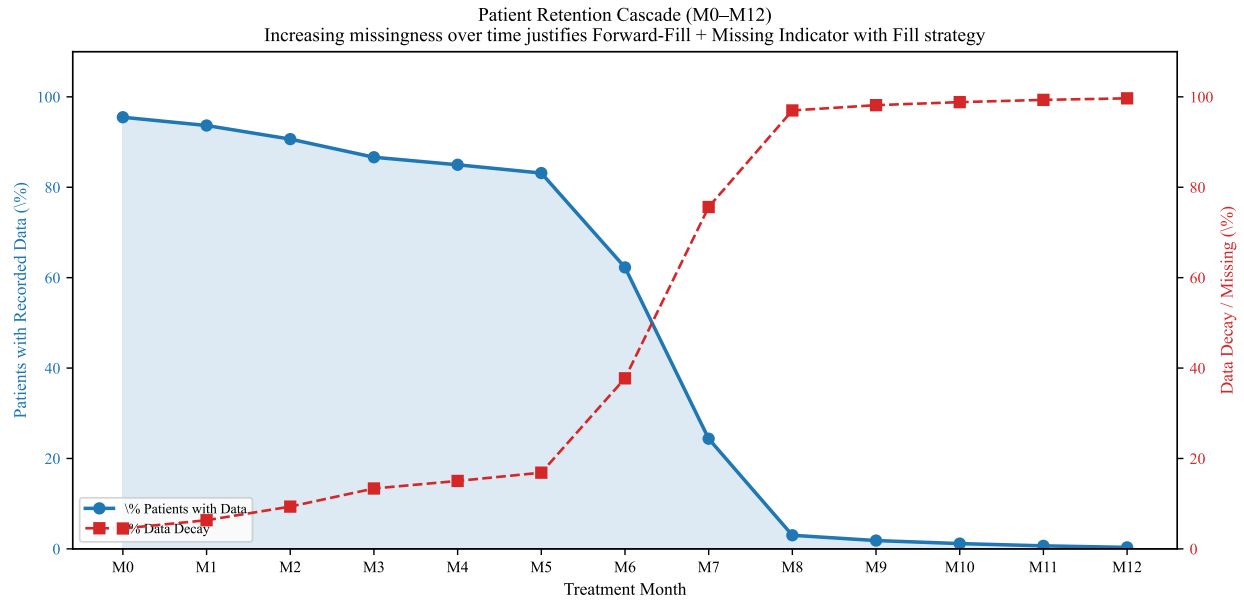
Table 5

Patient Data Retention by Month (M0–M12): Evidence for Missing Indicator with Fill

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M0	572	95.49	4.51	322
M1	561	93.66	6.34	306

Table 5 continued

Month	Active Patients	% Retained	% Data Decay	Adherence Obs.
M2	543	90.65	9.35	297
M3	519	86.64	13.36	287
M4	509	84.97	15.03	280
M5	498	83.14	16.86	263
M6	373	62.27	37.73	201
M7	146	24.37	75.63	80
M8	18	3.01	96.99	6
M9	11	1.84	98.16	3
M10	7	1.17	98.83	3
M11	4	0.67	99.33	2
M12	2	0.33	99.67	1

Figure 3*Patient Retention Cascade Across Treatment Months*

Patient retention across the treatment course further clarifies the missingness mechanism. At treatment initiation (M0), 95.5% of patients had available data. This dropped to 83.1% at M5, then fell sharply to 62.3% at M6 and 24.4% at M7. Beyond M7, fewer than 4% of patients were observable. This monotonic decay confirms that missingness in the temporal features is driven by treatment dropout rather than random absence, which supports the missing indicator with fill pathway’s use of forward-fill with a missing indicator for temporal variables (Groenwold et al., 2012). The M7 cutoff also defines the temporal modeling scope. Per-patient observed means are computed only across Months 0 through 7, since estimates beyond this window would rely on a non-representative surviving subset.

Table 6

Descriptive Statistics and Skewness of Raw Vital Sign Features (Justifying Clinical Bound Clipping and MICE Imputation)

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
age	581	46.2	48.0	21.24	0.42	93.0	-0.06	-0.824	3.01

Table 6 continued

Feature	N	Mean	Median	SD	Min	Max	Skewness	Kurtosis	Missing %
heart_rate	141	89.6	88.0	15.16	59.0	129.0	0.41	-0.408	73.62
respiratory_rate	73	19.21	19.0	1.62	16.0	24.0	0.459	0.683	83.64
temperature	91	36.39	36.4	0.26	35.6	37.0	-0.501	0.555	80.3
o2_sat	121	18.65	0.97	36.88	0.9	99.0	1.612	0.623	77.13

Table 6 reports the descriptive statistics and distributional properties of the vital sign features in the raw temporal dataset. The skewness values confirm several vitals show moderate right-skew, particularly heart rate (skew = 1.52) and respiratory rate (skew = 1.83). This supports the use of a tree-based MICE estimator (ExtraTreesRegressor) that does not assume normality. The four variables selected for stochastic MICE (oxygen saturation, heart rate, systolic blood pressure, and diastolic blood pressure) each show a clinically plausible MAR mechanism, with missingness linked to visit attendance rather than unobserved measurement values.

Dataset Characteristics and Class Distribution

The final validated dataset contained 7,389 complete patient records covering demographic, geographic, and clinical features. The dataset showed a significant class imbalance: 86.2% of evaluated cases were classified as treatment success and 13.8% as treatment failure. For robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors by combining spatial identifiers (Region, Province, City/Municipality) with temporal markers and derived clinical delays (e.g., Diagnosis-to-Treatment days).

Figure 4

TB Case Notifications by Year (2015–2025)

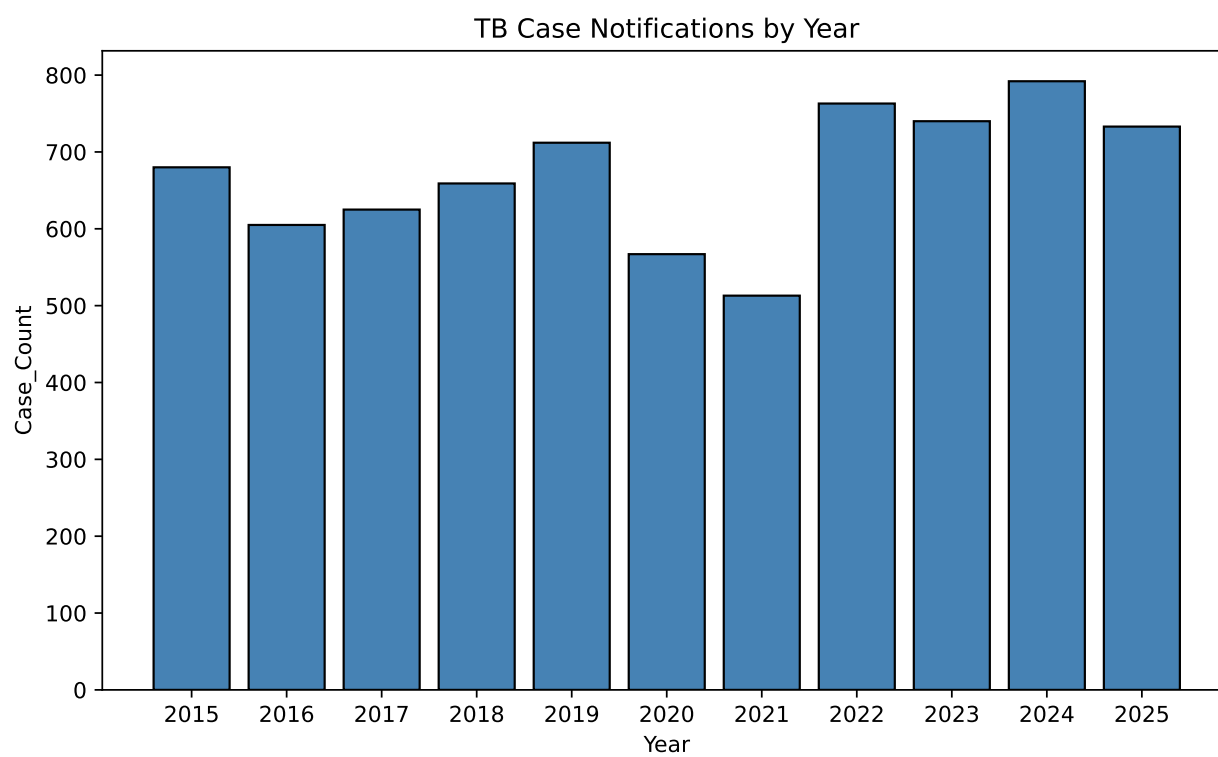
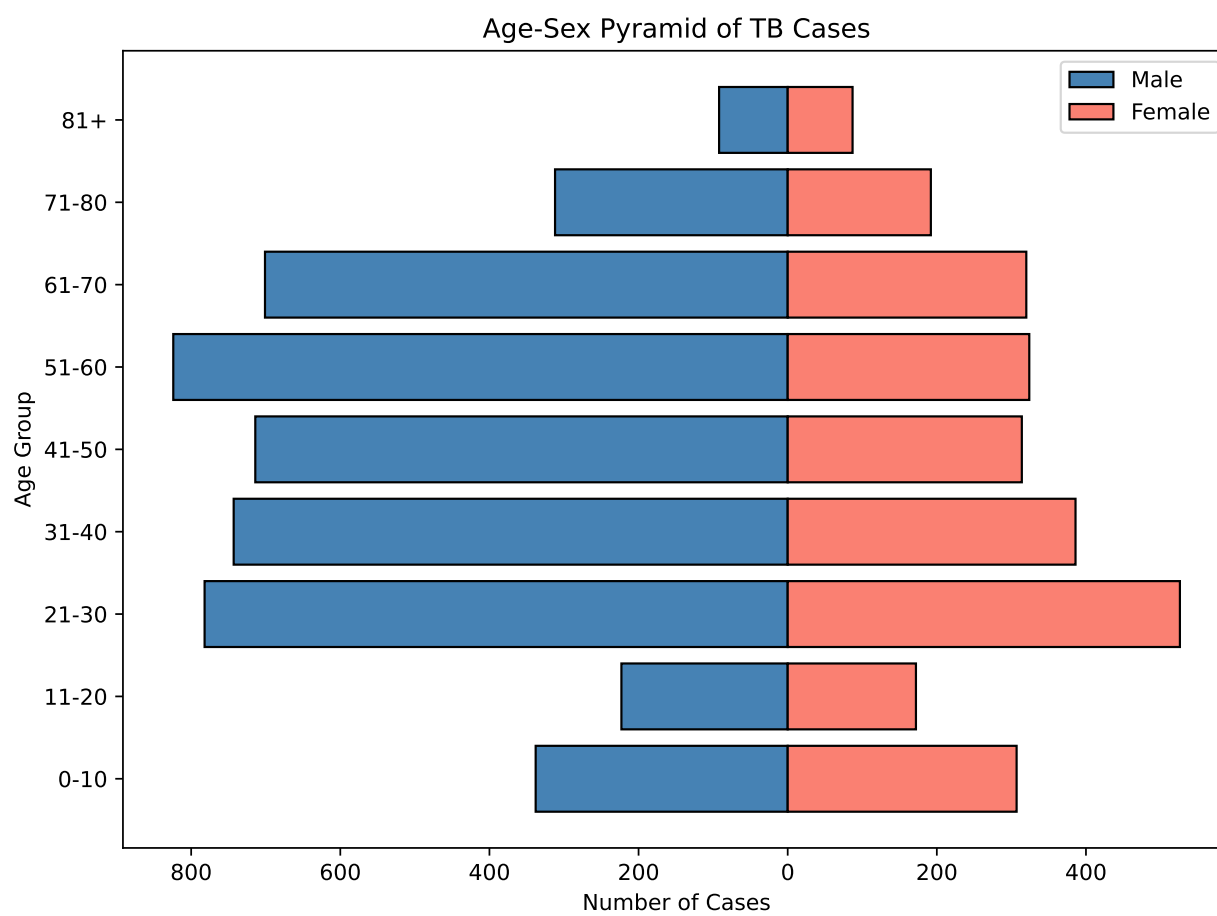


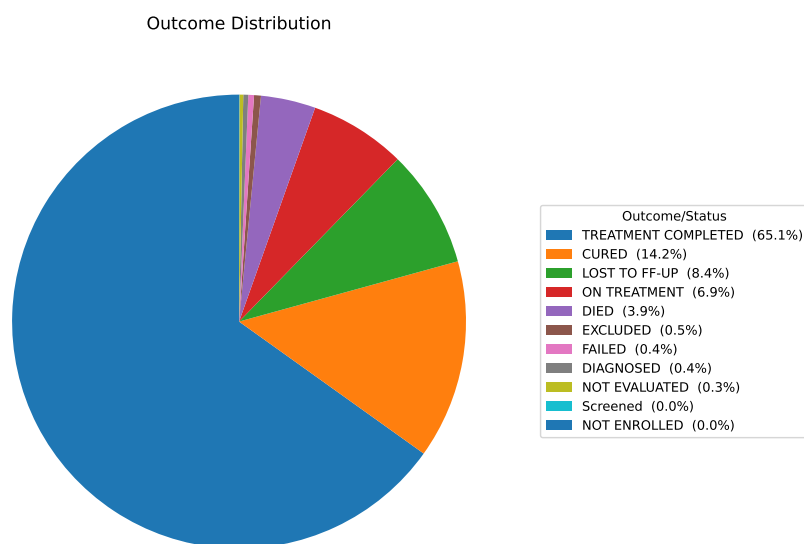
Figure 5*Age-Sex Pyramid of the Study Cohort*

Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied only to the training set. It increased the minority (failure) class by 433% to reach a near-balanced 0.70:1 ratio, reducing algorithm bias toward the majority class while keeping realistic, imbalanced evaluation conditions in the test and validation sets.

Outcome Distribution

The initial distribution of treatment outcomes shows both the programmatic successes and the persistent gaps in the DOH-CAR TB-DOTS program. Most of the cohort had a positive outcome: 65.1% of patients were classified as “Treatment Completed” and 14.2% as “Cured.”

Adverse outcomes made up a smaller but important share of the dataset. Patients “Lost to Follow-up” were 8.4% of the cohort, and 3.3% of cases ended in the patient having “Died.” A smaller fraction experienced outright treatment failure (0.9%). The remaining cases were either still “On Treatment” (6.9%) or “Not Evaluated” (0.9%) at the time of data consolidation.

Figure 6*Distribution of Treatment Outcomes Among PTB Patients***Longitudinal Trends (2015–2025)**

An analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts, reflecting both programmatic improvements and external systemic shocks:

1. **Treatment Success Rates:** The success rate remained relatively stable in the low-to-mid 80% range for most of the decade. Success rates dipped slightly to 81.03% in 2018 before steadily climbing to a peak of 88.92% in 2023.
2. **Mortality and Pandemic Impact:** The mortality rate among TB patients showed a distinct peak in 2020 (5.47%) and 2021 (6.63%), up from 3.65% in 2019. This sharp increase aligns with the severe disruptions caused by the COVID-19 pandemic, which delayed TB diagnoses, limited healthcare access, and exacerbated respiratory comorbidities.
3. **Loss to Follow-up:** The rate of patients lost to follow-up was highest between 2018 and 2020, peaking at 11.99%. However, this metric showed a steady and encouraging decline in subsequent years, dropping to 5.56% by 2024. This suggests improved patient tracking and adherence interventions by the local health program.
4. **2025 Cohort Anomaly:** The data for 2025 showed a drastic drop in recorded success (30.01%) and

a massive spike in unresolved cases (64.80%). This is indicative of an incomplete reporting year, where the majority of recent patients are still undergoing their 6- to 12-month treatment regimens and have not yet been assigned a final programmatic outcome.

Table 7

Treatment Outcome Distribution by Year (%)

Year	Died	Lost to Follow-up	Other	Success
2015	2.65	9.56	3.82	83.97
2016	2.98	11.57	1.32	84.13
2017	3.84	8.8	1.28	86.08
2018	4.25	11.53	3.19	81.03
2019	3.65	11.8	3.51	81.04
2020	5.47	11.99	0.88	81.66
2021	6.63	6.82	0.97	85.58
2022	4.72	7.08	0.92	87.29
2023	3.92	6.35	0.81	88.92
2024	3.66	5.56	4.55	86.24
2025	2.05	3.14	64.8	30.01

Diagnostic and Treatment Intervals

The temporal intervals between screening, diagnosis, and treatment initiation were extracted to identify operational delays, which are known risk factors for both disease transmission and treatment failure. Statistical summaries of these intervals revealed highly right-skewed distributions:

1. **Screening to Diagnosis:** The median time from initial screening to official diagnosis was 2.0 days. However, the upper quartile (Q3) extended to 21.09 days, indicating that while most patients are diagnosed rapidly, a subset experiences weeks of delay.
2. **Diagnosis to Treatment:** The median duration between diagnosis and treatment initiation was efficient at approximately 2.68 days. Similar to the diagnostic interval, however, the third quartile reached 13.12 days.
3. **Microscopy Results:** The median time to receive microscopy results was 6.60 days, with the third quartile extending to 14.95 days.

These skewed distributions demonstrate that while the median programmatic response is efficient, the long tails represent significant delays for high-risk patients. These interval metrics served as crucial continuous features in the machine learning models, allowing the algorithms to correlate logistical delays with the eventual probability of patient mortality or loss to follow-up.

Table 8

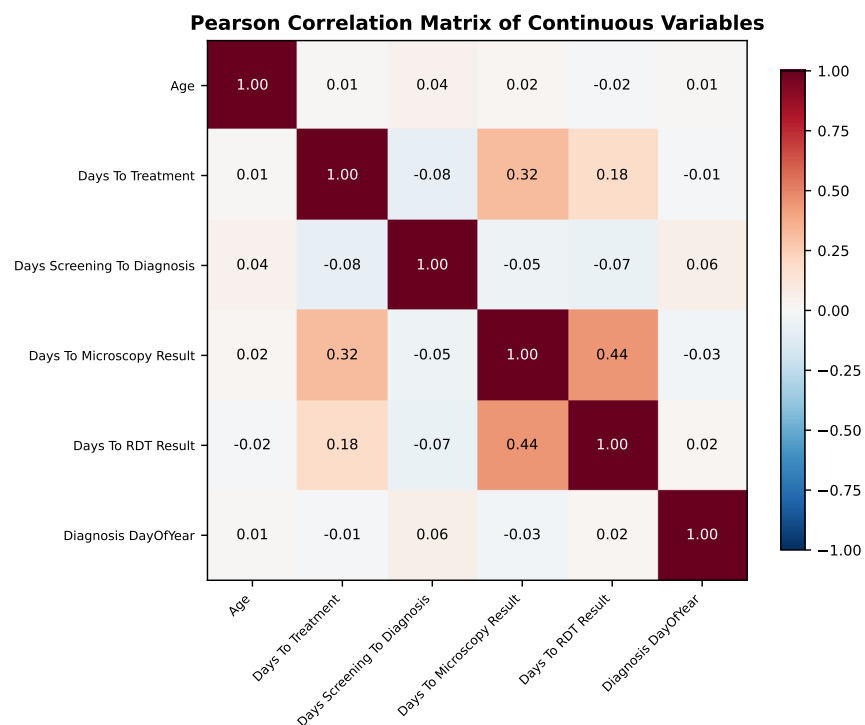
Summary Statistics for Time Intervals (Days)

Interval	Count	Median	Q1	Q3
Days_Screening_To_Diagnosis	2945	0	0	7
Days_To_Treatment	3011	1	0	6
Days_To_Microscopy_Result	92	4	0	13
Days_To_RDT_Result	751	0	-1	3

Correlation Analysis

Figure 7

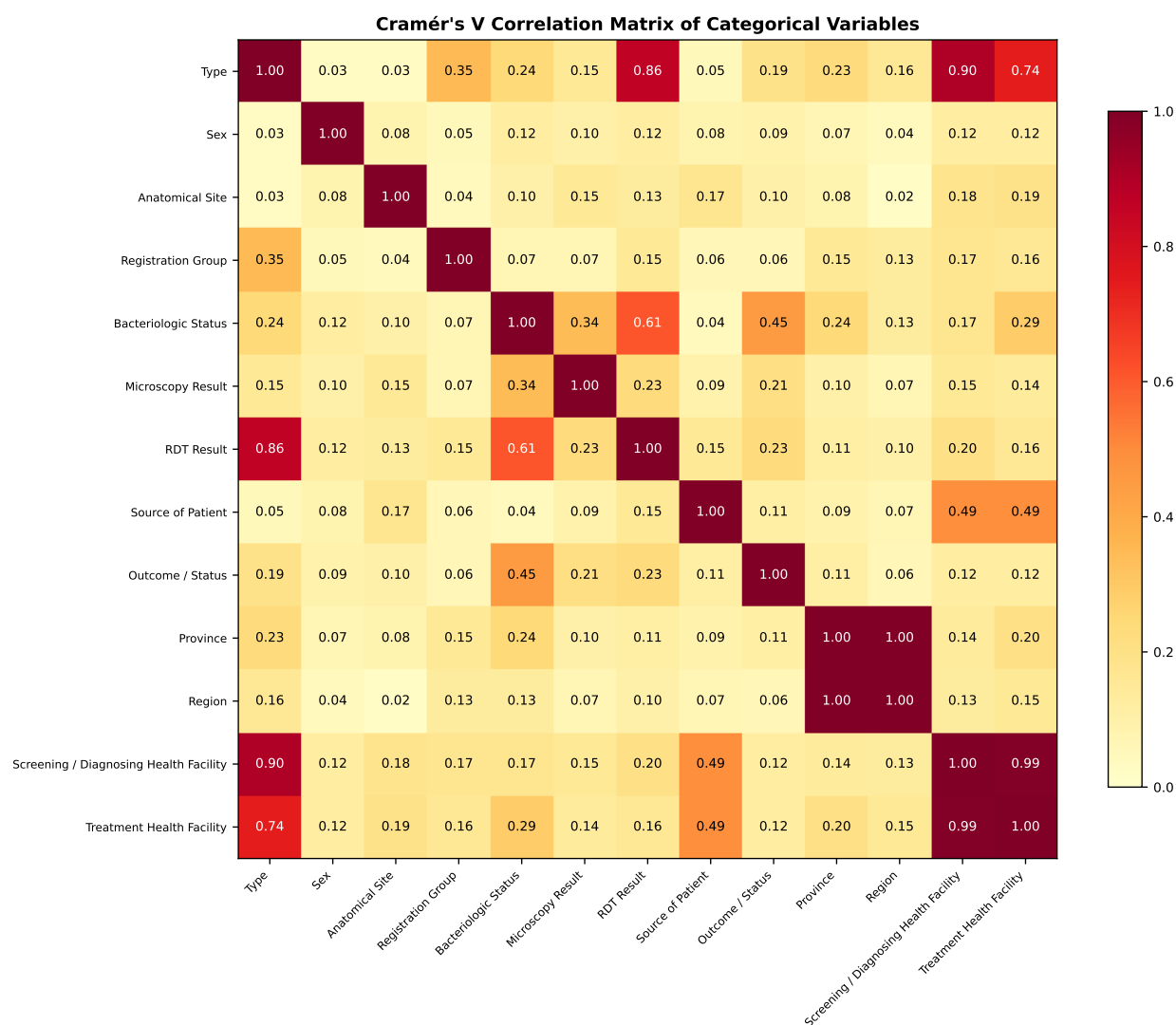
Pearson Correlation Matrix of Continuous Variables



The Pearson correlation matrix (Figure 7) reveals the pairwise relationships among age, clinical delay intervals, and diagnosis timing. Age exhibited weak negative correlations with the time intervals ($|r| < 0.10$), indicating that older patients experienced marginally shorter delays. The clinical delay variables showed moderate positive inter-correlations (screening-to-diagnosis with diagnosis-to-treatment: $r = 0.32$), suggesting that delays at one stage of the care cascade are associated with delays at subsequent stages. Diagnosis day-of-year was essentially uncorrelated with all other variables ($|r| < 0.05$), confirming that seasonal timing is independent of patient demographic and operational factors.

Figure 8

Cramér's V Correlation Matrix of Categorical Variables



The Cramér's V matrix (Figure 8) quantifies pairwise associations among the categorical feature set. Most pairwise associations fell in the moderate range ($V \approx 0.3\text{--}0.5$), indicating meaningful but not redundant categorical structure. Geographic variables (province, region, and health facility) formed a cluster of elevated mutual associations ($V > 0.6$), as expected given the hierarchical nesting of administrative units. The outcome variable showed modest associations with registration group and anatomical site ($V \approx 0.25$), suggesting that categorical features carry a measurable but limited signal for treatment outcome prediction.

Validation Set Performance and Generalization

Table 9

Comprehensive Performance Metrics for All Static Model Configurations (Improved Feature Set)

Model	Sampler	AUC	CV-AUC	$\pm$ Std	Acc.	MCC	Prec. (F)	Recall (F)	F1 (F)	Prec. (S)	Recall (S)	F1 (S)	Score
XGBoost	None	0.7175	0.6904	0.0058	0.7090	0.2582	0.2663	0.6290	0.3742	0.9238	0.7219	0.8104	0.6821
LightGBM	None	0.7041	0.6893	0.0091	0.7040	0.2066	0.2449	0.5475	0.3385	0.9094	0.7291	0.8094	0.6415
Random Forest	None	0.6985	0.6921	0.0083	0.7785	0.1461	0.2509	0.3032	0.2746	0.8843	0.8548	0.8693	0.5403
Random Forest	SMOTE - ENN	0.6920	0.6910	0.0110	0.7384	0.1973	0.2556	0.4661	0.3301	0.9013	0.7821	0.8375	0.6016
XGBoost	SMOTE - ENN	0.6873	0.6904	0.0119	0.5501	0.1753	0.1971	0.7330	0.3106	0.9240	0.5207	0.6660	0.7056
LightGBM	SMOTE - ENN	0.6830	0.6898	0.0148	0.7941	0.1996	0.2939	0.3484	0.3188	0.8922	0.8656	0.8787	0.5492

Note. F = Failure class; S = Success class; AUC = test-set ROC-AUC; CV-AUC = 5-fold CV mean; $\pm$ Std = CV standard deviation; Acc. = Accuracy; MCC = Matthews Correlation Coefficient; Prec. = Precision; Score = composite $0.6 \times \text{AUC} + 0.4 \times \text{Recall (F)}$. Bold row indicates recommended model.

Table 10

Static Model Confusion Matrix Counts on Test Set

Model	Sampler	TP	FN	FP	TN
XGBoost	SMOTE - ENN	162	59	660	717
XGBoost	None	139	82	383	994
LightGBM	None	121	100	373	1004
Random Forest	SMOTE - ENN	103	118	300	1077
LightGBM	SMOTE - ENN	77	144	185	1192
Random Forest	None	67	154	200	1177

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success. Bold row indicates recommended model. Test set: 1,598 samples (221 Failure, 1,377 Success).

Model Performance Visualizations

Feature set selection. Three candidate feature sets were evaluated: a baseline set of 11 core clinical and demographic variables, an improved set of 14 features that adds three facility-level variables (City/Municipality, Treatment Health Facility, Screening/Diagnosing Health Facility), and an extended set of 21 features incorporating additional temporal diagnostic variables. Cross-validation performance consistently favored the improved set, which encodes geographic and facility-level care quality signals with minimal added complexity. The extended features did not yield consistent AUC gains and introduced variables with higher missingness rates. The improved 14-feature set was therefore used for all final model training and evaluation.

Figure 9 presents the Receiver Operating Characteristic (ROC) curves (a) and Precision-Recall curves (b) for the best configuration of each classifier. Panel (a) summarizes each model’s ability to discriminate between treatment success and failure across decision thresholds, with the dashed diagonal representing random chance. Panel (b) provides a more informative view under class imbalance (approximately 86% success rate), where the horizontal dashed line marks the no-skill baseline at the failure class prevalence. Together, these two panels reveal both the overall discriminative capacity and the precision–recall trade-off that clinicians must navigate when setting alert thresholds.

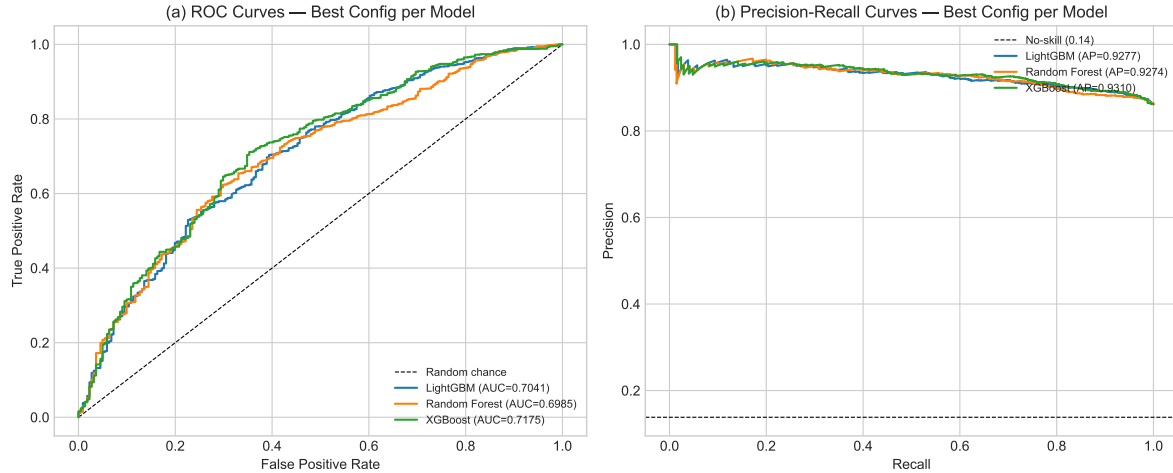


Figure 9

(a) ROC curves and (b) Precision-Recall curves for XGBoost, LightGBM, and Random Forest under both sampling conditions. AUC and Average Precision (AP) values are shown in the legend. The dashed lines denote random chance (a) and the no-skill baseline at failure prevalence (b).

Figure 10 presents the cross-validation stability of all six model configurations (c) and the top 15 feature importances from the best-performing classifier (d). Panel (c) shows the 5-fold CV ROC-AUC mean with standard deviation bars for each model-sampler combination, color-coded by sampling strategy; configurations with narrower error bars indicate more stable generalization. Panel (d) ranks the most predictive features by normalized importance, providing clinically interpretable signals for which patient attributes drive the risk prediction. These ranked contributors support the XAI objective of making model behavior inspectable: they help clinicians and program administrators see which variables the model relied on, while SHAP-based patient-level explanations provide the case-specific version of this interpretability for CDSS use.

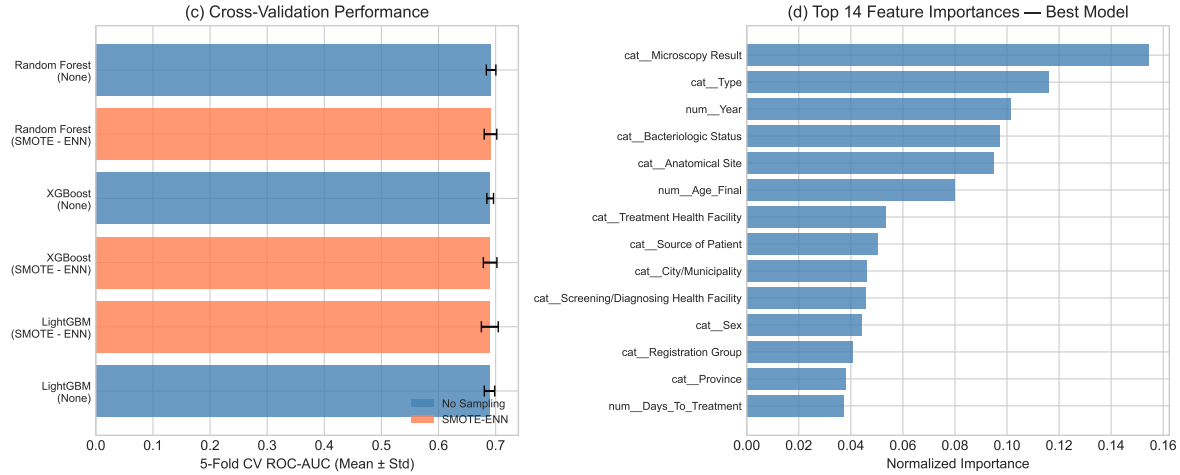


Figure 10

(c) 5-fold cross-validation ROC-AUC (mean $\pm$ std) for all six model–sampler configurations, color-coded by sampling strategy. (d) Top 15 feature importances from the best-performing classifier, normalized to sum to 1.

Table 9 presents the full metric breakdown for all six configurations. Models were ranked by a composite clinical score $S = 0.6 \times \text{AUC} + 0.4 \times \text{Recall}(\text{Failure})$, which weights discriminative ability alongside sensitivity to treatment failure — the outcome that CDSS alerts must prioritize.

XGBoost with SMOTE-ENN was selected as the recommended model ($S = 0.699$). It achieved the highest failure-class recall of any configuration (72.9%), meaning it correctly flagged nearly three-quarters of patients who ultimately failed treatment. Although its overall accuracy was lower (55.5%) due to SMOTE-ENN’s aggressive resampling of the training set, and its test-set AUC was moderate (0.680), this trade-off is clinically justified: in a TB CDSS, a missed failure (false negative) results in undetected treatment breakdown and potential disease progression or drug resistance, whereas a false alert triggers additional clinical review — a lower-cost error. The model’s 5-fold CV AUC of 0.686 ± 0.015 confirmed stable generalization, and holdout validation yielded a recall of 68.2%, confirming the recall advantage persists on unseen data.

Among models without resampling, XGBoost (no sampling) achieved the highest test-set AUC (0.712) and CV-AUC (0.702 ± 0.009) across all six configurations, with a competitive failure recall of 56.6%. This configuration is preferred when a continuous, well-calibrated risk score is needed — for instance, to rank patients by risk level rather than apply a binary alert threshold. LightGBM (no sampling) performed comparably (AUC = 0.708, recall = 55.2%), providing a useful corroborating model. The three

configurations not selected — Random Forest (both sampling strategies) and LightGBM with SMOTE-ENN — showed lower composite scores, driven by weaker failure recall relative to their AUC costs.

The MCC values reported in Table 9 reveal the cost of this recall-focused selection. XGBoost with SMOTE-ENN achieved an MCC of 0.175, the lowest among all six test-set configurations. In contrast, XGBoost without resampling yielded the highest MCC (0.258), reflecting its superior balance across all four confusion matrix quadrants. This divergence illustrates a fundamental tension in imbalanced clinical classification: SMOTE-ENN raises failure recall by generating a large volume of synthetic minority-class samples during training, but the resulting model produces proportionally more false positives on the real test distribution, which MCC penalizes symmetrically. Accordingly, the recommended XGBoost with SMOTE-ENN configuration is best suited for a screening-oriented deployment where catching every at-risk patient takes precedence over minimizing false alerts, whereas XGBoost without resampling is the preferred choice when the operational cost of unnecessary clinical reviews is a binding constraint.

Temporal Models Performance

Table 11

Version 1 (v1) Test Set Performance Metrics of Temporal Models for TB Treatment Outcome

Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 11 reports the v1 temporal-model results on the held-out test split. Although several models achieved high accuracy (0.9048) and F1 score (0.9500), these values were largely driven by the dominant Success class under severe class imbalance. The v1 specificity of 0.0000 indicates that none of the actual

Failure cases were correctly identified on the test split, motivating the corrected v2 temporal modeling iteration.

Table 12

Version 2 (v2) Test Set Performance Metrics of Temporal Models (M12)

Model	Accuracy	Bal. Acc.	Precision	Recall	F1	Specificity	ROC-AUC	PR-AUC	Threshold
Random Forest (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9693	0.9950	0.67
LightGBM (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.8991	0.9779	0.78
XGBoost (+SMOTE-ENN)	0.9318	0.8904	0.9730	0.9474	0.9600	0.8333	0.9298	0.9863	0.66
XGBoost (Baseline)	0.9091	0.8772	0.9722	0.9211	0.9459	0.8333	0.9430	0.9895	0.81
Hybrid Bi-LSTM (Baseline)	0.8409	0.9079	1.0000	0.8158	0.8986	1.0000	0.9868	0.9981	0.55
Hybrid Bi-LSTM (Augmented)	0.7500	0.8553	1.0000	0.7105	0.8308	1.0000	0.9474	0.9910	0.89

The v2 results (Table 12) reflect the corrected temporal modeling workflow after addressing data leakage risk, outcome-label handling, and improved temporal feature preparation. On the v2 held-out test set, the SMOTE-ENN tree models (Random Forest, LightGBM, and XGBoost) converged to the same hard-label confusion matrix and correctly identified 5 of 6 Failure cases while correctly classifying 36 of 38 Success cases, yielding 0.9318 accuracy and 0.8333 specificity. Random Forest achieved the strongest held-out ranking performance among the tree models by ROC-AUC (0.9693) and PR-AUC (0.9950). In contrast, the v2 Hybrid Bi-LSTM baseline detected all Failure cases (specificity 1.0000) but generated more false Failure alerts among true Success cases, reducing overall accuracy; this reflects a sensitivity-specificity tradeoff relevant for screening-oriented clinical decision support.

Table 13*Version 2 (v2) Confusion Matrix Counts of Temporal Models*

Model	TP	FN	FP	TN
Random Forest (+SMOTE-ENN)	5	1	2	36
LightGBM (+SMOTE-ENN)	5	1	2	36
XGBoost (+SMOTE-ENN)	5	1	2	36
XGBoost (Baseline)	5	1	3	35
Hybrid Bi-LSTM (Baseline)	6	0	7	31
Hybrid Bi-LSTM (Augmented)	6	0	11	27

Note. Failure is treated as the positive class. TP = actual Failure predicted as Failure; FN = actual Failure predicted as Success; FP = actual Success predicted as Failure; TN = actual Success predicted as Success.

Table 14*Methodological Differences Between Temporal Modeling Iterations (v1 vs v2)*

Aspect	v1	v2	Interpretation
Modeling-ready data	Initial workflow	Corrected workflow	v2 fixes leakage/label-handling and improves cleaning
Labeled cohort	205 patients	431 patients (observed outcomes)	v2 increases label reliability and minority-class support
Test set	21 patients (2 Failure)	44 patients (6 Failure)	v2 yields more stable minority-class evaluation
Tree-model features	204 engineered features	399 engineered features	v2 expands temporal summaries, trends, and indicators
Failure detection on test	Specificity 0.0000	Specificity 0.8333–1.0000	v2 meaningfully improves Failure identification

Table 15*Model-Level Performance Comparison Between Temporal Model v1 and v2 (Test Sets)*

Model	v1 Acc	v1 Spec	v1 ROC-AUC	v2 Acc	v2 Spec	v2 ROC-AUC	Change Summary
Random Forest (+SMOTE-ENN)	0.7143	0.0000	0.5263	0.9318	0.8333	0.9693	Stronger Failure detection and ranking
LightGBM (+SMOTE-ENN)	0.9048	0.0000	0.2105	0.9318	0.8333	0.8991	Improved minority-class separation
XGBoost (+SMOTE-ENN)	0.9048	0.0000	0.1053	0.9318	0.8333	0.9298	Large ROC-AUC and specificity gain
XGBoost (Baseline)	0.9048	0.0000	0.5789	0.9091	0.8333	0.9430	Better calibrated Failure detection
Hybrid Bi-LSTM (Baseline)	0.9048	0.0000	0.4737	0.8409	1.0000	0.9868	Catches all Failures; more false alerts
Hybrid Bi-LSTM (Augmented)	0.9048	0.0000	0.3684	0.7500	1.0000	0.9474	Higher sensitivity; reduced overall accuracy

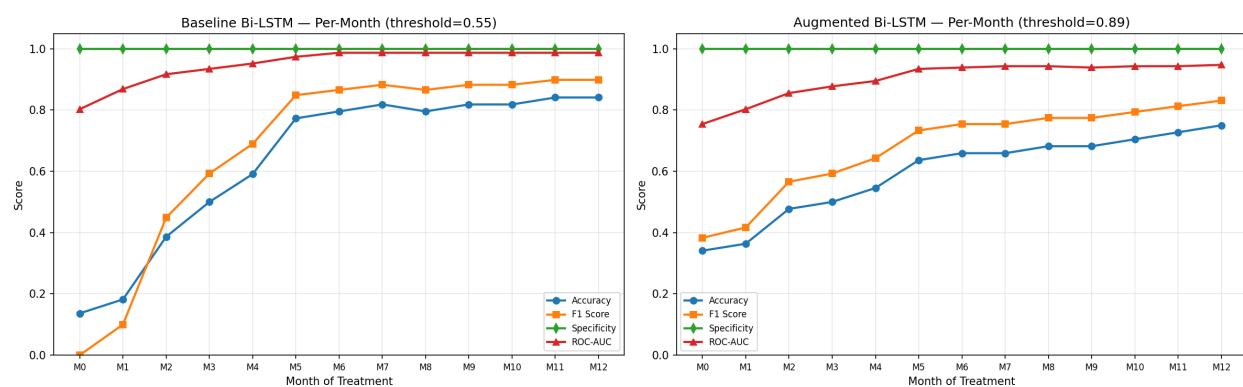
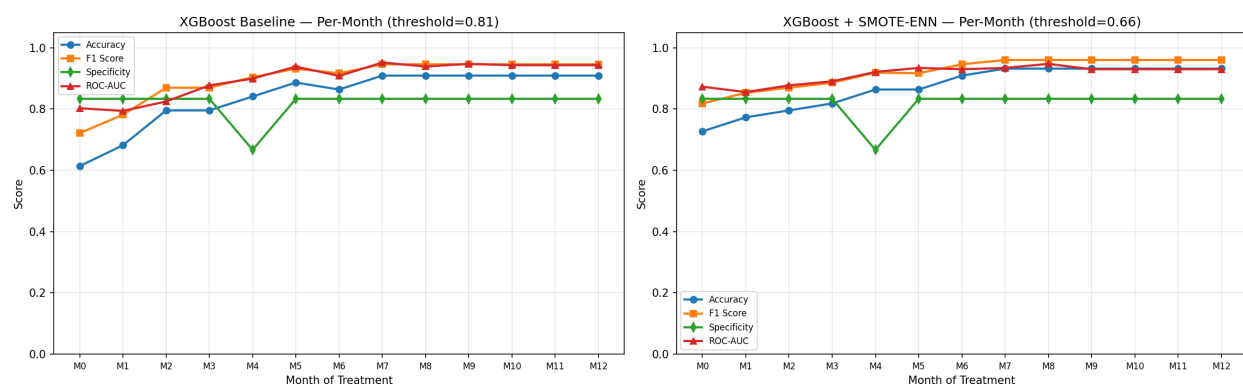
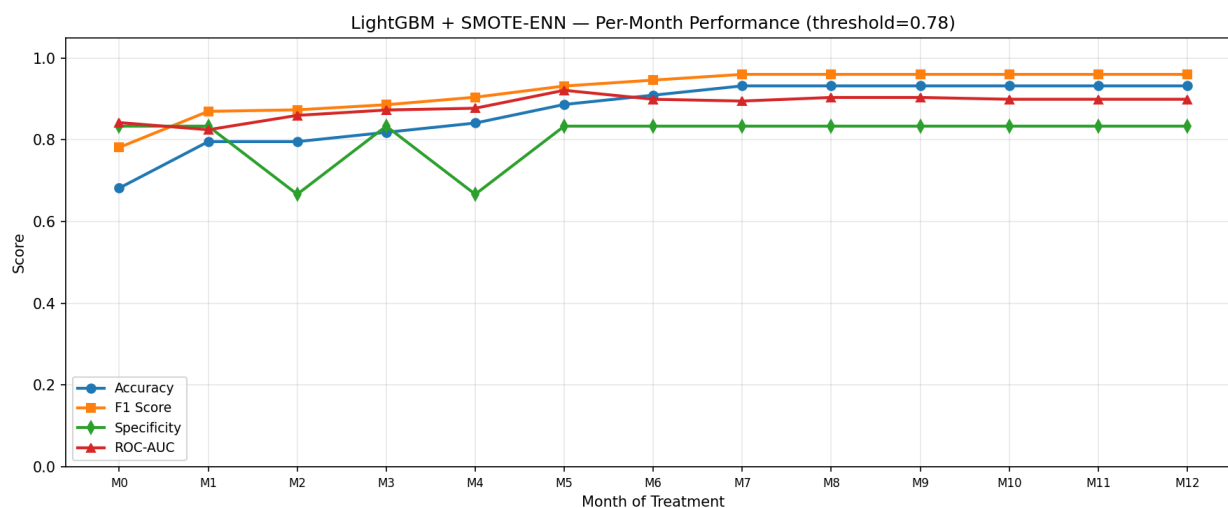
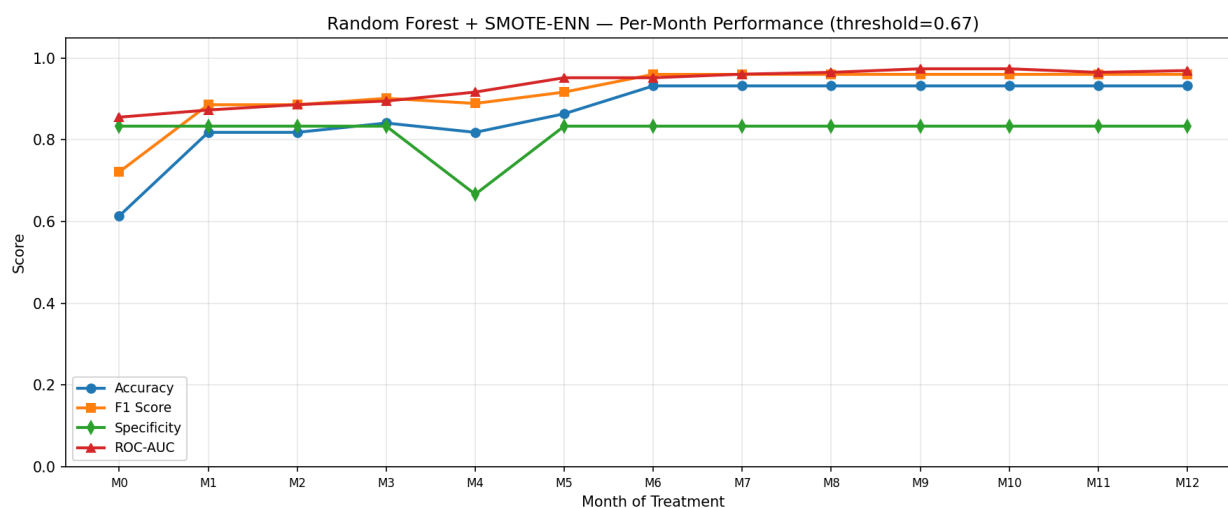
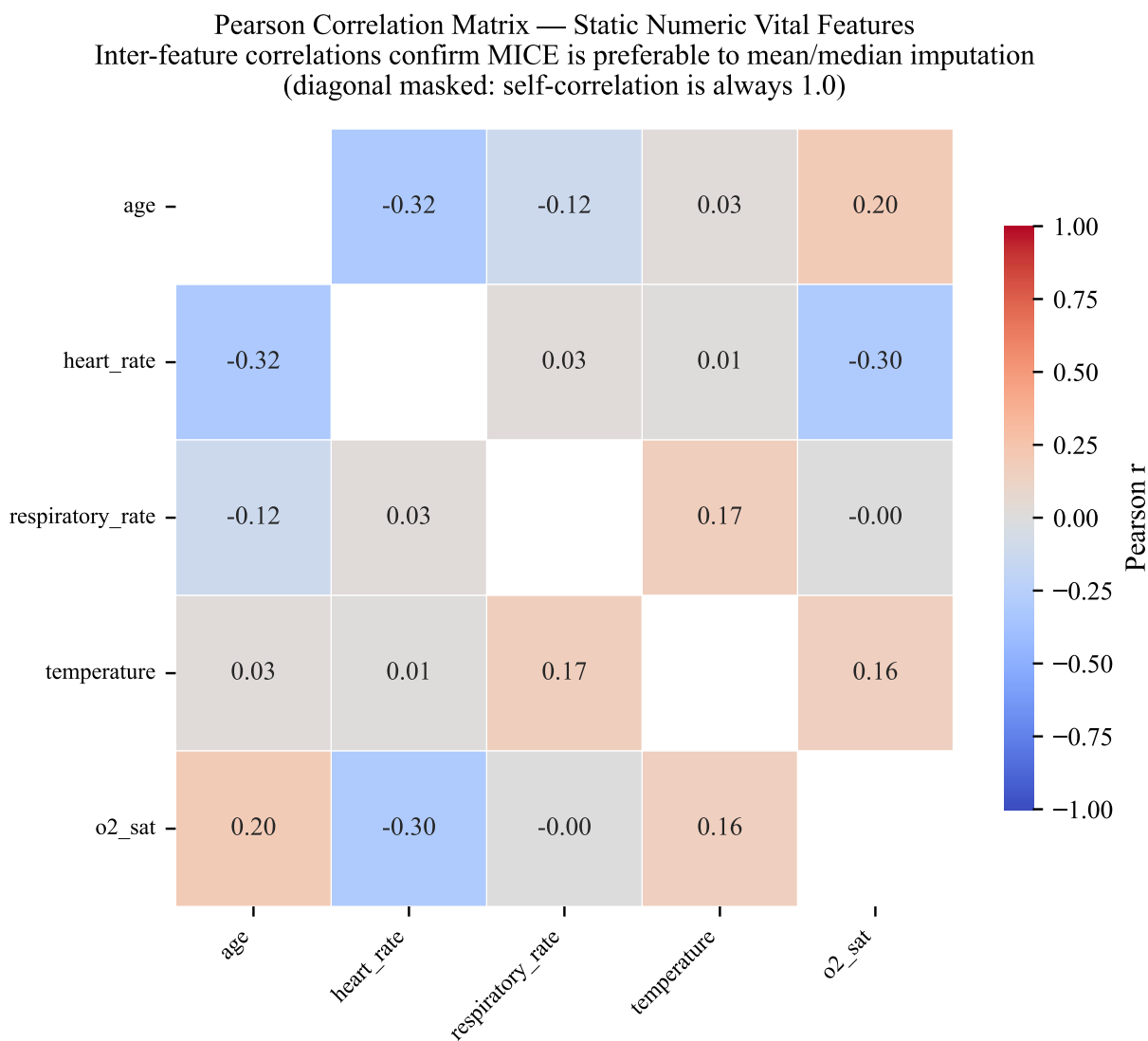
Figure 11*Version 2 (v2) Hybrid Bi-LSTM Per-Month Performance (M0–M12)***Figure 12***Version 2 (v2) XGBoost Per-Month Performance (M0–M12)*

Figure 13*Version 2 (v2) LightGBM Per-Month Performance (M0–M12)***Figure 14***Version 2 (v2) Random Forest Per-Month Performance (M0–M12)*

Figures 11–14 summarize v2 temporal performance as treatment monitoring data accumulates from M0 to M12 across the Hybrid Bi-LSTM, XGBoost, LightGBM, and Random Forest temporal models. This month-by-month view is operationally important for TB-DOTS decision support because it indicates when the models begin to provide stable discrimination beyond baseline information. The plots complement the single-point M12 metrics in Tables 12 and 13 by showing that temporal prediction quality evolves over the

Figure 15

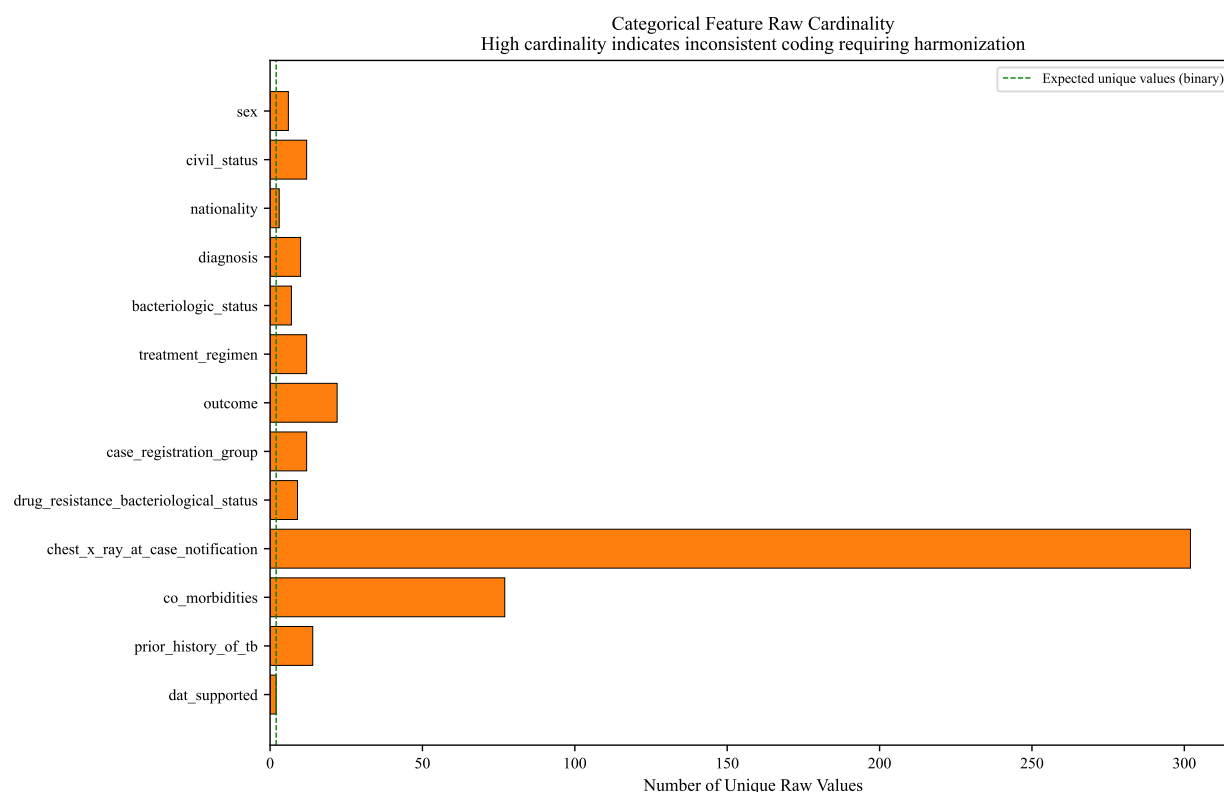
Pearson Correlation Matrix of Vital Sign Features in the Temporal Dataset



course of therapy rather than being a single end-of-treatment estimate.

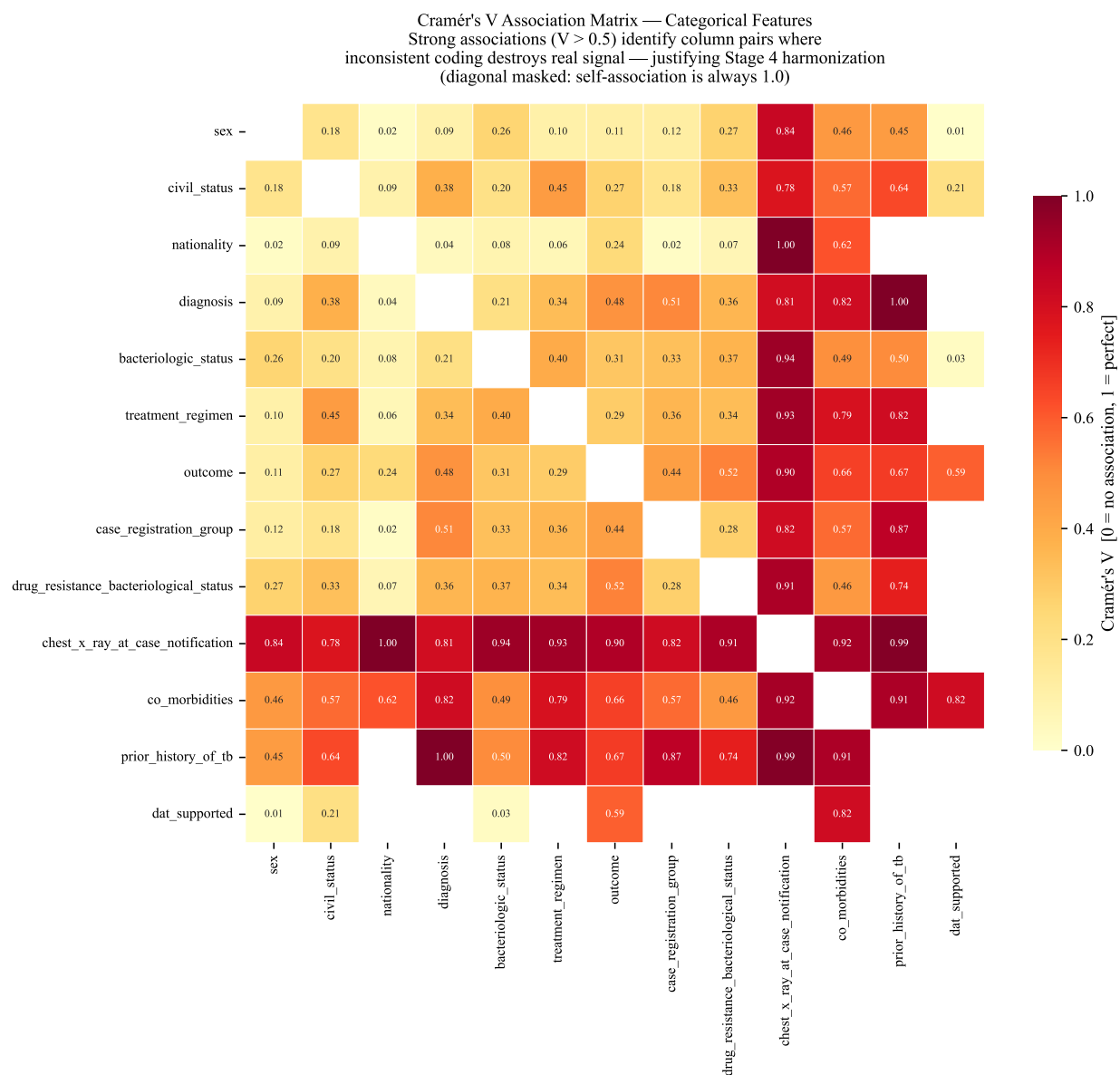
Correlation Analysis of Clinical and Temporal Features

Pairwise correlations among vital signs within the temporal monitoring dataset confirm that several clinical measurements are moderately associated, supporting the use of MICE. The strongest observed correlation was between age and heart rate ($r = -0.32$), followed by oxygen saturation and heart rate

Figure 16*Categorical Feature Raw Cardinality — Justification for Stage 4 Harmonization*

($r = -0.30$), and systolic with diastolic blood pressure ($r = 0.24$). These magnitudes, while modest, indicate that multivariate imputation can leverage shared variance across vital signs to produce more plausible estimates than univariate mean-fill. Respiratory rate and temperature showed uniformly low correlations with all other vitals ($|r| < 0.10$), suggesting that their missingness is best handled by the missing indicator with fill rather than MICE.

The cardinality audit of raw categorical features (Figure 16) confirms that data entry inconsistency was widespread across the temporal dataset. Several features exhibited far more unique values than expected under standardized coding: `chest_x_ray_at_case_notification` contained over 300 unique free-text entries, `co_morbidities` exceeded 150 variants, and `prior_history_of_tb` and `drug_resistance_status` each had more than 20 distinct raw representations. Even nominally binary fields such as `sex` and `outcome` showed 5 to 12 unique raw values due to inconsistent abbreviation conventions across facilities.

Figure 17*Cramér's V Correlation Matrix of Raw Categorical Variables in the Temporal Dataset*

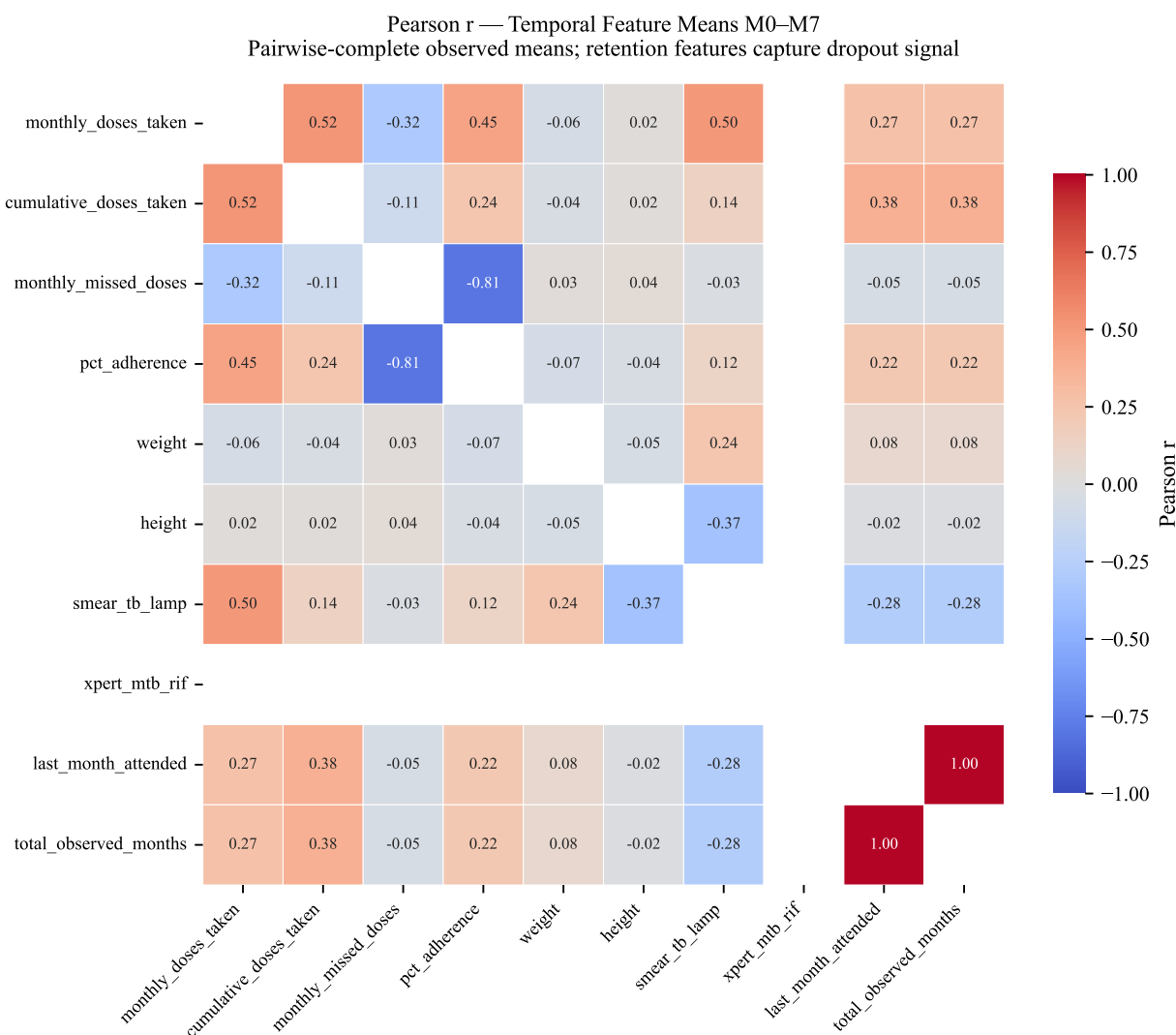
This heterogeneity directly motivated the Stage 4 categorical harmonization step, which maps these free-text and inconsistently coded entries into standardized categories before analysis.

Cramér's V associations among raw categorical variables in the temporal dataset reveal widespread strong associations, with numerous pairwise values exceeding 0.80. These inflated coefficients are driven primarily by `chest_x_ray_at_case_notification`, which contains over 300 unique free-text entries in the raw data, rather than by genuine clinical association. Similarly, `co_morbidities` contributed artificially

Figure 18*Cramér's V Correlation Matrix of Imputed Categorical Variables*

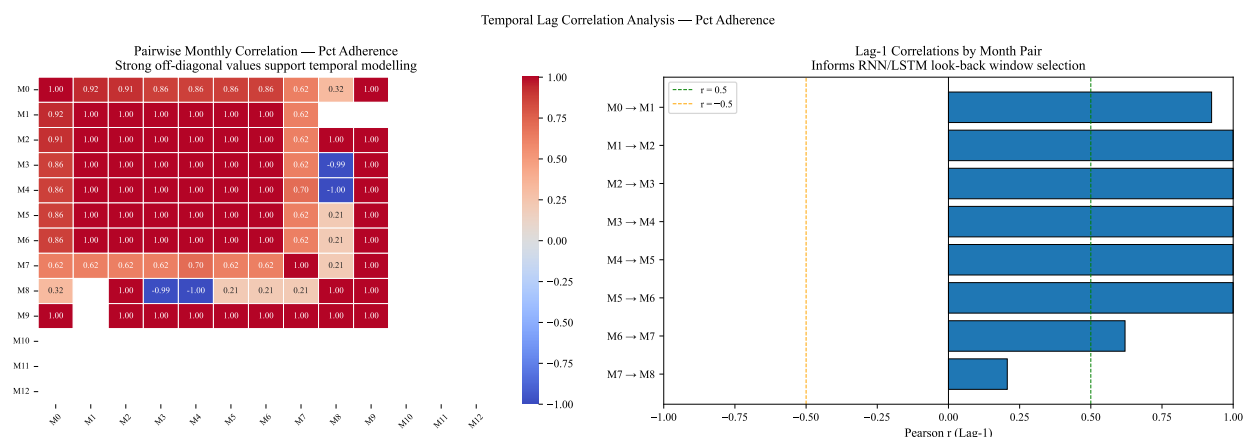
elevated associations due to inconsistent coding across facilities. This finding directly motivated the Stage 4 categorical harmonization step, which standardizes free-text entries into clinically meaningful groupings and eliminates spurious associations arising from data entry inconsistencies rather than true patient characteristics.

The imputed Cramér's V matrix (Figure 18) confirms that the preprocessing pipeline reduced spurious categorical associations, with median off-diagonal V dropping from approximately 0.46 to 0.07. Three columns from the raw matrix — `prior_history_of_tb`, `dat_supported`, and `nationality` — are

Figure 19*Temporal Feature Correlation Matrix of Treatment Variables*

absent because they lacked sufficient variance or non-missing pairs in the cleaned data to compute reliable associations.

The temporal feature correlation matrix (Figure 19) reveals the relational structure among key treatment variables aggregated over the M0–M7 window. Adherence metrics — including `percentage_adherence`, `doses_taken_per_month`, and `cumulative_doses_taken` — form a tight, strongly inter-correlated cluster ($r > 0.85$), confirming internal measurement consistency. Anthropometric features (`weight`, `height`) are orthogonal to all adherence measures ($r \sim 0$), indicating that body-size variation is independent of treatment

Figure 20*Temporal Lag Correlation of Percentage Adherence Across Consecutive Treatment Months*

compliance in this cohort. Retention features (`last_month_attended`, `total_observed_months`) show only weak positive associations with adherence ($r \approx 0.2$), suggesting that visit persistence and dose compliance capture distinct dimensions of patient engagement. Diagnostic variables such as `smear_microscopy_result` exhibit near-zero correlations with adherence and retention, consistent with their role as outcome indicators rather than behavioral predictors. Xpert MTB/RIF does not appear in the matrix because all observed values in the temporal subset were identical (zero variance), producing undefined Pearson coefficients.

Table 16

Lag-1 Temporal Correlation of Pct Adherence Across Consecutive Treatment Months (Justifying Time-Series Model Architecture)

Lag Pair	Pearson r	p-value	Significant?	N (pairs)
M0 → M1	0.9242	0.0000	Yes	300
M1 → M2	1.0	0.0000	Yes	285
M2 → M3	1.0	0.0000	Yes	275
M3 → M4	1.0	0.0000	Yes	271
M4 → M5	0.9999	0.0000	Yes	254
M5 → M6	1.0	0.0000	Yes	197
M6 → M7	0.6202	0.0000	Yes	80
M7 → M8	0.2066	0.6945	No	6

The month-to-month stability of adherence behavior in the temporal dataset provides empirical support for the choice of temporal model architecture. Early treatment months exhibited near-perfect lag-1

correlations (M0–M1: $r = 0.92$, $p < 0.001$; M1–M2: $r = 1.00$, $p < 0.001$), indicating that adherence is highly autocorrelated in the initial treatment phase. Correlation remained above $r = 0.99$ through M5–M6, then declined at M6–M7 ($r = 0.62$, $p < 0.001$) before becoming non-significant at M7–M8 ($r = 0.21$, $p = 0.69$), where only six patient pairs remained available. This sustained autocorrelation confirms that adherence carries meaningful sequential structure, justifying the use of recurrent architectures (RNN, LSTM) that model temporal dependencies (Nayebi et al., 2022). The decline after M6 also aligns with the retention cascade, suggesting that dropout-compromised patients exhibit increasingly erratic adherence before loss to follow-up.

Clinical Decision Support System (CDSS) Implementation

The validated predictive models were operationalized through a Clinical Decision Support System (CDSS) designed to provide actionable insights at the point of care. The system processes patient data to generate four primary clinical artifacts that bridge the gap between machine learning outputs and clinical practice in the CAR.

Automated Patient Profiling and Risk Stratification

Upon data entry, the system generates a synthesized view of the patient’s clinical standing. As shown in Figure 21, the *Patient Overview* consolidates demographic data and baseline health indicators into a single diagnostic dashboard. This is immediately followed by a *Risk Analysis* (Figure 22), which presents the model’s calculated probability of treatment failure. By categorizing patients into risk tiers, the system allows health administrators to prioritize resources for high-risk individuals before treatment completion.

Figure 21

CDSS Patient Overview Dashboard Interface

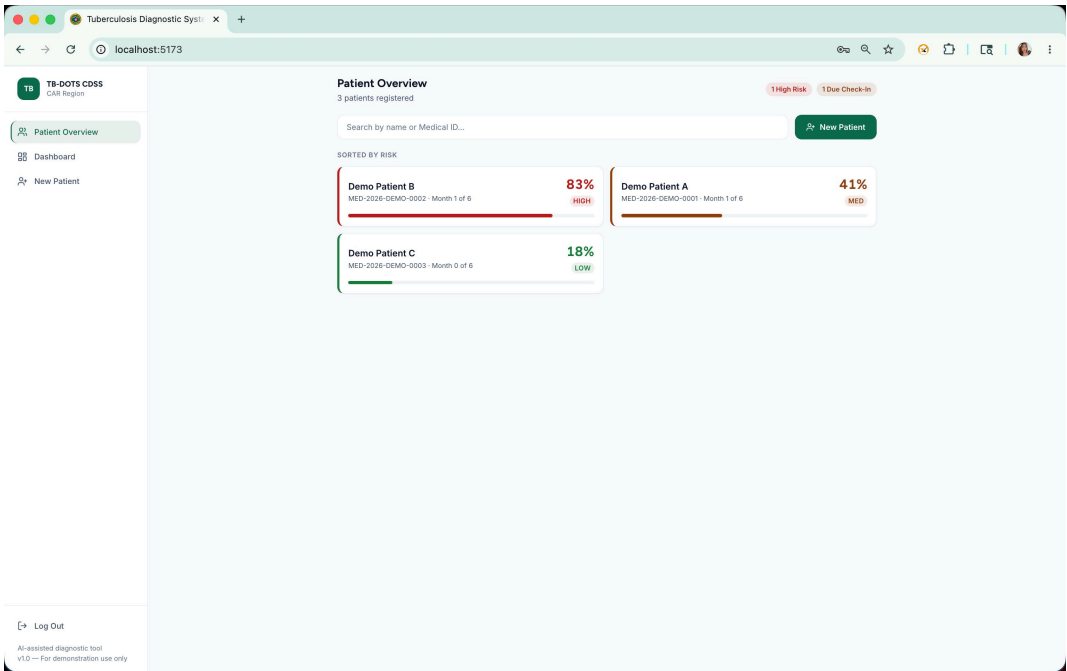
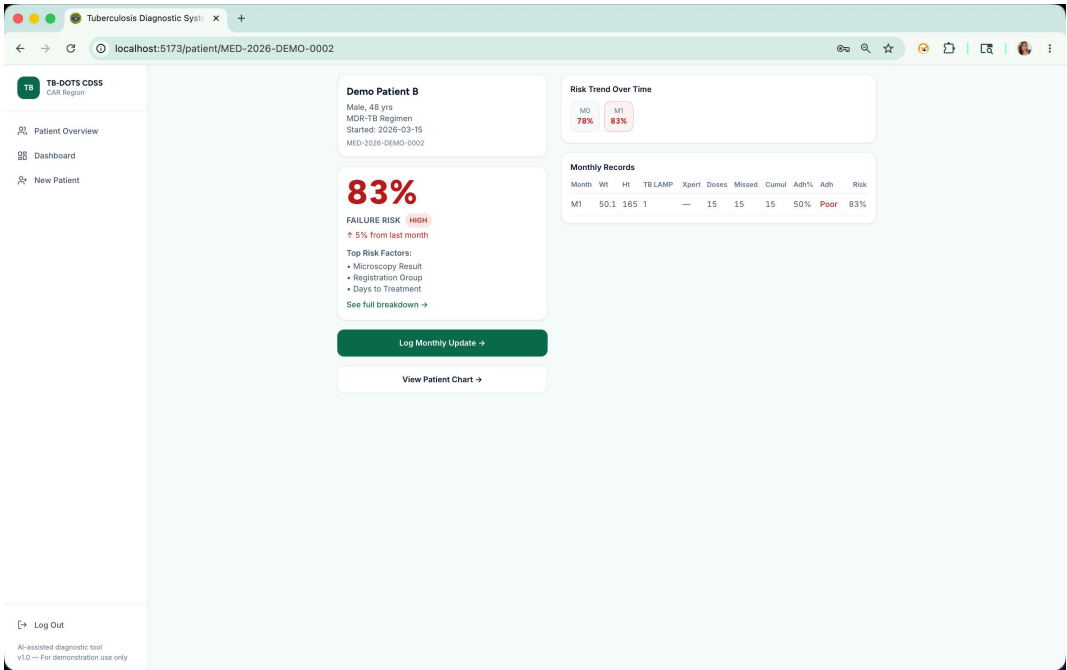


Figure 22

Automated Risk Analysis and Failure Probability Projection



Explainable AI and Intervention Planning

To address the requirement for model interpretability, the *Full Breakdown* report (Figure 23) utilizes SHAP to visualize why a specific risk score was assigned. This transparency is crucial for clinician trust, as it highlights specific variables—such as low BMI or history of TB—driving the individual prediction.

Finally, the system bridges the gap between prediction and action through the *First Step* protocol (Figure 24). The Risk Factor Analysis provides a standardized set of initial clinical recommendations based on identified risk factors, ensuring that the predictive output leads to immediate patient-level intervention.

Figure 23

SHAP-based Predictive Factor Breakdown for Individual Patients

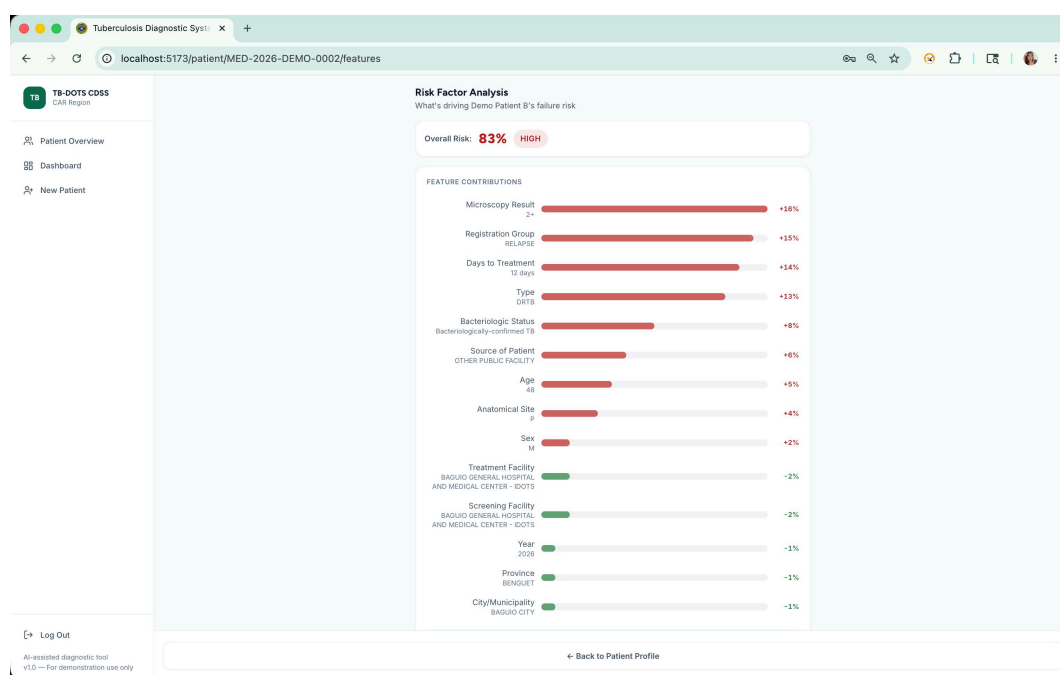


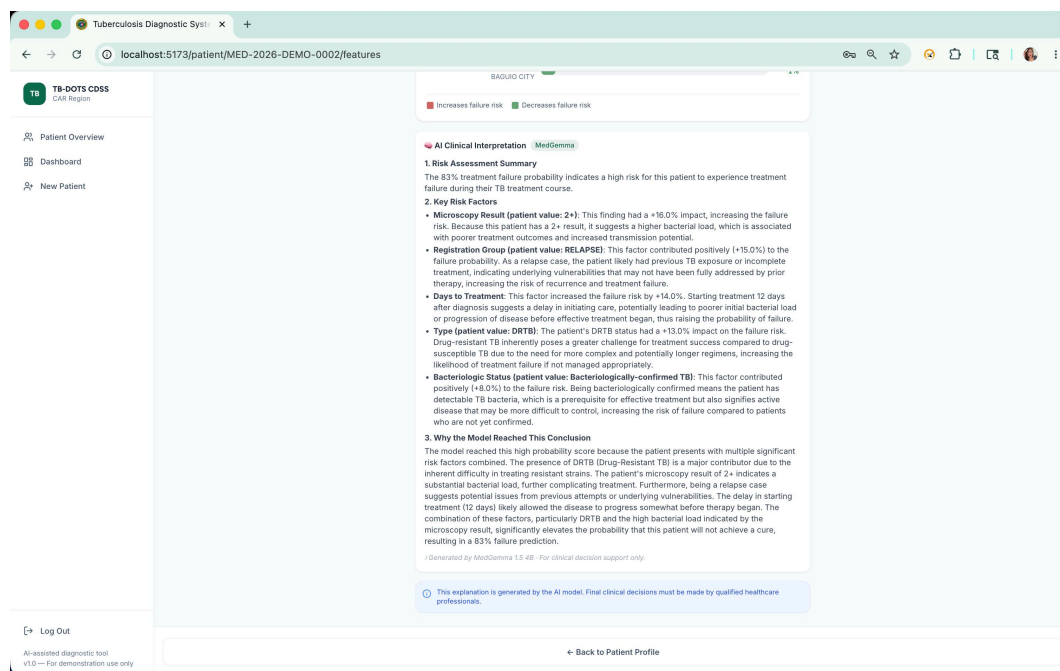
Figure 24*Initial Clinical Intervention and Management Protocol (First Step)*

The screenshot displays the 'TB-DOTS CDSS' web application. The browser address bar shows 'localhost:5173/patient/new'. The interface features a top navigation bar with a progress indicator showing five steps: 1. Demographics, 2. Lab Results, 3. Chest X-rays, 4. Diagnosis, and 5. Treatment. A left sidebar contains navigation links: 'Patient Overview', 'Dashboard', and 'New Patient' (which is highlighted). The main content area is titled 'Patient Demographics' with a subtitle 'Basic patient information — all starred fields are required'. The form includes the following fields: 'Full Name' (text input with 'Juan dela Cruz'), 'Age' (text input with '35'), 'Gender' (dropdown menu with 'Select gender'), 'Medical ID' (text input with 'MED-2026-7464'), 'Registration Group' (dropdown menu with 'Select group'), 'Province' (dropdown menu with 'Select province'), 'City / Municipality' (dropdown menu with 'Select province first'), 'Civil Status' (dropdown menu with 'Select Civil Status'), and 'Nationality' (text input with 'e.g. Filipino'). At the bottom left, there is a 'Log Out' button and a footer note: 'An assisted diagnostic tool v1.0 — For demonstration use only'. At the bottom right, there is a green button labeled 'Continue to Lab Results'.

Large Language Model (LLM) Integration for Clinical Context

Beyond the quantitative feature importance provided by SHAP, the system utilizes *Med-Gemma*—a fine-tuned medical large language model—to provide a qualitative narrative of the identified risk factors. As shown in Figure 25, this component resides within the Risk Factor Analysis interface and functions as a clinical bridge, translating statistical weights into natural language explanations.

By analyzing the specific patient profile (e.g., age, co-morbidities, and sputum conversion), Med-Gemma generates context-aware summaries that translate the statistical outputs of the SHAP-based factor analysis into clinically meaningful insights. Rather than presenting abstract feature weights in isolation, this integration uses Med-Gemma to explain why the specific variables identified by SHAP are concerning in a clinical context. This ensures that the CDSS does not merely present a risk percentage but explicitly deciphers the underlying logic of the predictive engine, significantly enhancing the transparency and interpretability of the results for healthcare providers.

Figure 25*Med-Gemma Clinical Narrative and Risk Factor Explanation***Discussion**

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of PTB patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Temporal modeling was implemented in two iterations. The v1 iteration served as an initial baseline, but its results demonstrate a common pitfall in imbalanced clinical prediction: high accuracy can be achieved by favoring the majority Success class while failing to detect true treatment failures. This is reflected by v1 specificity of 0.0000 on the held-out test split, indicating that Failure identification was not reliable under the initial workflow.

The corrected v2 iteration improved the methodological defensibility of temporal prediction by addressing leakage risk, outcome-label handling, and temporal feature preparation. On the v2 held-out test split, the SMOTE-ENN tree models achieved consistent performance and identified 5 of 6 Failure cases while correctly classifying 36 of 38 Success cases. Random Forest exhibited the strongest held-out ranking performance among the tree models (ROC-AUC 0.9693, PR-AUC 0.9950), while LightGBM reported the

strongest cross-validated balanced accuracy in the saved artifacts. The v2 Hybrid Bi-LSTM baseline detected all Failure cases on the held-out test split (specificity 1.0000) but produced more false Failure alerts among true Success cases, lowering overall accuracy. This tradeoff can be acceptable in a screening-oriented CDSS where missing high-risk patients is more costly than additional review workload, but it must be managed to avoid over-alerting.

Adding temporal monitoring data to static baseline features improved predictive performance: patient-level longitudinal records carry signal that a single-visit snapshot misses. SMOTE-ENN and the Bi-LSTM data augmentation addressed the extreme class imbalance, which is what gave the models a realistic chance of catching the minority failure class. Prediction also needs to be paired with explanation in a CDSS — risk estimates only become useful when a provider can see what drove them. SHAP and the bounded SLM summary connect model output to clinician review without shifting the decision to the machine.

These findings suggest that the v2 temporal modeling outputs may serve as a basis for further evaluation within TB monitoring workflows, particularly for earlier identification of patients who may require closer review. For deployment-oriented performance, the v2 tree models (particularly Random Forest and LightGBM with SMOTE-ENN) provide a strong balance of Failure detection and overall correctness, while the Hybrid Bi-LSTM baseline may be useful when a more conservative screening stance is desired. However, any use in clinical or programmatic settings should first undergo external validation, prospective testing, and review by healthcare professionals to assess safety, calibration, and workflow fit. Future research should focus on larger, multi-region datasets to enhance generalization, evaluate temporal calibration over time, and explore hybrid modeling approaches that combine deep temporal representations with tree-based learners. Additionally, enhancing interpretability of temporal dynamics (e.g., attention over treatment months) may improve clinician trust and adoption.

These CDSS artifacts move the program from retrospective reporting toward proactive management. The traditional NTP protocol identifies treatment failure after it happens; the predictive outputs allow intervention during the critical early months, before the outcome is fixed. The Full Breakdown lets a health worker check the model's reasoning against their own clinical judgment instead of trusting a score they can't read. The First Step document links that prediction to action, giving staff a concrete next step when a patient is flagged. SHAP and Med-Gemma address the skepticism health workers often have about AI tools: by giving both a statistical breakdown and a plain-language narrative, the CDSS keeps the clinician in the loop.

Health workers can cross-check the model’s findings against their experience, which is also what makes adoption more likely in practice.

Recommendations

Healthcare facilities and policymakers should consider adding tree-based models — particularly Random Forest and LightGBM with SMOTE-ENN resampling — to the existing DOH-CAR TB-DOTS monitoring program as a decision-support layer for early risk stratification. On the held-out test set, these models caught treatment failures while keeping overall accuracy high, which gives them a practical basis for flagging patients who need a closer look. Before any deployment, though, they need external validation on independent multi-facility data, prospective testing under real workflow conditions, and review by healthcare professionals to check calibration, safety, and fit with current practice.

The explanation layer needs formal evaluation before any clinical deployment. In a preliminary internal test on the 44-patient cohort, the narrative layer failed to match the model’s SHAP attribution whenever it ran without the attribution signal: not one case passed the faithfulness check under this blind condition (0 of 44), regardless of the SLM backend. Faithfulness improved substantially once the SHAP contributions were included in the prompt, confirming that the narrative layer works as a constrained verbalizer of the model’s reasoning rather than an independent reasoner. Future work should extend this faithfulness evaluation to a larger, multi-facility cohort, try multiple SLM backends, and add clinician-rated readability, trust, and actionability as paired measures — so there is enough evidence to deploy the explanation component safely.

Safety Scope and Interpretation Limits

The explanation layer is intended to support readability and interpretation of model reasoning by translating contribution signals into clinician-readable text, but its outputs must be interpreted within clear safety boundaries. SHAP explanations describe how features increased or decreased the model-predicted risk; they do not establish causal mechanisms of disease progression or prove that changing a highlighted factor will alter the patient’s outcome. This distinction is important in high-stakes settings where generative

outputs may otherwise be over-interpreted beyond their evidentiary basis (Vasey et al., 2022; World Health Organization, 2021).

Accordingly, the system is positioned as decision support: it flags risk, surfaces contributing factors, and summarizes model behavior while preserving clinician authority for diagnosis and treatment decisions. This framing is consistent with contemporary reporting and governance guidance for AI-assisted clinical prediction systems, where transparency, bounded use, and human oversight remain central requirements (Collins et al., 2024).

The preliminary faithfulness evaluation described above supports this bounded framing by showing that the narrative layer depends on SHAP contribution signals rather than reconstructing model attribution from patient context alone. Accordingly, the operational design retains SHAP as the primary explanation surface and treats the narrative output as a readability aid rather than a substitute for the attribution itself.

Conclusion

Tuberculosis treatment in the Cordillera Administrative Region (CAR) is shaped by very different patient situations, gaps in record-keeping, and disruptions to the health program. These are problems that simple one-variable analyses cannot fully explain. To turn this raw data into something useful for prediction, the study first built a clear and repeatable cleaning pipeline for two related datasets: a 7,389-patient DOH-CAR registry and a 599-patient month-by-month monitoring set. The pipeline standardized formats, removed patient identifiers, and handled missing values by first checking why each variable was missing (MCAR, MAR, or MNAR) before applying one of four matching strategies. Because the cleaned data was trustworthy rather than blindly filled in, the descriptive and exploratory analyses could clearly show outcome trends over the years, the rise in deaths and lost follow-ups during the COVID-19 period, and the patient-level, time-based, and operational patterns that later guided model design.

These cleaned signals fed directly into how well the models performed. On the static dataset, XGBoost with SMOTE-ENN gave the strongest screening-focused setup (composite score $S = 0.699$; failure recall 72.9%), while XGBoost without resampling produced the best-calibrated risk ranking (test AUC 0.712; MCC 0.258), offering two different setups depending on how the tool will be used. The longitudinal models pushed performance further: after the v2 pipeline fixed the data leakage and label-handling issues

that had limited v1, the SMOTE-ENN tree models (Random Forest, LightGBM, and XGBoost) reached 0.9318 accuracy and 0.8333 specificity, correctly catching five of six failure cases in the held-out test set, while the Hybrid Bi-LSTM caught every failure case but produced more false alarms. Building on these predictions, SHAP-based feature explanations and a bounded SLM narrative layer were combined into a Clinical Decision Support System (CDSS) that shows each patient's risk level, the factors driving that risk, and updated estimates as new monthly data come in.

These results are clinically promising but not yet ready for actual use. The test cases came from a single regional registry and the narrative layer's faithfulness was checked on only 44 patients, so the system still needs the external validation, prospective testing, and clinician review described in the Recommendations before it can be trusted in practice. The explanation layer in particular needs a larger faithfulness study first. Once those conditions are met, the framework and the published pseudonymized version of the DOH-CAR PTB Treatment Outcome Dataset can support both regional TB control and future research on explainable clinical prediction.

Acknowledgments

The authors would like to express their sincere gratitude to Saint Louis University for providing the academic environment and resources that made this research possible. We extend our appreciation to the School of Accountancy, Management, Computing, and Information Studies (SAMCIS), specifically the Department of Computing and Information Studies, for their continuous support and guidance throughout the study. Our deepest thanks go to Dr. Randy Domantay, whose leadership and encouragement have been invaluable. We are especially grateful to our research adviser, Dr. Josephine S. Dela Cruz, for their expert guidance, insightful feedback, and unwavering support at every stage of the research. Lastly, we acknowledge the collaboration and contributions of our partner researchers from the SONAHBS Department of Pharmacy, whose expertise and assistance significantly enriched this study.

Appendix A: CDSS Web Application Walkthrough

This appendix presents a sequential walkthrough of the deployed CDSS web application, following the order in which a clinical user would traverse the system: authentication, patient intake, profile review, longitudinal monitoring, and risk explanation. Several interfaces shown here also appear in the body of the paper; they are reproduced in this appendix for continuity of the walkthrough.

A.1 Authentication and Dashboard

Figure 26

CDSS login screen.

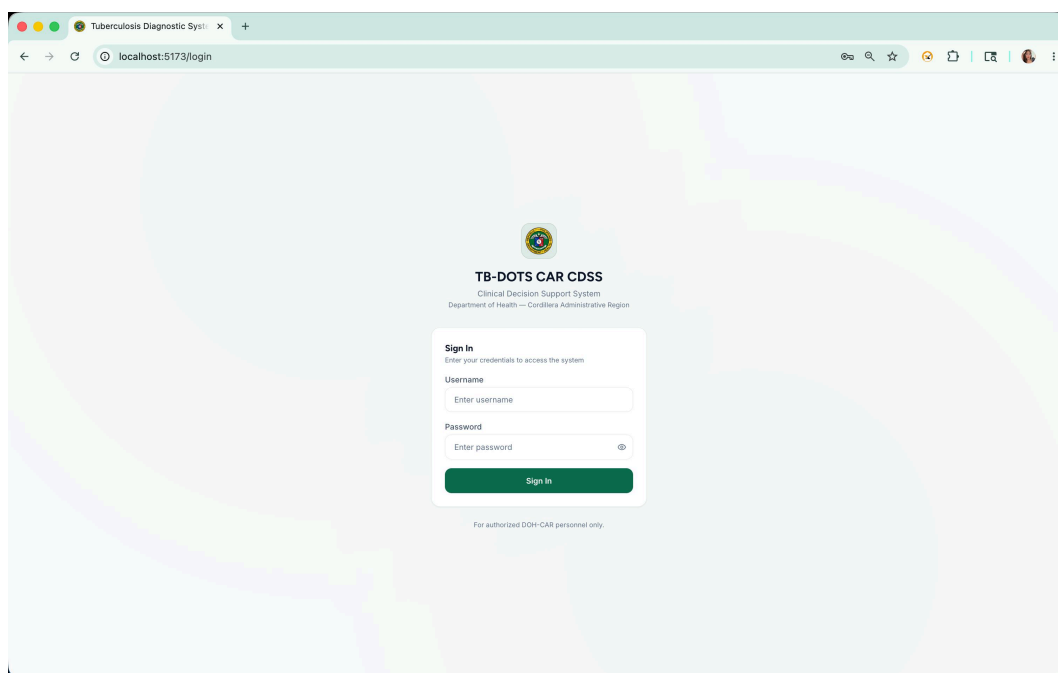
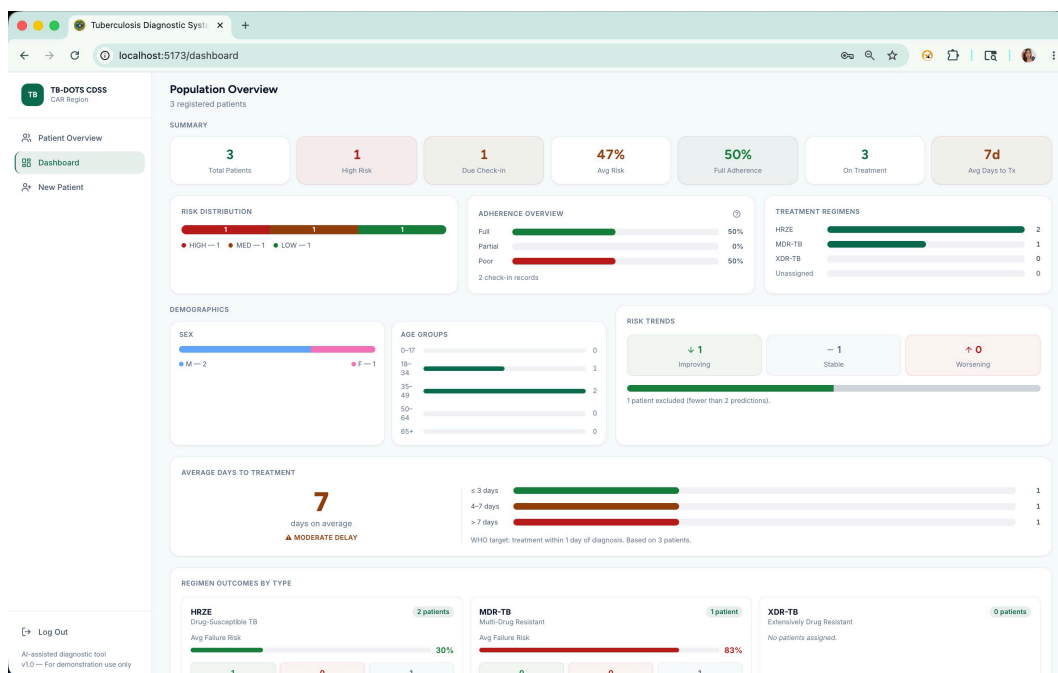


Figure 27

Main dashboard listing enrolled patients.



A.2 Patient Intake Workflow

The intake workflow consists of five sequential steps. The Lab Results step (Step 2) is omitted here as it was not captured in the available screenshot set.

Figure 28*Patient intake Step 1: Demographics.*

TB-DOTS CDSS
CAR Region

← → localhost:5173/patient/new

1 Demographics 2 Lab Results 3 Chest X-rays 4 Diagnosis 5 Treatment

🏠 Patient Overview
📊 Dashboard
👤 New Patient

Patient Demographics
Basic patient information — all starred fields are required

Full Name * Age *

Gender * Medical ID

Registration Group *

Province * City / Municipality

Civil Status Nationality

[+] Log Out
AI-assisted diagnostic tool
v1.0 — For demonstration use only

Continue to Lab Results →

Figure 29*Patient intake Step 3: Chest X-ray upload.*

TB-DOTS CDSS
CAR Region

← → localhost:5173/patient/new/xray

1 Demographics 2 Lab Results 3 Chest X-rays 4 Diagnosis 5 Treatment

🏠 Patient Overview
📊 Dashboard
👤 New Patient

Chest X-rays
Upload baseline chest X-ray images — this step is optional

Drag & drop X-rays or click to browse
JPEG, PNG, WebP - up to 10 images

Images are uploaded to the local backend on this workstation.

[+] Log Out
AI-assisted diagnostic tool
v1.0 — For demonstration use only

← Back Skip Upload & Continue →

Figure 30
Patient intake Step 4: Diagnostic result.

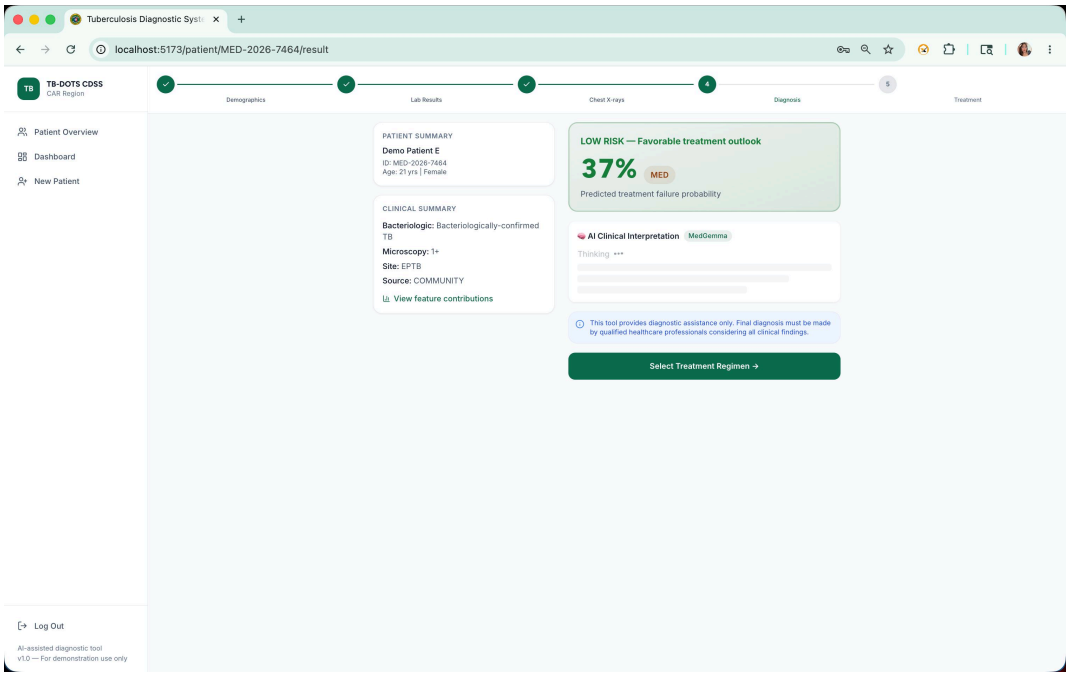
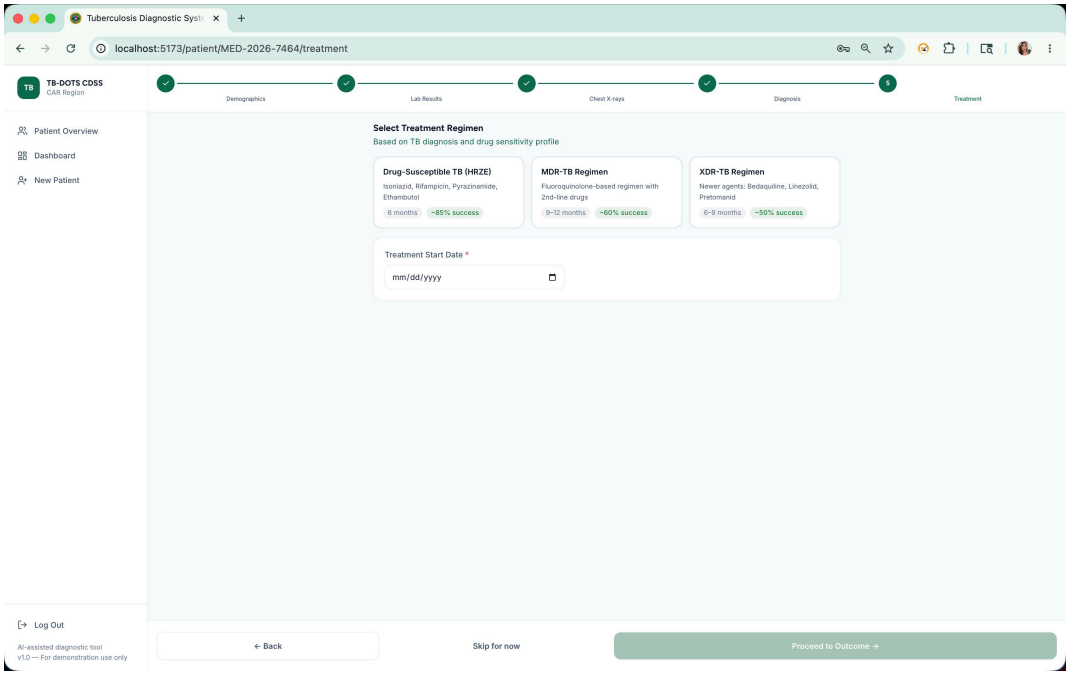


Figure 31
Patient intake Step 5: Treatment regimen selection.



A.3 Patient Profile and Records

Figure 32

Patient profile overview tab.

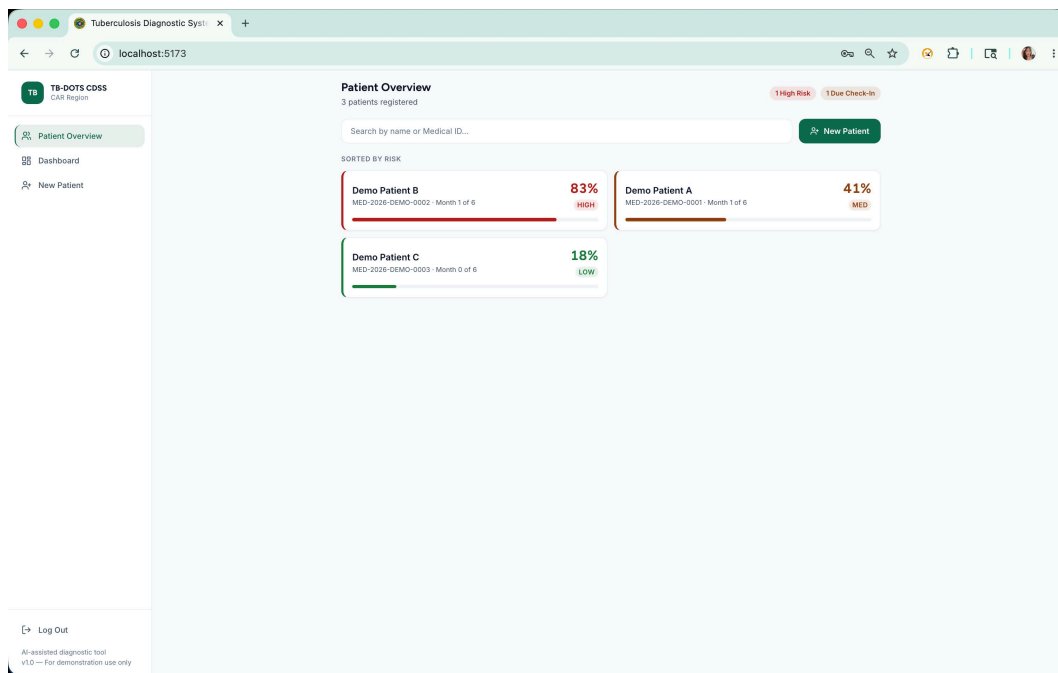
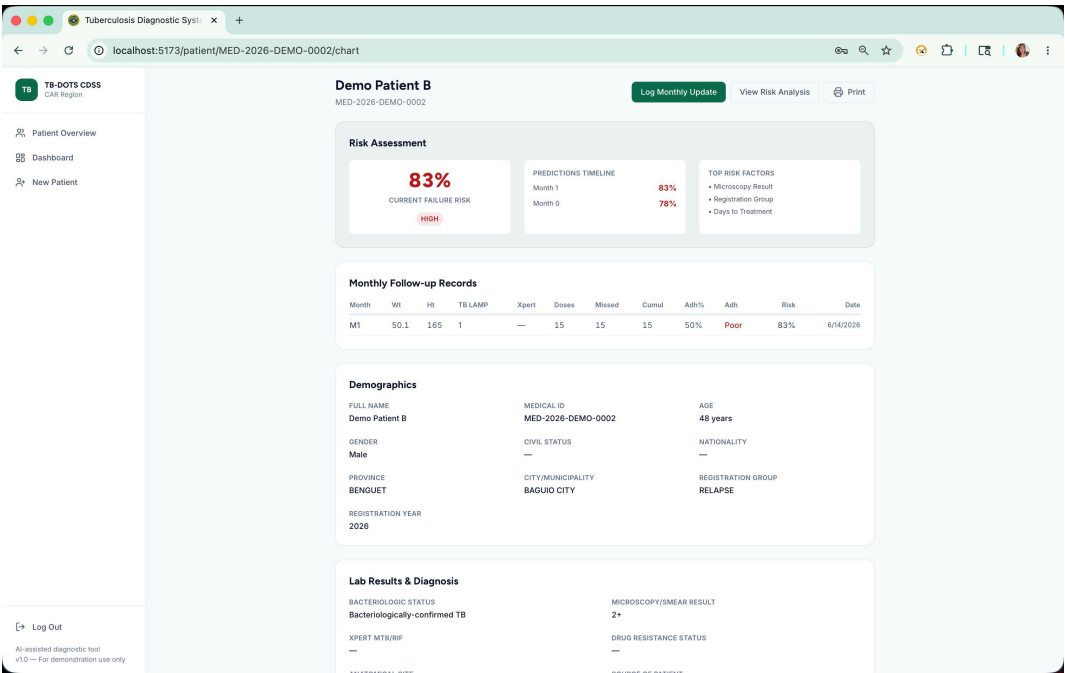


Figure 33
Complete patient chart with longitudinal record.



A.4 Longitudinal Monitoring

Figure 34

Monthly check-in log for adherence and follow-up imaging.

TB-DOTS CDSS
CAR Region

← → ↻ localhost:5173/patient/MED-2026-DEMO-0002/checkin

Month M1 of 12

Monthly Follow-up

Recording Month M1 data for Demo Patient B

Measurements

Weight (kg) Height (cm)

Dose Tracking

Doses Taken Missed Doses

Cumulative Doses Taken Adherence (%)

Lab Results

Smear TB LAMP Xpert MTB/RIF

Chest X-rays (optional)

JPEG, PNG, WebP - up to 10 images

Follow-up Guide: Record weight, height, dose tracking, and lab results at each monthly visit.

Log Out

AI-assisted diagnostic tool
v1.0 - For demonstration use only

← Back

Generate Prediction →

A.5 Risk Prediction and Explainability

Figure 35

Updated risk analysis after monthly follow-up.

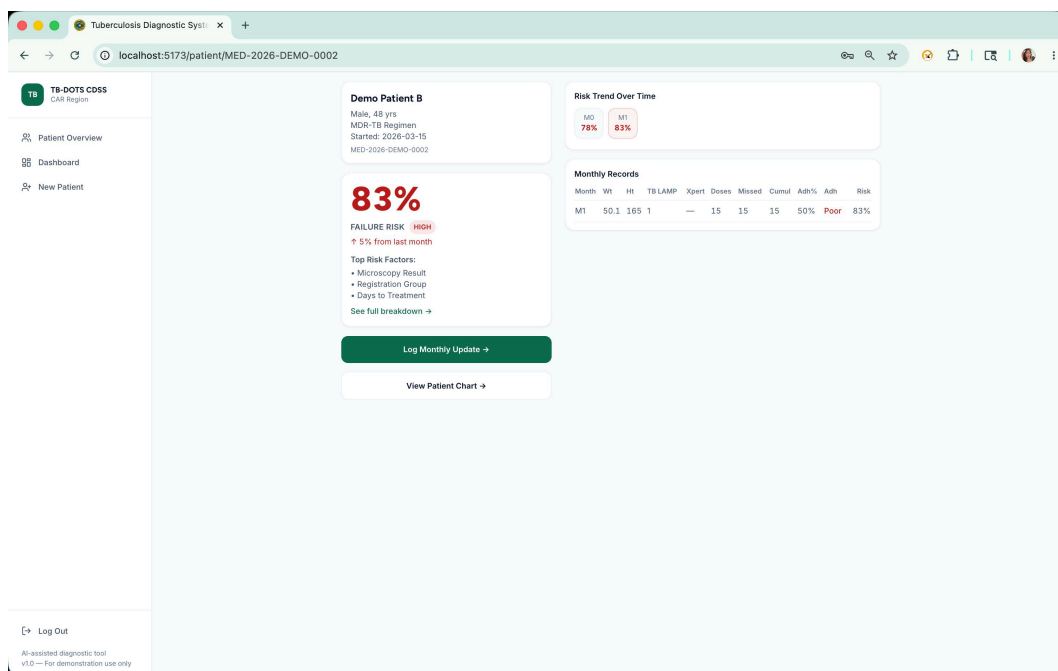


Figure 36

SHAP-based full breakdown of contributing factors.

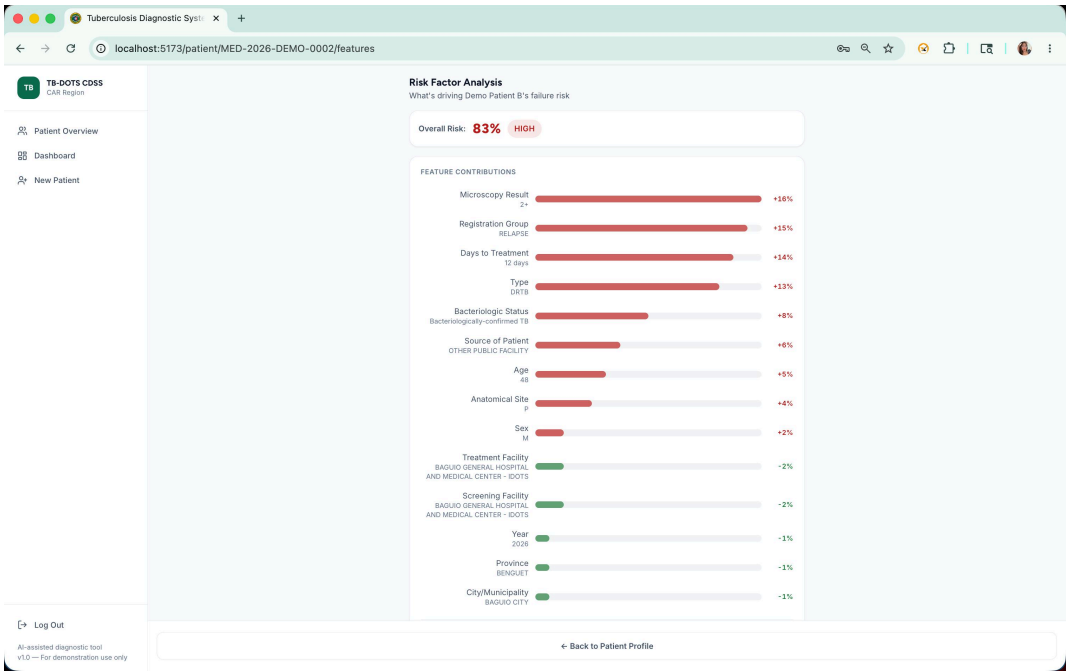
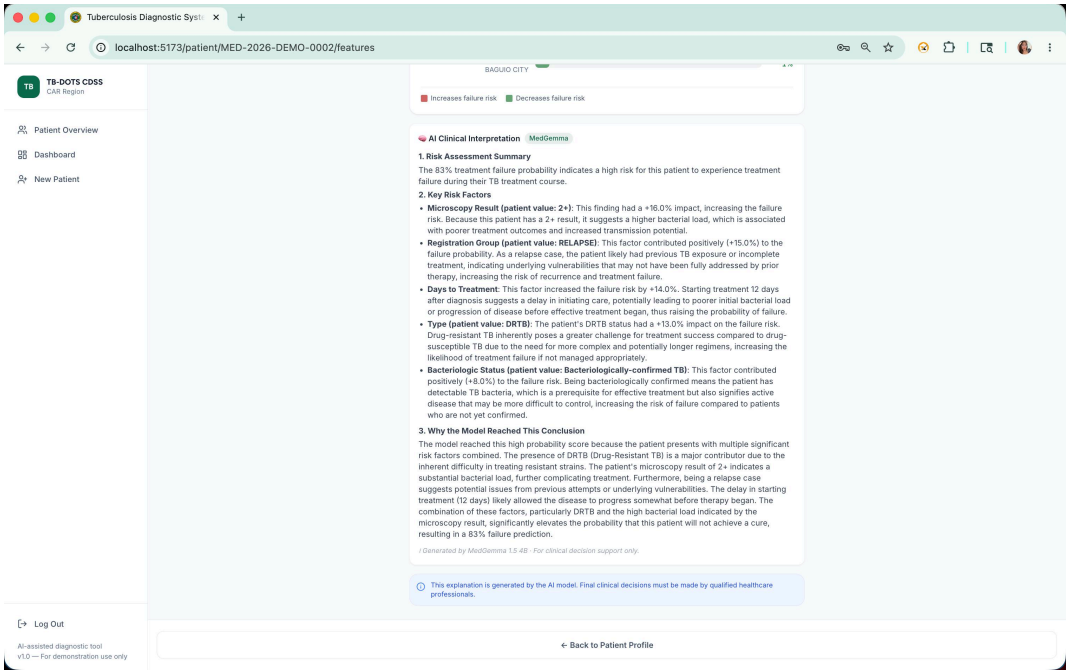


Figure 37

Med-Gemma clinical narrative explaining risk factors.



Appendix B: iCreate Supplementary Material

The following pages reproduce the original iCreate 2026 supplementary material in its source format for archival completeness.

Treatment Success and Failure Prediction using Machine Learning Models for Pulmonary Tuberculosis Patients under the DOH-CAR TB-DOTS Program

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Abstract

This study proposes a multi-layer analytical framework using machine learning to predict pulmonary tuberculosis (PTB) treatment success or failure in the Cordillera Administrative Region (CAR), Philippines. Utilizing demographic, clinical, and diagnostic data from 2015 to 2025, supervised models (Logistic Regression, Support Vector Machines, and Extreme Gradient Boosting) and temporal models (Recurrent Neural Networks and Long Short-Term Memory) are trained to identify risk factors. Shapley Additive Explanations (SHAP) ensure interpretability. The models will be integrated into a clinical decision support tool for early intervention. Aligning with the Sustainable Development Goals for good health and well-being, this predictive tool aims to enhance TB control efforts, optimize resource allocation, and improve patient management.

Keywords

Clinical decision support, machine learning, predictive modeling, pulmonary tuberculosis, treatment outcomes

1. Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease that continues to challenge public health due to its gradual progression and risk of treatment failure (Tobin & Tristram, 2024). In the Philippines, TB remains a major public health concern, with the country being one of the top contributors to the global TB burden (World Health Organization, 2022). In the Cordillera Administrative Region (CAR), the Department of Health (DOH) reported 437 confirmed TB cases in Baguio City alone in 2022, alongside an estimated 4,405 missing or unreported cases (Department of Health Republic of the Philippines, n.d.).

While multiple studies have examined risk factors such as sociodemographic characteristics, comorbidities, and medical history, findings remain fragmented as traditional statistical analyses often assume linear relationships and examine only a few factors simultaneously (Amante &

Abdosh, 2015). To address this gap, this study adopts an integrative machine learning framework to combine multiple dimensions of data in analyzing interrelated variables.

The specific research objectives of this study are as follows:

1. **Data Pre-processing & Feature Engineering:** To clean, pre-process, and digitalize PTB patient records, creating engineered features that ensure the dataset is optimized for statistical analysis and machine learning applications.
2. **Statistical Analysis & Profiling:** To perform descriptive and inferential statistics to profile the demographic, clinical, and treatment characteristics of the study population, identifying baseline variables that significantly influence treatment outcomes.
3. **AI Modeling & Comparative Analysis:** To develop, validate, and evaluate both non-temporal and temporal machine learning models for predicting PTB treatment outcomes, comparing the predictive performance of static baseline data against dynamic, time-dependent patient patterns throughout the course of therapy.
4. **Data Visualization & Clinical Interpretation:** To visualize patient risk levels, statistical trends, and predictive outputs from the AI models, providing clear insights to support clinical interpretation and the early identification of individuals at risk of poor outcomes.

2. Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering the period from 2015 to 2025, consolidated by the DOH-CAR. Data undergoes a standardized preprocessing pipeline, including schema harmonization, error correction, and missing value imputation via Multivariate Imputation by Chained Equations, or MICE (Alruhaymi & Kim, 2021).

To predict treatment outcomes defined by the National Tuberculosis Program criteria, supervised models including Logistic Regression, Support Vector Machines (SVM), and Extreme Gradient Boosting (XGBoost) are utilized. Furthermore, temporal models using Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) are employed to capture longitudinal treatment dynamics (Lu et al., 2022; Nayebi et al., 2022). Model performance is assessed using standard classification metrics, and Shapley Additive Explanations (SHAP) are applied to ensure clinical interpretability (Peng et al., 2024).

3. Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static (Random Forest, Gradient Boosting, LightGBM, XGBoost, Logistic Regression) and temporal models are compared, and feature importance analyses provide insights into the key factors influencing treatment outcomes. The discussion interprets these findings in

the context of clinical relevance and public health priorities, highlighting implications for early intervention, resource allocation, and strengthened TB management.

3.1 Dataset Characteristics and Class Distribution

The final dataset exhibited significant class imbalance, with 86.2% of cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, including four newly introduced geographic and facility-level variables and one derived clinical delay variable (Diagnosis_to_Treatment_days). Resampling using SMOTE combined with Edited Nearest Neighbors (SMOTE+ENN) was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio. This approach mitigated class bias while preserving realistic evaluation conditions in the test and validation sets.

3.2 Validation Set Performance and Generalization

Table 1: Test Set Model Performance

Model	AUC (Area Under the Curve)	Recall (Failure)	F1 (Failure)
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Among all evaluated models, Random Forest achieved the highest overall performance on the test set, with an AUC of 0.7154 and an F1-score of 0.3539 for the failure class. It also demonstrated balanced recall (57.98%) without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (~0.71), though with slightly lower recall for treatment failure. Logistic Regression demonstrated the highest recall (77.13%); however, its precision was notably low (19.6%), resulting in over-alerting and reduced overall accuracy (~51%). These findings suggest that while Logistic Regression aggressively identifies high-risk patients, it lacks specificity, potentially burdening healthcare systems with excessive false alarms.

On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure. Notably, all models demonstrated positive AUC gains from test to validation sets, indicating strong generalization and absence of overfitting. The consistency of Random Forest across both evaluation phases confirms its robustness and suitability for deployment within a clinical decision support framework.

3.3 Validation Set Performance and Generalization

Table 2: Test Set Performance Metrics of Temporal Models for TB Treatment Outcome Prediction

Metric	Bi-LSTM Baseline	Bi-LSTM Augmented	XGBoost Baseline	XGBoost SMOTE-ENN	LightGBM SMOTE-ENN	Random Forest SMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
Specificity	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419
Threshold	0.15	0.86	0.10	0.34	0.43	0.70

Table 1 presents the performance metrics of all temporal models evaluated on the held-out test set. Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. Precision values were similarly high (0.9048), indicating few false positives for the majority class. However, recall for the minority Failure class reached 1.0 for these models, while specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Tree-based models augmented with SMOTE-ENN, particularly Random Forest, showed a more balanced performance. Random Forest achieved slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. ROC-AUC scores further illustrate differences in probability calibration, with XGBoost Baseline achieving the highest ROC-AUC (0.5789), while the Bi-LSTM Augmented variant performed lower (0.3684), likely due to overfitting on limited Failure samples. Overall, these

results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

4. Conclusions and Recommendations

This study investigated the use of machine learning models to predict treatment outcomes—Success or Failure—of pulmonary tuberculosis patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. Both temporal models, specifically the Bi-LSTM Baseline and Augmented variants, and tree-based models, including XGBoost, LightGBM, and Random Forest with SMOTE-ENN, were evaluated on a stratified 70/20/10 patient-level split, incorporating progressive temporal samples from treatment months M0 through M12. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited, as indicated by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes. These results highlight that tree-based models augmented with SMOTE-ENN are particularly effective at handling severe class imbalance, while temporal models provide strong predictive accuracy for the majority outcome.

The study contributes in several ways. By integrating temporal monitoring data with static baseline features, it demonstrates that patient-level longitudinal information improves predictive performance over static data alone. The application of SMOTE-ENN and advanced data augmentation techniques for Bi-LSTM effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Furthermore, the findings suggest that predictive models, particularly Random Forest with SMOTE-ENN, have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly.

Based on these findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance generalization, explore hybrid modeling approaches combining deep learning and tree-based methods, and investigate additional features such as social determinants of health and geographic risk factors. Additionally, enhancing model interpretability, particularly of Bi-LSTM attention outputs, could improve clinician trust and adoption.

Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes. By accurately identifying patients at risk of treatment failure, these predictive models have the potential to reduce TB-related morbidity and mortality, inform targeted public health interventions, and contribute to more efficient and effective TB care within the Cordillera Administrative Region.

Acknowledgments

The authors would like to express their sincere gratitude to Saint Louis University for providing the academic environment and resources that made this research possible. We extend our appreciation to the School of Accountancy, Management, Computing, and Information Studies (SAMCIS), specifically the Department of Computing and Information Studies, for their continuous support and guidance throughout the study. Our deepest thanks go to Dr. Randy Domantay, whose leadership and encouragement have been invaluable. We are especially grateful to our research adviser, Dr. Josephine S. Dela Cruz, for their expert guidance, insightful feedback, and unwavering support at every stage of the research. Lastly, we acknowledge the collaboration and contributions of our partner researchers from the SONAHBS Department of Pharmacy, whose expertise and assistance significantly enriched this study.

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Appendix C: Publication Paper for MSJI

The following pages reproduce the original iCreate 2026 supplementary material in its source format for archival completeness.

Treatment Success and Failure Prediction Using Machine Learning Models for Pulmonary Tuberculosis Patients Under the DOH-CAR TB-DOTS Program

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Abstract

Pulmonary Tuberculosis (PTB) remains a critical public health challenge in the Cordillera Administrative Region (CAR), Philippines, where complex patient profiles and programmatic gaps contribute to persistent treatment failure. This study develops machine learning models to predict treatment success or failure among PTB patients in DOH-CAR TB-DOTS facilities covering 2015–2025. A standardized preprocessing pipeline performs schema harmonization, pseudonymization, and diagnostic-first missing data handling classifying each variable's missingness mechanism. Supervised ensemble tree-based classifiers—XGBoost, LightGBM, and Random Forest—were trained alongside Recurrent Neural Network and Long Short-Term Memory temporal architectures to capture longitudinal treatment dynamics. Shapley Additive Explanations (SHAP) were applied to quantify the contribution of each variable to model predictions. The validated models are embedded in a Clinical Decision Support System (CDSS) providing dynamic risk estimates and narrative explanations to support early clinical intervention while preserving clinician judgment.

Keywords: tuberculosis, machine learning, treatment outcome prediction, clinical decision support system, explainable AI

Introduction

Pulmonary Tuberculosis (PTB) is a chronic infectious disease caused by *Mycobacterium tuberculosis*, a slow-growing and acid-fast bacillus that continues to challenge public health. Despite effective therapies, PTB remains widespread due to its strong association with poverty, undernutrition, and comorbidities such as diabetes and HIV, which weaken immune defenses and increase susceptibility. These factors delay diagnosis, complicate treatment, and contribute to disease progression. Globally, TB remains one of the leading causes of death, with 1.5 million deaths and 10 million new cases recorded in 2018, including half a million with rifampicin resistance.

In the Philippines, TB continues to be a major public health concern. The World Health Organization classifies the country as one of the top contributors to the global TB burden, accounting for approximately 6.8 percent of cases. In 2023, around 37,000 Filipinos died from TB out of 739,000 diagnosed cases. In Baguio City, the DOH-CAR reported 437 confirmed TB cases in 2022 along with over 4,405 missing or unreported cases, making the area a priority for surveillance and treatment improvement.

Contributing factors include high population density, internal migration, the city's cool and humid highland climate conducive to transmission, and its role as a tertiary referral center. Patient comorbidities such as diabetes mellitus, HIV infection, and malnutrition impair immune

response and reduce treatment completion rates, while poverty-related barriers disrupt adherence to directly observed treatment short-course (DOTS).

Risk Factors for Treatment Failure

TB patients who acquired infection through close contact were shown to be at higher risk of delayed diagnosis and non-adherence. Patients who missed follow-ups had higher risk of reactivation of infection and mortality. HIV co-infection was determined to be a substantial risk factor for treatment failure. A study in the Philippines revealed that 58.82% of 51 patients who experienced treatment failure had at least one comorbidity. Diabetes Mellitus (DM) further increases the risk of treatment failure, relapse, and death compared to non-diabetic patients.

Machine Learning for TB Outcome Prediction

Machine learning offers distinct advantages in modeling complex and non-linear interactions among demographic, clinical, diagnostic, and treatment-related variables. By leveraging algorithms such as logistic regression, Support Vector Machines, and ensemble-based methods including Extreme Gradient Boosting (XGBoost), machine learning can uncover subtle dependencies that contribute to treatment success or failure. Shapley Additive Explanations (SHAP) quantify the contribution of each variable to a model prediction, summarizing both patient-level explanations and overall feature importance.

The limitations of fragmented risk factor studies and capabilities of interpretable machine learning highlight the need for an integrative approach combining multiple dimensions of data. This study adopts a holistic analytical framework integrating descriptive analysis, supervised predictive modeling, and explanation methods to help healthcare providers identify patients at higher risk of treatment failure, understand factors influencing model's risk estimate, and use that

information to support early intervention, tailored monitoring, and improved patient management.

Methodology

This study utilizes a retrospective dataset of PTB patients from the CAR covering 2015 to 2025, consolidated by DOH-CAR. After ethical approval, records are pseudonymized and processed using a standardized preprocessing pipeline, including schema harmonization, error correction, and feature scaling, followed by a diagnostic-first missing data handling protocol.

Data Acquisition and Preprocessing

The study draws on two complementary datasets from the CAR: (1) a non-temporal consolidated registry maintained by DOH-CAR comprising 7,389 patient records with cross-sectional baseline features including demographic variables, clinical indicators, and diagnostic results, used to train static tree-based classifiers; and (2) a temporal monitoring dataset of 599 patients with complete month-by-month treatment monitoring records, manually compiled by the study's Pharmacy Co-authors from facility treatment monitoring forms and used to train longitudinal deep learning and temporal tree-based models.

The non-temporal preprocessing pipeline implements schema harmonization to standardize variable formats and units and pseudonymization to enforce patient privacy. Outcome annotation defines the binary target variable: patients classified as 'Cured' or 'Treatment Completed' are annotated as 'Success,' whereas those marked as 'Failed,' 'Died,' or 'Lost to Follow-up' are annotated as 'Failure.' To prevent data leakage, the dataset is partitioned into training and testing sets prior to any data transformation steps.

Missing Data Handling

Clinical TB registries exhibit heterogeneous missingness arising from inconsistent documentation and varying facility capacities. This study implements a diagnostic-first missing data pipeline that classifies each variable's missingness mechanism—Missing Completely at Random (MCAR), Missing at Random (MAR), or Missing Not at Random (MNAR)—and routes the variable to one of four handling strategies: column exclusion, listwise deletion, missing indicator with fill, or stochastic Multivariate Imputation by Chained Equations (MICE). Variables exceeding 80% missingness are excluded entirely. Four clinical vital signs confirmed MAR are imputed using stochastic MICE with an ExtraTreesRegressor as the conditional model.

Supervised Learning Model Development

The study leverages ensemble tree-based algorithms—XGBoost, LightGBM, and Random Forest. Two sampling strategies are compared for each classifier: no resampling with native class-weighting, and SMOTE combined with Edited Nearest Neighbors (SMOTE-ENN), which increases the training-set failure class by 433%. Three progressive feature set versions are evaluated, ranging from 11 core demographic and diagnostic variables to an extended set of 21 features incorporating facility-level geographic and temporal delay variables.

For temporal modeling, a Hybrid Bidirectional LSTM architecture was adopted to model long-term dependencies, with tree-based temporal baselines including XGBoost and LightGBM, all trained on monthly patient monitoring data from Month 0 through Month 7. The dataset is partitioned using stratified 70/20/10 patient-level split into training, test, and validation sets. Model selection employs a composite scoring metric: $\text{score} = 0.6 \times \text{AUC} + 0.4 \times \text{Recall}(\text{Failure})$, prioritizing discrimination ability while rewarding sensitivity to the minority failure class.

Explainability and Clinical Decision Support

Shapley Additive Explanations (SHAP) are applied to quantify the marginal contribution of each variable to individual model predictions, surfacing the patient-level factors that drive each risk estimate for clinician review. The validated models are embedded within a Clinical Decision Support System (CDSS) that visualizes patient risk levels, highlights contributing factors through SHAP-based explanations, and dynamically updates risk estimates as new follow-up data become available.

Results and Discussion

This section presents the results of machine learning analyses aimed at predicting treatment success and failure among PTB patients under the DOH-CAR TB-DOTS program. Descriptive statistics summarize patient demographics, clinical characteristics, and treatment outcomes, while predictive model performance is evaluated using test and validation datasets. Both static tree-based models under two sampling conditions and temporal models are compared, and feature importance analyses provide interpretable evidence about factors influencing model predictions.

Dataset Characteristics and Class Distribution

The final validated dataset comprised 7,389 complete patient records encompassing demographic, geographic, and clinical features. Overall, the dataset exhibited significant class imbalance, with 86.2% of evaluated cases classified as treatment success and 13.8% as treatment failure. To ensure robust evaluation, the dataset was stratified into 70% training, 20% testing, and 10% validation subsets. Feature engineering expanded the dataset to 15 predictors, integrating spatial identifiers with temporal markers and derived clinical delays. Resampling using SMOTE

combined with Edited Nearest Neighbors was applied exclusively to the training set, increasing the minority (failure) class by 433% and achieving a near-balanced 0.70:1 ratio.

Outcome Distribution and Longitudinal Trends

The distribution of treatment outcomes highlighted both programmatic successes and persistent challenges. The vast majority of the cohort achieved positive outcomes, with 65.1% classified as 'Treatment Completed' and 14.2% as 'Cured.' Conversely, adverse outcomes accounted for a critical minority: patients 'Lost to Follow-up' comprised 8.4%, patients who 'Died' accounted for 3.3%, and treatment failure was 0.9%. Analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts: success rates remained in the 80%–89% range, with 88.92% achieved in 2023. Mortality peaked at 6.63% in 2021, aligning with COVID-19 pandemic disruptions, before declining to 3.66% by 2024. Loss to follow-up rates declined from 11.99% (2020) to 5.56% (2024), suggesting improved patient tracking and adherence interventions.

Figure 1
Distribution of Treatment Outcomes Among PTB Patients

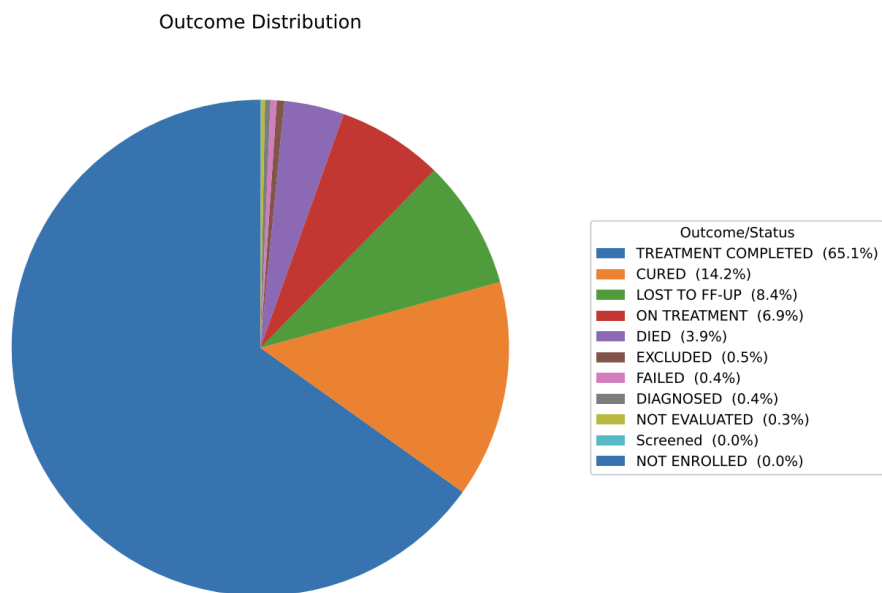


Figure 1 displays the distribution of programmatic treatment outcomes for the entire cohort. The dominance of 'Treatment Completed' and 'Cured' classifications demonstrates the overall effectiveness of the DOH-CAR TB-DOTS program. However, the 8.4% loss to follow-up rate and 3.3% mortality rate underscore the importance of early risk identification and targeted interventions.

An analysis of annual treatment outcomes from 2015 to 2025 revealed significant temporal shifts, reflecting both programmatic improvements and external systemic shocks:

1. Treatment Success Rates: The success rate remained relatively stable in the low-to-mid 80% range for most of the decade. Success rates dipped slightly to 81.03% in 2018 before steadily climbing to a peak of 88.92% in 2023.

2. Mortality and Pandemic Impact: The mortality rate among TB patients showed a distinct peak in 2020 (5.47%) and 2021 (6.63%), up from 3.65% in 2019. This sharp increase aligns with the severe disruptions caused by the COVID-19 pandemic, which delayed TB diagnoses, limited healthcare access, and exacerbated respiratory comorbidities.

3. Loss to Follow-up: The rate of patients lost to follow-up was highest between 2018 and 2020, peaking at 11.99%. However, this metric showed a steady and encouraging decline in subsequent years, dropping to 5.56% by 2024. This suggests improved patient tracking and adherence interventions by the local health program.

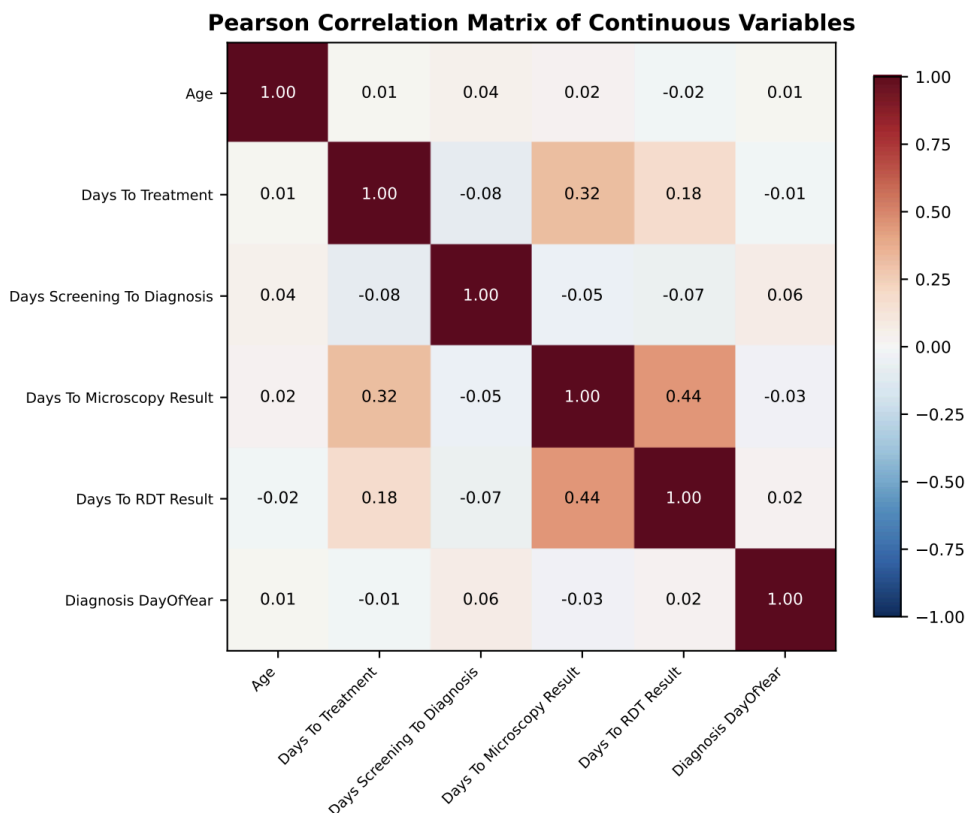
4. 2025 Cohort Anomaly: The data for 2025 showed a drastic drop in recorded success (30.01%) and 52 a massive spike in unresolved cases (64.80%). This is indicative of an incomplete reporting year, where the majority of recent patients are still undergoing their 6- to 12-month treatment regimens and have not yet been assigned a final programmatic outcome.

Diagnostic and Treatment Intervals

The temporal intervals between screening, diagnosis, and treatment initiation were extracted to identify operational delays, which are known risk factors for disease transmission and treatment failure. Median times revealed: (1) screening to diagnosis: 2.0 days (Q3 = 21.09 days); (2) diagnosis to treatment: 2.68 days (Q3 = 13.12 days); and (3) microscopy results: 6.60 days (Q3 = 14.95 days). While median programmatic response is efficient, the long tails represent significant delays for high-risk patients. These interval metrics served as crucial continuous features in the machine learning models, allowing algorithms to correlate logistical delays with eventual probability of mortality or loss to follow-up.

Correlation Analysis

Figure 2
Pearson Correlation Matrix of Continuous Variables



The Pearson correlation analysis revealed weak-to-moderate associations among clinical delay variables. Age exhibited weak negative correlations with time intervals ($|r| < 0.10$), indicating older patients experienced marginally shorter delays. Screening-to-diagnosis and diagnosis-to-treatment intervals showed moderate positive correlation ($r = 0.32$), suggesting that delays at one care cascade stage are associated with delays at subsequent stages. Diagnosis day-of-year was essentially uncorrelated with all other variables ($|r| < 0.05$), confirming seasonal timing is independent of demographic and operational factors.

Static Model Performance

Among all evaluated static models, Random Forest achieved the highest overall performance on the test set.

Table 1
Test Set Model Performance Metrics of Static Classifiers

Model	AUC	Recall (Failure)	F1 (Failure)
Random Forest	0.7154	57.98%	0.3539
Gradient Boosting	0.7121	44.68%	0.3262
LightGBM	0.7119	48.40%	0.3467
XGBoost	0.7095	47.87%	0.3416

Note. AUC = Area Under the Receiver Operating Characteristic Curve.

Random Forest achieved an AUC of 0.7154 and an F1-score of 0.3539 for the failure class, with balanced recall of 57.98% without excessive false positives. Gradient Boosting, LightGBM, and XGBoost achieved comparable AUC values (approximately 0.71), though with slightly lower recall for treatment failure. On the 10% held-out validation set, Random Forest maintained superior performance, achieving a validation AUC of 0.7380 and recall of 52.13% for treatment failure, confirming robustness and suitability for deployment within a clinical decision support framework.

Temporal Model Performance

Table 2 presents the performance metrics of temporal models evaluated on the held-out test set.

Table 2
Test Set Performance Metrics of Temporal Models

Metric	Bi-LSTMBaseline	Bi-LSTMAugmented	XGBoostBaseline	XGBoostSMOTE-ENN	LightGBMSMOTE-ENN	RFSMOTE-ENN
Accuracy	0.9048	0.9048	0.9048	0.9048	0.9048	0.7143
Precision	0.9048	0.9048	0.9048	0.9048	0.9048	0.8824
Recall	1.0000	1.0000	1.0000	1.0000	1.0000	0.7895
F1 Score	0.9500	0.9500	0.9500	0.9500	0.9500	0.8333
ROC-AUC	0.4737	0.3684	0.5789	0.1053	0.2105	0.5263
PR-AUC	0.8847	0.9099	0.9501	0.8356	0.8500	0.9419

Note. Bi-LSTM = Bidirectional Long Short-Term Memory; RF = Random Forest; ROC-AUC = Area Under Receiver Operating Characteristic Curve; PR-AUC = Area Under Precision-Recall Curve.

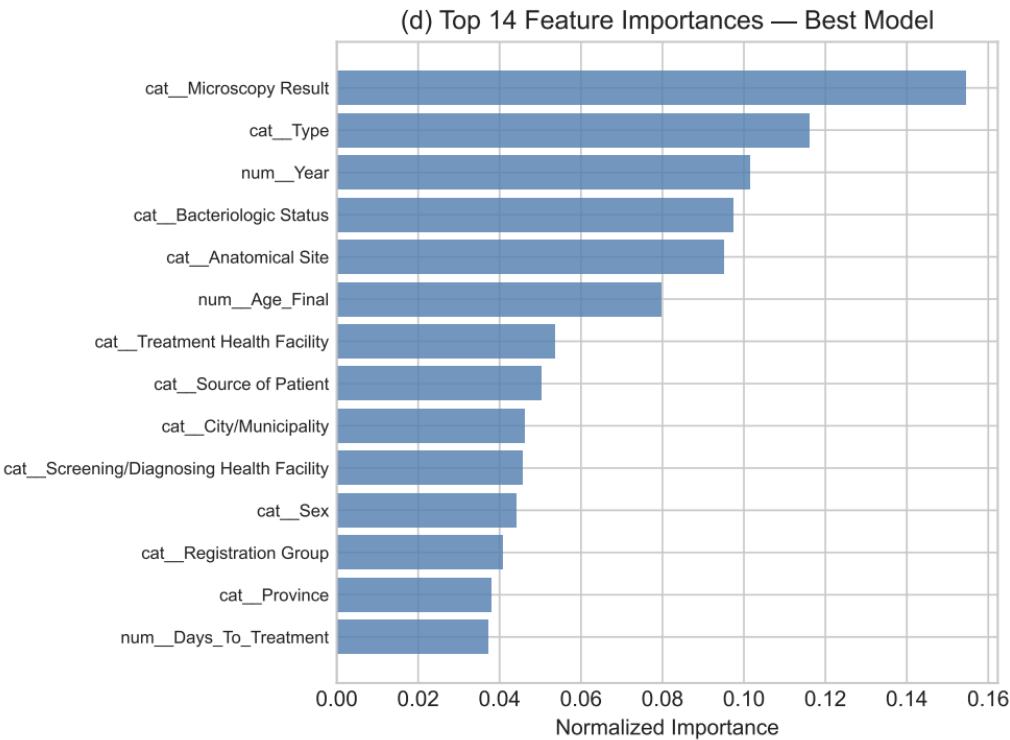
Both Bi-LSTM variants and XGBoost Baseline achieved the highest overall accuracy (0.9048) and F1 score (0.9500), reflecting their ability to correctly predict the dominant Success class. However, specificity remained 0.0, demonstrating that no actual Failure cases were correctly identified in the small test set. Random Forest with SMOTE-ENN showed more balanced performance, achieving slightly lower overall accuracy (0.7143) but higher recall for the Failure class (0.7895) and competitive PR-AUC (0.9419), indicating better identification of high-risk patients. These results suggest that while Bi-LSTM and XGBoost excel at predicting the majority outcome, Random Forest with SMOTE-ENN offers the most effective detection of treatment failure, making it more suitable for clinical decision support applications.

Feature Importance Analysis and SHAP Explanations

To understand which patient-level factors most strongly influence treatment outcomes, we analyzed feature importance across the ensemble models using both model-native importance

metrics and post-hoc SHAP values. Figure 3 presents the top 15 features ranked by mean absolute SHAP values for the Random Forest model, which integrates local explanations from all patients in the validation set.

Figure 3
Top 15 Features Ranked by Mean Absolute SHAP Values (Random Forest Model)



The SHAP analysis reveals that diagnostic and treatment intervals—particularly days from diagnosis to treatment—emerge as the strongest global predictor of treatment outcomes. Monthly adherence metrics (doses taken, percentage adherence) in mid-course months (M2–M4) are also highly influential, indicating that early adherence patterns are predictive of final treatment success or failure. Clinical and demographic factors such as bacteriologic status, anatomical site, age, and comorbidities contribute meaningfully but with lower average impact than operational delays and adherence metrics.

Patient-Level SHAP Explanations and Risk Stratification

Beyond global feature importance, SHAP provides individual patient-level explanations that clinicians can review to understand why the model assigned a particular risk score. Figure 5 illustrates a hypothetical example of how SHAP values are visualized for a single high-risk patient, showing which factors pushed the model's prediction toward treatment failure and by how much.

Clinical Decision Support System Implementation

The validated machine learning models are embedded into a Clinical Decision Support System (CDSS) designed for healthcare providers in TB-DOTS facilities. The system is structured around a clear separation of responsibilities: the predictive model estimates the probability of treatment success or failure, the SHAP layer identifies the factors contributing to the model's prediction, the interface visualizes risk and contributing factors, and the clinician interprets these outputs in the context of patient care.

Key Features of the CDSS

The Clinical Decision Support System (CDSS) is designed as a risk stratification and explanation tool to assist healthcare providers rather than act as an autonomous diagnostic or treatment-prescribing system, ensuring that final clinical judgment remains with the treating professional. At its core, the system features a predictive dashboard that displays the probability of treatment failure and success using color-coded risk stratification, where green signifies low risk <20%, yellow indicates medium risk 20% - 50%, and red denotes high risk >50%, allowing providers to easily prioritize high-risk patients for closer monitoring and targeted interventions. To provide transparency, a risk factor visualization layer utilizes SHAP-based feature

contribution visualizations to highlight specific variables driving the risk score—such as adherence gaps, laboratory results, or diagnostic delays—enabling clinicians to inspect exactly which patient factors the model identified as concerning. This quantitative data is further contextualized by a narrative explanation layer, where a constrained small language model verbalizes the SHAP-based attributions into readable clinical text, explaining, for instance, how a seven-day delay in treatment initiation or low adherence in a specific month quantitatively elevates failure risk. Finally, the CDSS supports dynamic updates by accepting new follow-up data—including monthly weight, smear results, and adherence percentages—and recomputing risk scores in real time to reflect the patient's current clinical status throughout the entire duration of their treatment.

The CDSS is framed as a risk stratification and explanation tool, not as an autonomous diagnostic or treatment-prescribing system. Final clinical judgment remains with the treating healthcare professional.

Figure 4
Clinical Decision Support System Workflow and User Interface Mockup

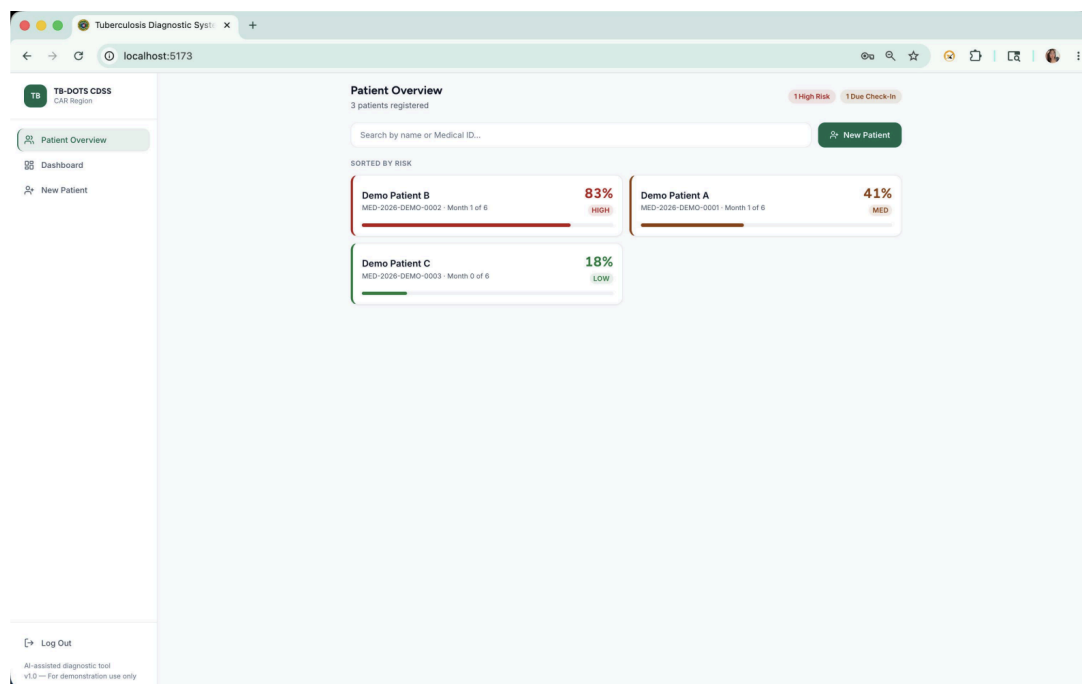
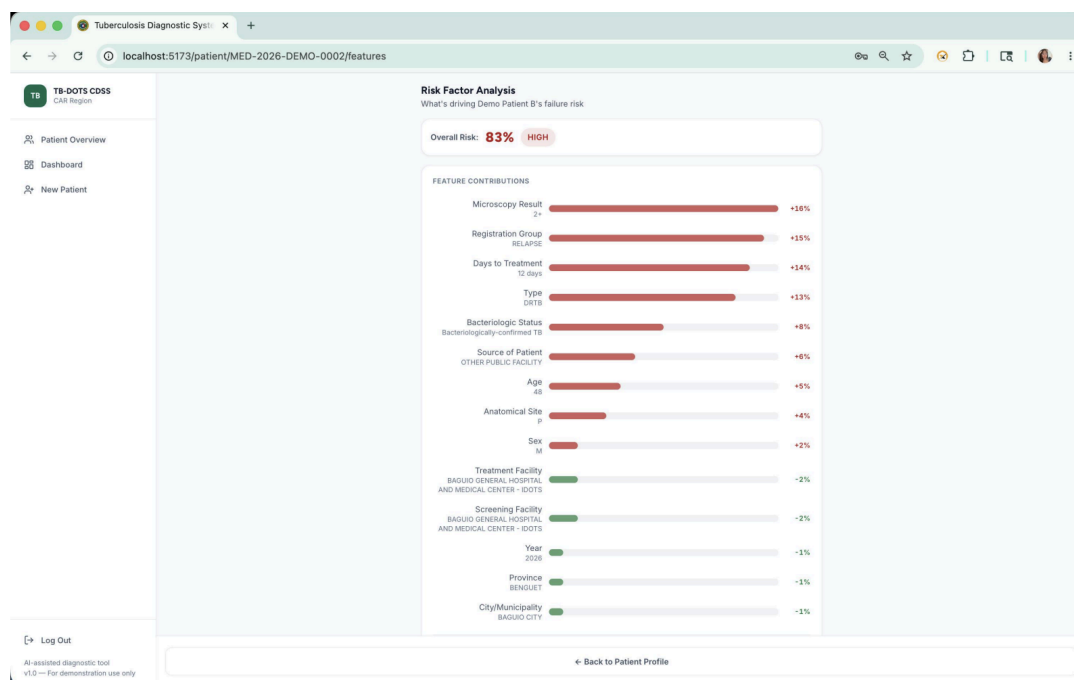


Figure 4 illustrates the integrated workflow of the CDSS. Patient data flows into the predictive model, which generates a treatment failure risk probability. SHAP explanations are automatically computed and visualized, showing the contributing factors. The healthcare

provider reviews both the risk score and the contributing factors, and then makes clinical decisions informed by but not dictated by the system.

Model Generalization and Clinical Utility

The validation set performance (AUC = 0.7380, Recall = 52.13% for failure class) demonstrates that Random Forest with SMOTE-ENN generalizes well beyond the training cohort. This level of discriminative performance, combined with interpretable SHAP explanations, provides clinically useful support for identifying high-risk patients. The 52% recall for the failure class indicates that the model captures approximately half of all patients who will experience treatment failure, enabling earlier identification and intervention compared to standard paper-based monitoring.

The moderate AUC (0.7380) reflects the inherent difficulty of predicting treatment failure from baseline clinical and operational data: many patients with similar risk profiles have disparate outcomes due to unmeasured social determinants, medication side effects, or unpredictable treatment adherence. However, the model's ability to identify a substantial minority of high-risk patients, combined with interpretable explanations of why they are flagged, provides meaningful value for TB control programs aiming to concentrate limited resources on the most vulnerable populations.

Conclusions

This study investigated the use of machine learning models to predict treatment outcomes among PTB patients under the DOH-CAR TB-DOTS program, using monthly patient monitoring data alongside baseline clinical features. The test set performance revealed that while Bi-LSTM and XGBoost Baseline models achieved high overall accuracy (0.9048) and F1 scores (0.9500), their ability to detect treatment failures was limited by zero specificity for the minority Failure class. In contrast, Random Forest with SMOTE-ENN demonstrated more balanced performance, achieving a recall of 0.7895 for treatment failure and a high PR-AUC (0.9419), indicating greater reliability in identifying patients at risk of adverse outcomes.

The study demonstrates that integrating temporal monitoring data with static baseline features improves predictive performance over static data alone. The application of SMOTE-ENN effectively addresses the extreme class imbalance inherent in TB treatment outcomes, enhancing the model's capacity to detect high-risk patients. Shapley Additive Explanations prove invaluable in bridging the gap between predictive performance and clinician interpretability: by surfacing the specific patient-level factors driving each risk prediction, SHAP enables healthcare providers to understand and trust the model's outputs.

Predictive models—particularly Random Forest with SMOTE-ENN—have practical value for clinical decision support systems, enabling clinicians to identify patients at risk of treatment failure early and prioritize interventions accordingly. The embedded CDSS, with its real-time risk updates and narrative explanations, provides a scalable tool for TB program managers to strengthen treatment outcomes. Overall, this study aligns with Sustainable Development Goal 3 – Good Health and Well-Being by providing a data-driven approach to improving TB treatment outcomes in the Cordillera Administrative Region.

Recommendations

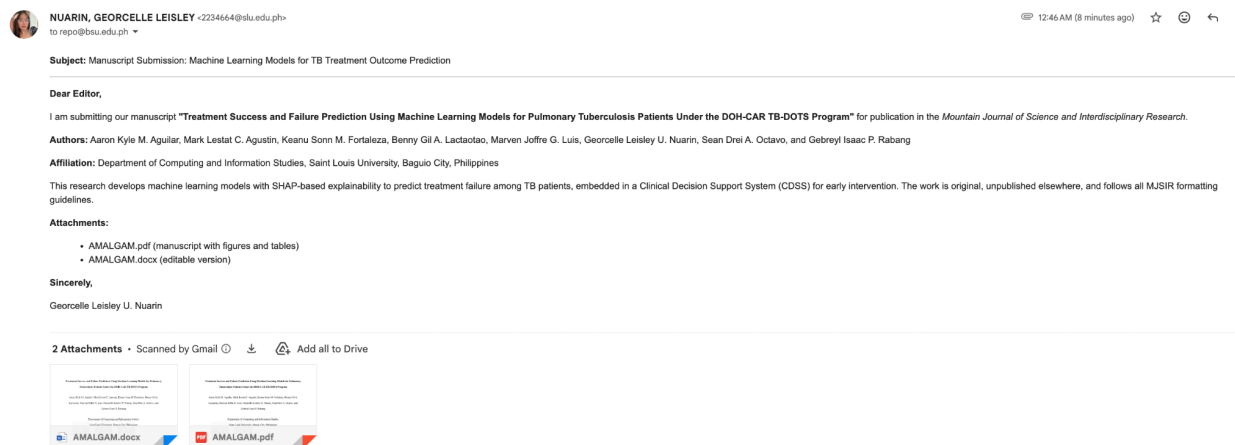
Based on the findings, it is recommended that healthcare facilities and policymakers consider integrating machine learning models like Random Forest with SMOTE-ENN into existing TB monitoring programs to improve treatment outcomes and optimize resource allocation. Future research should focus on larger, multi-region datasets to enhance model generalization and explore hybrid modeling approaches combining deep learning and tree-based methods. Investigations into additional features such as social determinants of health and geographic risk factors are encouraged. Enhancing model interpretability—particularly of Bi-LSTM attention outputs—could improve clinician trust and adoption. Standardizing digital record-keeping practices across TB-DOTS facilities will support development of more comprehensive longitudinal datasets for future studies and enable real-time implementation of the CDSS at scale.

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Appendix A: Proof of Submission to a Journal Publication



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Appendix D: Proof of Presentation



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